# MountWizzard4

## Version 4.0.0a38

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# Welcome to MountWizzard4!

MountWizzard4 is a general utility for 10micron users for improving the workflow for astronomy work. It runs on Windows11, Windows10, Mac OSX (beginning from 10.12 to 26.x) and Linux (Ubuntu from 16.04 to 26.04). If you have some knowledge around Raspberry Pi's and other SOC, you might be able to install MountWizzard4 on a RPi4 or RPi5.

The PDF version of the documentation could be found here: https://mworion.github.io/MountWizzard4/mountwizzard4.pdf

**Hint**

MountWizzard4 version 4 is a major release with many new features and improvements. Please check the changelog for details: https://mworion.github.io/MountWizzard4/changelog/changelog.html. Actually this is a beta version ! Please check the known limitations and report any issues you find. Install it to a new workdir and do not overwrite your existing v3 installation. You can have both versions installed at the same time.

# Before starting

First let us have a look to the basic architecture: MountWizzard4 is an application installed on your external computer which is connected to the mount computer via an IP connection. The best choice is to use a wired connection. As the 10micron mounts also support a serial line, please be reminded MountWizzard4 does not! Many of the features are handled on the mount computer itself and MountWizzard4 does the GUI frontend for the user by using the command protocol provided by 10micron.

The basic idea is that MountWizzard4 will try to generate “digital twin” for the mount. All parameter changes for the mount will be sent to it and changes of it’s state are polled to make status visible in MountWizzard4. Therefore regular polling of data is needed.

# Overview

Beside this documentation there is a youtube channel available with descriptions, previews, explanations:

https://www.youtube.com/channel/UCJD-5qdLEcBTCugltqw1hXA

For full operation MountWizzard4 supports several frameworks: INDI / INDIGO, ASCOM, Alpaca and in addition Sequence Generator Pro and N.I.N.A. as camera device.

# Known limitations

MountWizzard4 v4 does support python 3.11 - 3.14 right now. The reason for that is the lack of precompiled packages. Some features are limited to windows as they need the original 10micron updater program for execution. As python moves with its releases forward, also MountWizzard4 will follow.

On windows please check if you are working in a 32bit or 64bit environment. You need to choose the ASCOM setup (drivers etc.) and the python install accordingly.

If you are using the 10micron updater features on windows, MountWizzard4 remote controls the updater application. Please do not interrupt this automation.

# Reporting issues

To have an eye on your setup here are some topics which you could check:

- Mount connection available and stable. Wifi might have performance problems. Look for right network settings in mount and local setup.
- Good counter check is review settings, status bars, message window if something is going wrong.

To improve quality and usability any feedback is highly welcome! To maintain a good transparency and professional work for my, please respect the following recommendations how to feed back.

**Note**

Please report issues / bugs here:

https://github.com/mworion/MountWizzard4/issues.

And if you have feature requests discussions or for all other topics of interest there is a good place to start here:

https://github.com/mworion/MountWizzard4/discussions

In case of a bug report please have a good description (maybe a screenshot if it's related to GUI) and add the log file(s). Normally you just could drop the log file (or PNG in case of a screen shot) directly to the webpage issues on GitHub. In some cases GitHub does not accept the file format (unfortunately for example FITs files). I this case, please zip them and drop the zipped file. This will work. If you have multiple files, please don't zip them to one file! I need them separated and zipped causes more work.

If changes are made due to a feedback, new releases will have a link to the closed issues on GitHub.

# Table of contents

## Features

For being fully operational, MountWizzard4 needs either:

- INDI server(s) (see: https://indilib.org) where your devices are connected to.
- INDIGO server(s) (see: http://www.indigo-astronomy.org) where your devices are connected to.
- ASCOM Alpaca remote server (see: https://ascom-standards.org/FAQs/Index.htm) abstracting your ASCOM devices or devices which speak native ASCOM Alpaca if you want to connect over IP with your environment.
- Running versions of Sequence Generator Pro or N.I.N.A. as frontend for camera device.
- For the core devices there is native ASCOM support (Windows platform only). Please be reminded, that ASCOM has 32bit and 64bit driver implementations and MountWizzard4 could also be installed in 32bit or 64 bit python environment. They must be not be mixed! 32bit python supports only 32bit drivers and vice versa. Normally this should not be an issue. I strongly recommend to use 64bit python and 64bit ASCOM drivers.
- In addition an internet connection is used for some services which might be very helpful.

It is recommended to use mount firmware 3.x or later as some of the functions don't work with older firmware versions. It should not be a problem using older versions. A HW pre2012 might also have some issues. MountWizzard4 supports also older firmwares from 2.x onwards, but with limited features and untested.

It is recommended to use mount firmware 2.16 or later as some of the functions don't work with older firmware versions.

Here is an overview of the functionality available in MountWizzard4:

- Many settings and features of the mount can be shown and changed.
- Control movement of the mount as well as tracking speeds.
- Coordinates in J2000 as well as in JNow.
- Virtual keypad

- Model building with different model setups and model generating capabilities. Sorting points for effective slew paths or dome situations.
- Model building is done in parallel threads (imaging, plate solving, slewing) to reduce time.
- Show the actual model and alignment error. Give hints on how to improve the raw polar alignment.
- Model optimisation: deleting points, automatic removing point for target RMS etc.
- Manage models stored in the mount (save, load, delete).
- Dome geometry integration (MountWizzard4 knows about target flip side and slews dome correctly as well as any geometrical constraints).
- Environment data: MountWizzard4 shows data from OpenWeatherMap, ClearOutside, External Sensors like MBox, Stickstation, UniHedronSQR as well as direct linked sensors like MGBox.
- Refraction handling external / internal from the above sources.
- Satellite: searching, displaying, programming, updating tracking.
- Tools: FITS Header renaming, Park positions, etc.
- Remote shutdown of MountWizzard4 and Mount via IP commands.
- Measurements and CSV saving for most environment and mount data
- Imaging: control of connected camera / cooler / filter.
- WOL (wake on LAN) boot for mount. MountWizzard4 catches MAC address automatically on first manual start.
- Audio signals for different events (end slew, finished modeling, alert, etc.)
- Updater for all MountWizzard4 functions.
- Generate / load / save as many profiles as you would like.
- Show alignment stars. Choose and automatically center for polar or orthogonal adjustments.
- Imaging: expose one or N images, auto solve or auto stack these images.
- Imaging: show distortion grid, astrometric calculations (flux, roundness, sharpness)

# Installation

**Hint**

MountWizzard4 version 4 is a major release with many new features and improvements. Please check the changelog for details: https://mworion.github.io/MountWizzard4/changelog/changelog.html. Actually this is a beta version ! Please check the known limitations and report any issues you find. Install it to a new workdir and do not overwrite your existing v3 installation. You can have both versions installed at the same time.

MountWizzard4 v4 (and also valid for v3) will change the way of installation. As it relies on python and uses a virtual environment the whole process got to complicated to maintain and was not easy for the users to follow. Therefore installation changes to an overall public free solution called **uv**.

**uv** is able to handle python installations and versioning, activating and deactivating of virtual environments, installing and versioning MountWizzard4 an that all with only 3 commands across all supported platforms (Windows, Linux and MacOS).

There are as well some videos to watch the installation process:

- Windows11: https://www.youtube.com/watch?v=iKBfWjA5I0Q
- RASPI: https://www.youtube.com/watch?v=7dPa8rjAssE

## Installing **uv**

I strongly recommend to install uv as a standalone program. The description could be found here: https://docs.astral.sh/uv/getting-started/installation/. To install do the following steps:

- open a terminal window on your system
- select from the install page your corresponding OS
- copy and paste the terminal command calls the install script and let it run

After that, uv is installed. To move on, your terminal is not aware of the new features of uv. The simplest way to do get to this point is to close the terminal windows and open a new one.

Please check the presence of uv with the command

```
uv --version
```

If you get the version number, everything is perfect.

## Installing python and the virtual environment

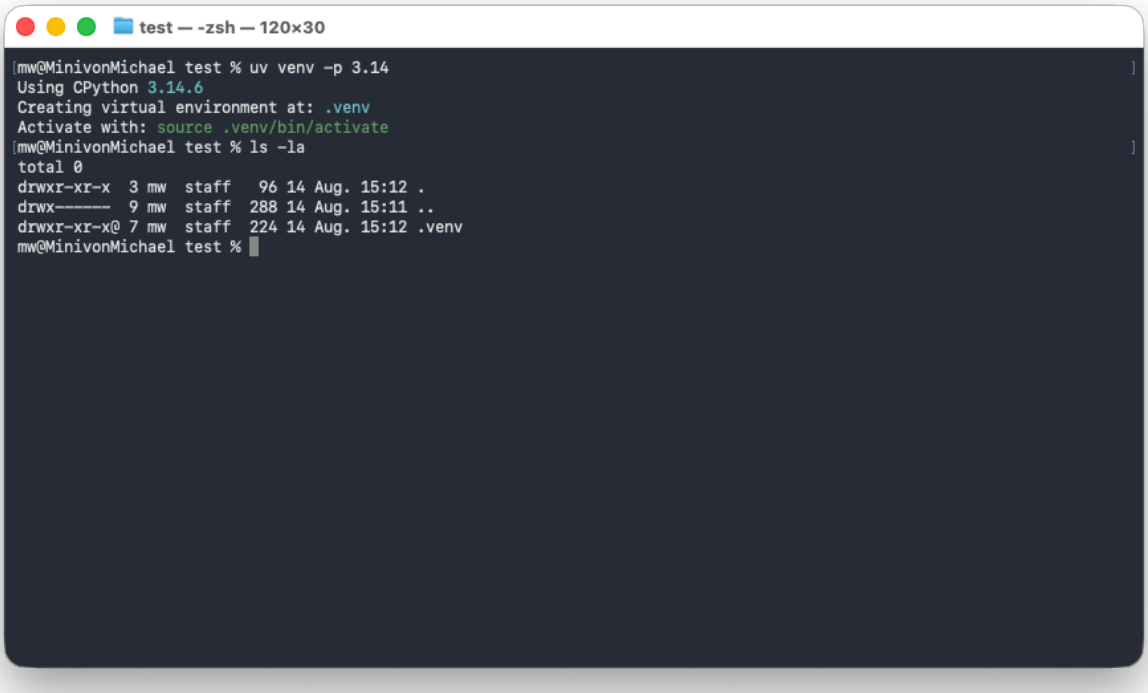
There is no need to have python installed on the forehand and there no dependency to any python installation already present with the use of uv. Mountwizzard4 v3 plays on python 3.8 - 3.10, so I would recommend to use the n ewest one 3.10. For installing, please choose a work folder for Mountwizzard4, open a terminal window and change directory to this work folder. the command

```
uv venv -p 3.10
```

does download the right python package (3.10) and installs a virtual environment in you work folder. The virtual environment is stored in a directory there called ".venv". If you go for testing the new v4 version, please install python 3.14 with the command:

```
uv venv -p 3.14
```

The resulting virtual environment is only for this work folder and does not interfere with any other python installation on your system. You can have as many virtual environments as you like on your system, but they are all separated from each other. This is the main advantage of using uv and virtual environments in general. In you work folder it should now be a new folder called ".venv" with the virtual environment:




## Installing MountWizzard4

The installation is now a simple command:

```
uv pip install mountwizzard4
```

Installation will download the necessary packages (visible) and installs them in the virtual environment. If you are using RASPI hardware, please check the RASPI information below as well!

If you want to have a dedicated version of MountWizzard4, just add the version number in the command line (e.g. 4.0.0) like

```
uv pip install mountwizzard4==4.0.0
```

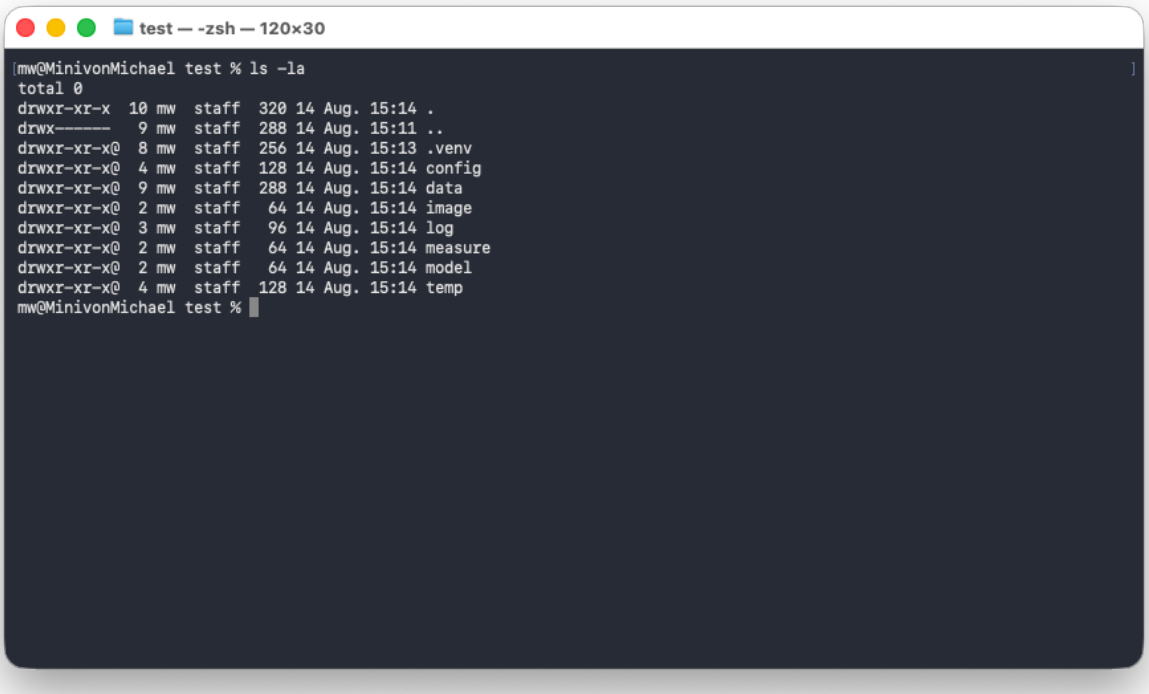
Changing versions upgrade and downgrade work the same way just with the appropriate version numbers.

## Run MountWizzard4

Once everything installed running MountWizzard4 is also a simple command:

```
uv run mw4
```

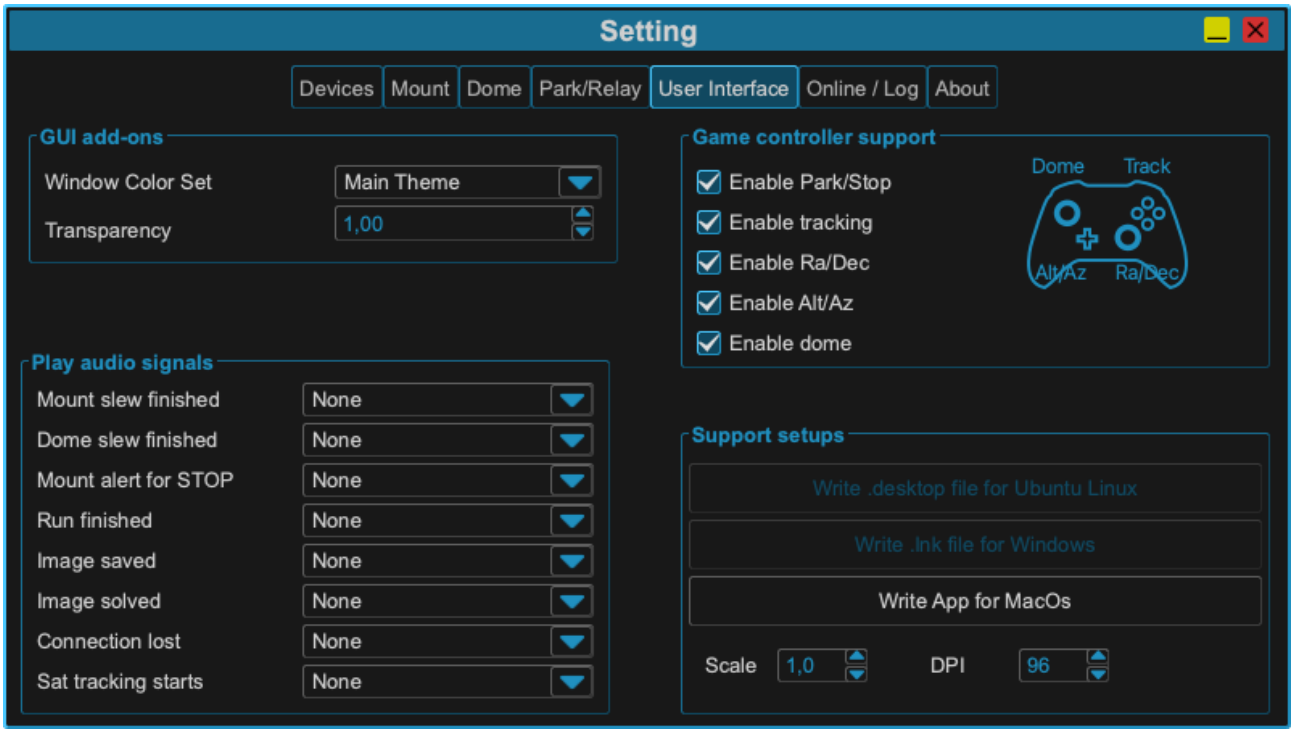
If you would like some options please use the command line parameters MountWizzard4 supports. These are setting for the appearance sizes on windows primary. If you look after the first run, your work folder should now contain new folders:



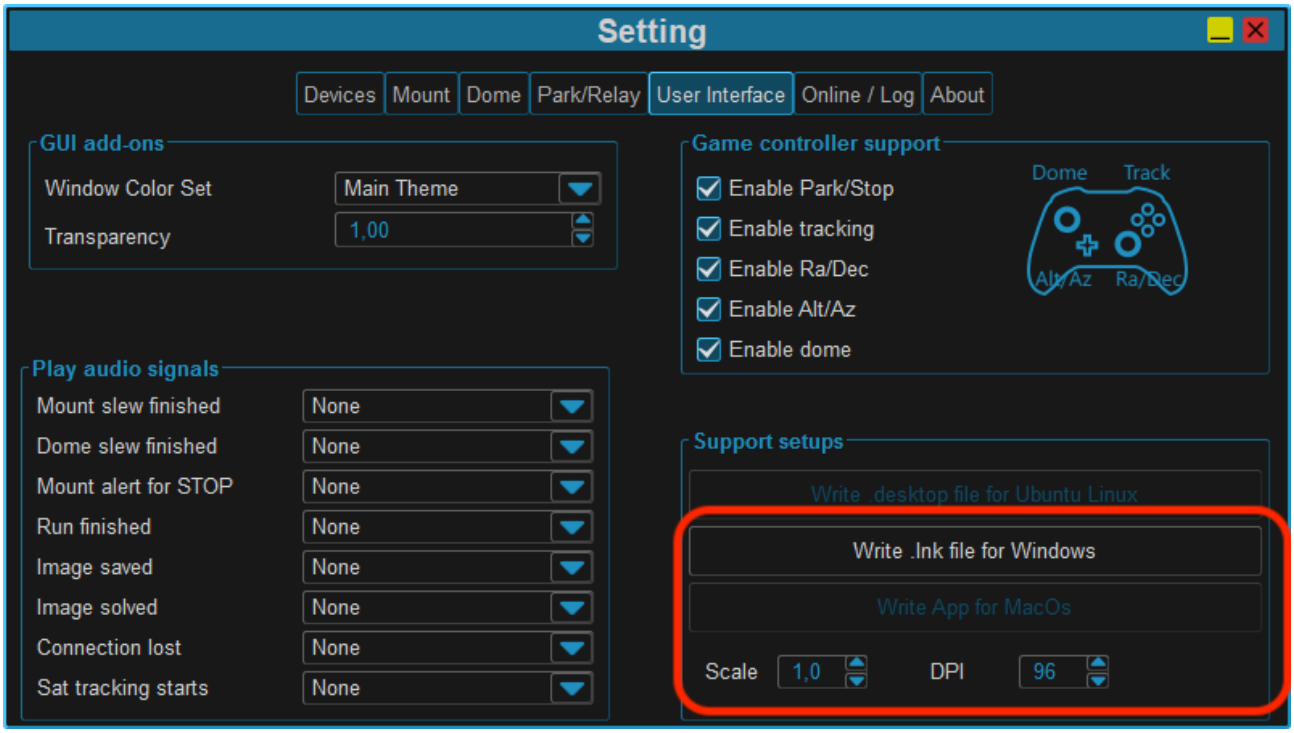

Basically that's it.

## Comfort setup for MW4 users

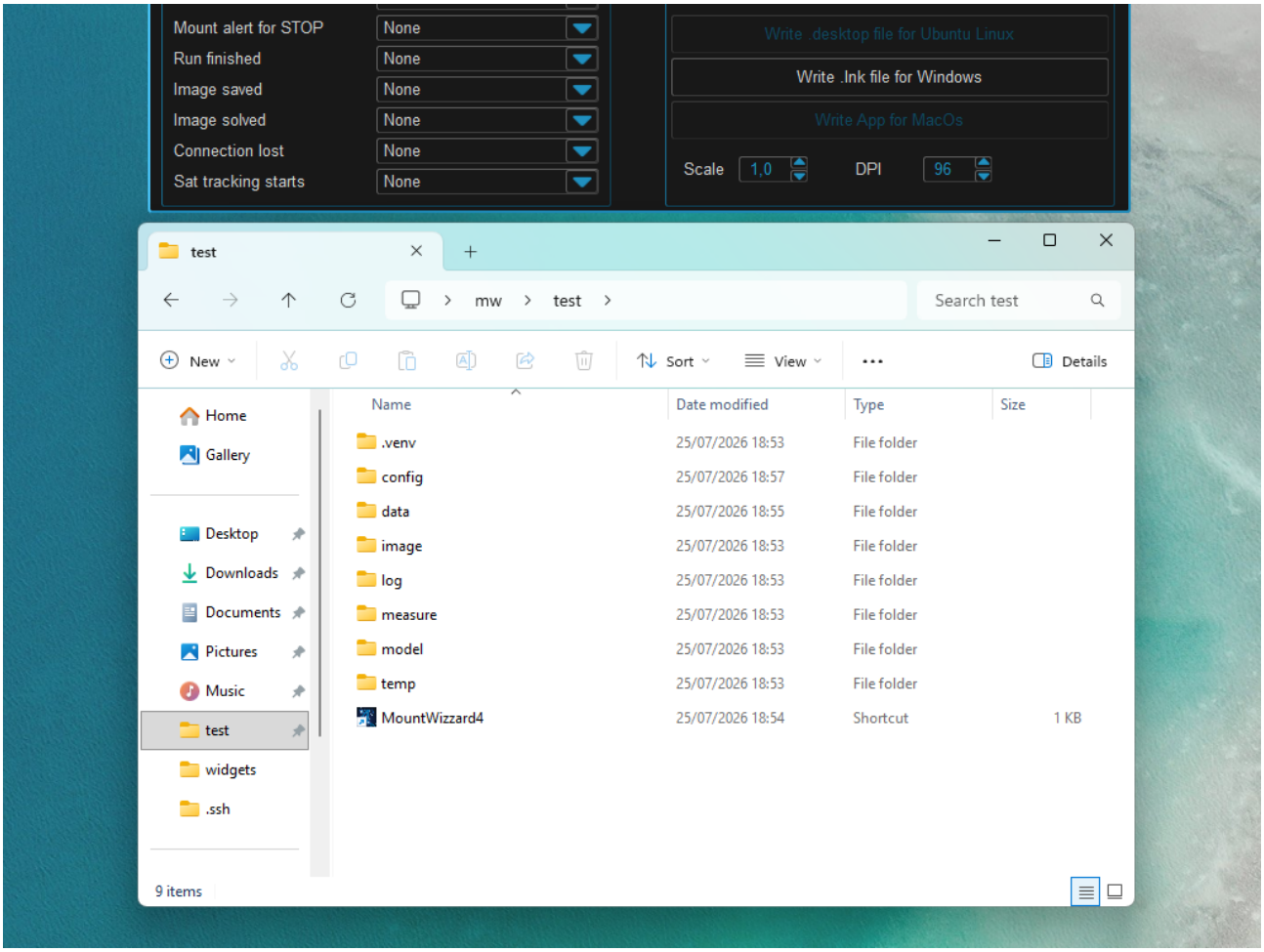
The first start has to be done manually with the command above. After that, you can use MountWizzard4 for optimizing the user experience. Please check the following tab underr setting:




MountWizzard4 offers a number setups for the user on windows, ubuntu and macOS. For each operating system dedicated setup files could be generated to be able to integrate MoutWizzard4 into the gui of the system.




For e.g. windows it generate a link with an icon to start MountWizzard4 with a double click and stores it in the work folder. From there you can move it to your desktop or any other location you like. This includes also the startup paremeters for the appearance of MountWizzard4: DPI and scale factor. For linux it generates a desktop file to integrate it, and for macOS it generates a .app bundle to integrate it into the system. I will look like this:




All these setups are generated in the work folder and could be moved to any other. The files generated coudl be manually edited.

**⚠ Warning**

When generating the setup files, please make sure that you have the right permissions to write into the work folder. If not, please change the permissions or move the work folder to a location where you have the right permissions. Existing files will be overwritten without any warning.

## Update MountWizzard4

Once everything installed running MountWizzard4 is also a simple command. You should run the command from the root of your work folder where the virtual environment is located. Run the command

```
uv pip install mountwizzard4 --upgrade
```

to update MountWizzard4 to the latest version. If you already have installed v3 of MountWizzard with uv, please do not use the internal update anymore. It will be removed in v4. UV handles the update / downgrade and the relations to python versions better the my (limited) attempt. As MountWizzard4 will follow newer libraries and therefore also python versions, you will have to update this in parallel in the old setup, but uv will manage this for you out of the box.

## Use MountWizzard4 beta versions

If you would like to tryout beta versions (or any dedicated version) of MountWizzard4, please use the standard install method. For the latest beta you just can add the following parameter:

```
uv pip install mountwizzard4 --pre
```

## Command line options

MountWizzard4 supports a number of command line options:

```
'-d', '--dpi'
```

Setting QT font DPI (+dpi = -fontsize, default=96)

```
'-s', '--scale'
```

Setting Qt DPI scale factor (+scale = +size, default=1)

```
'-h', '--help'
```

Getting this information.

## Integrating MountWizzard4 linux system

If you plan to run MountWizzard4 on a linux system, you can use the additional files for linux in the support_install folder.

https://github.com/mworion/MountWizzard4/tree/main/support_install

These are some support for starting MountWizzard4 with a desktop icon and some other features. Please check this website for further information to customize the content of the mountwizzard4.desktop file to your needs and how to install it:

https://itsfoss.gitlab.io/post/how-to-install-and-edit-desktop-files-on-linux-desktop-entries/

## Integrating MountWizzard4 Windows system

If you plan to run MountWizzard4 on a windows system, you can use the additional files for windows in the support_install folder for setting up a link with an icon to start MountWizzard4 with a double click.

## Additional for RaspberryPi (4, 5)

**Hint**

This is only necessary for v3. MountWizzard4 v4 brings all binaries directly. For v4 just proceed like above.

If you want to install MountWizzard4 v3 to an Raspi or ARM64 based computer, please follow the installation process. of uv, python and virtual environment. Before starting MountWizzard4 v3 you need to install some precompiled packages as they were not available for the necessary python 3.8-3.10 versions. For that use the same terminal window pointing to you work folder and add the two following commands:

```
uv pip install https://github.com/mworion/mountwizzard4/raw/main/
wheels/PyQt5-5.15.9-cp38.cp39.cp310-abi3-manylinux_2_17_aarch64.whl
```

After that, please proceed with the use of MountWizzard4 like described above.

## Installing on StellarMate

A good option to run MountWizzard4 on a linux based system is the use of StellarMate. The installation process is similar to the one described above, but there are some specific steps to follow. Please check the following link for detailed instructions: https://stellarmate.com/support/faq/stellarmate/how-to-install-mw4-on-sm-os-2-0.html

## Updating python in workdir

There might be some reasons for updating the python part due to security patches or simply to upgrade the version

```
uv venv -p <new python version string> --allow-existing
```

Please use the parameter –allow-existing to keep your current installations.

# Configuration

The configuration part of the documentation is divided into several sections, each of which describes a different aspect of the configuration of the observatory setup. There were several aspects of tuning MountWizzard4 to your needs, and this section will help you to understand how to do that. Still there is many more information in tooltips and help texts in the application itself.

**Hint**

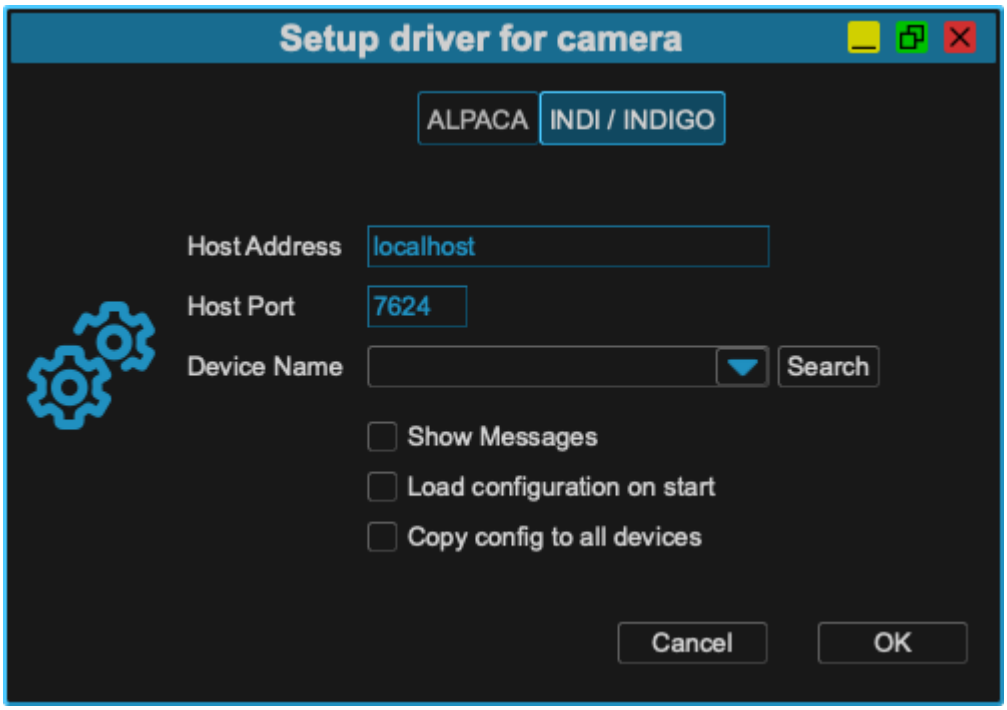
MountWizzard4 v4 is a major release with many new features and improvements. It comes with a new config file format it is more human readable in case of expert tuning topics. Please setup your configuration with the new version and do not overwrite your existing.

## Adding the camera

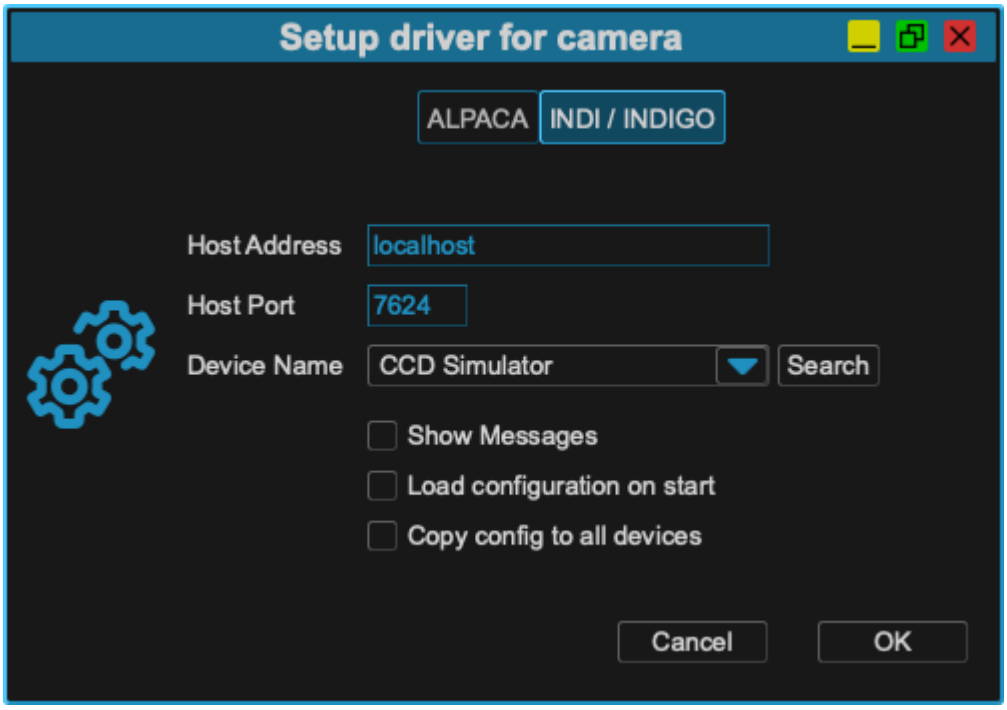
For adding devices like the camera select the Sett. / Conf. tab in the main menu and there select the devices tab. For the following explanation we would like to connect a camera, a filter and adding a link to the mount as well. The mount link is only used for reading the parameters of the mount driver of your setup (e.g. focal length, aperture).



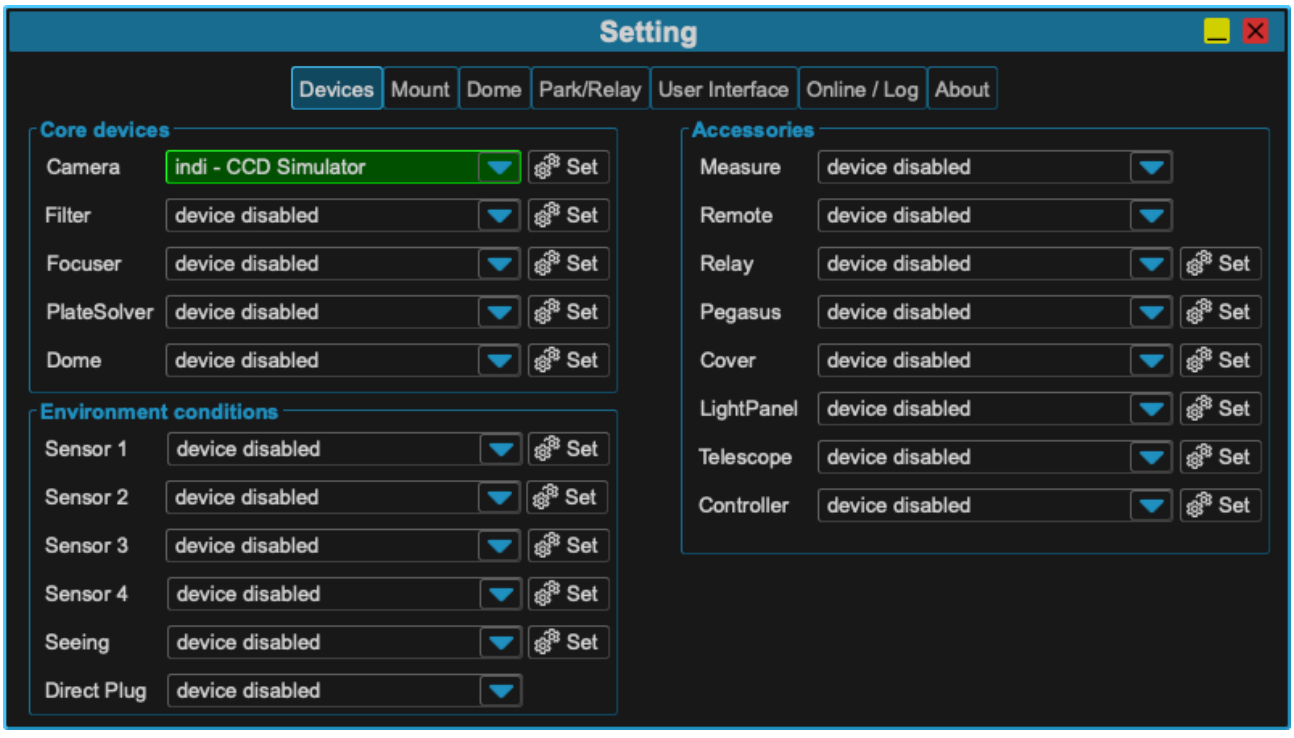

In core devices select Setup für Camera. A popup shows up. Please fill in the ip address of the INDI or ALPACA server, where your devices are connected to.




If you are using INDI, you search for INDI camera devices by clicking on the search button. Once you finished searching by pressing OK button, a list of available devices will be populated. From the list choose the device and finish the setup with OK button.




The selected camera will be highlighted green in the drop down menu and in the status as well. Once you configured the camera, the selection list will be stored for later use. If you want to disable the camera, please select device disabled in the menu.

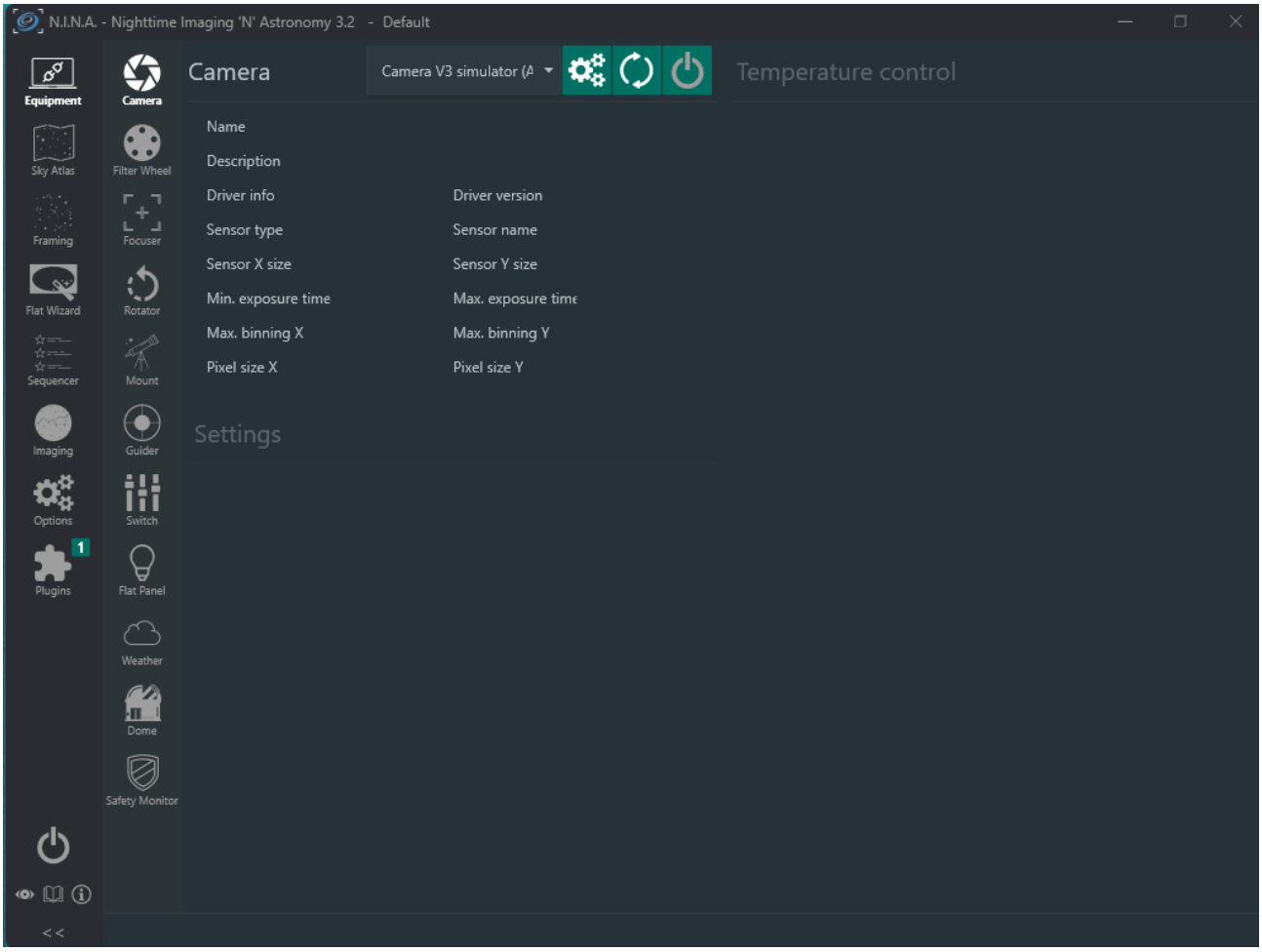



MountWizzard4 will now try to connect to the device and show green light whenever a connection is established.

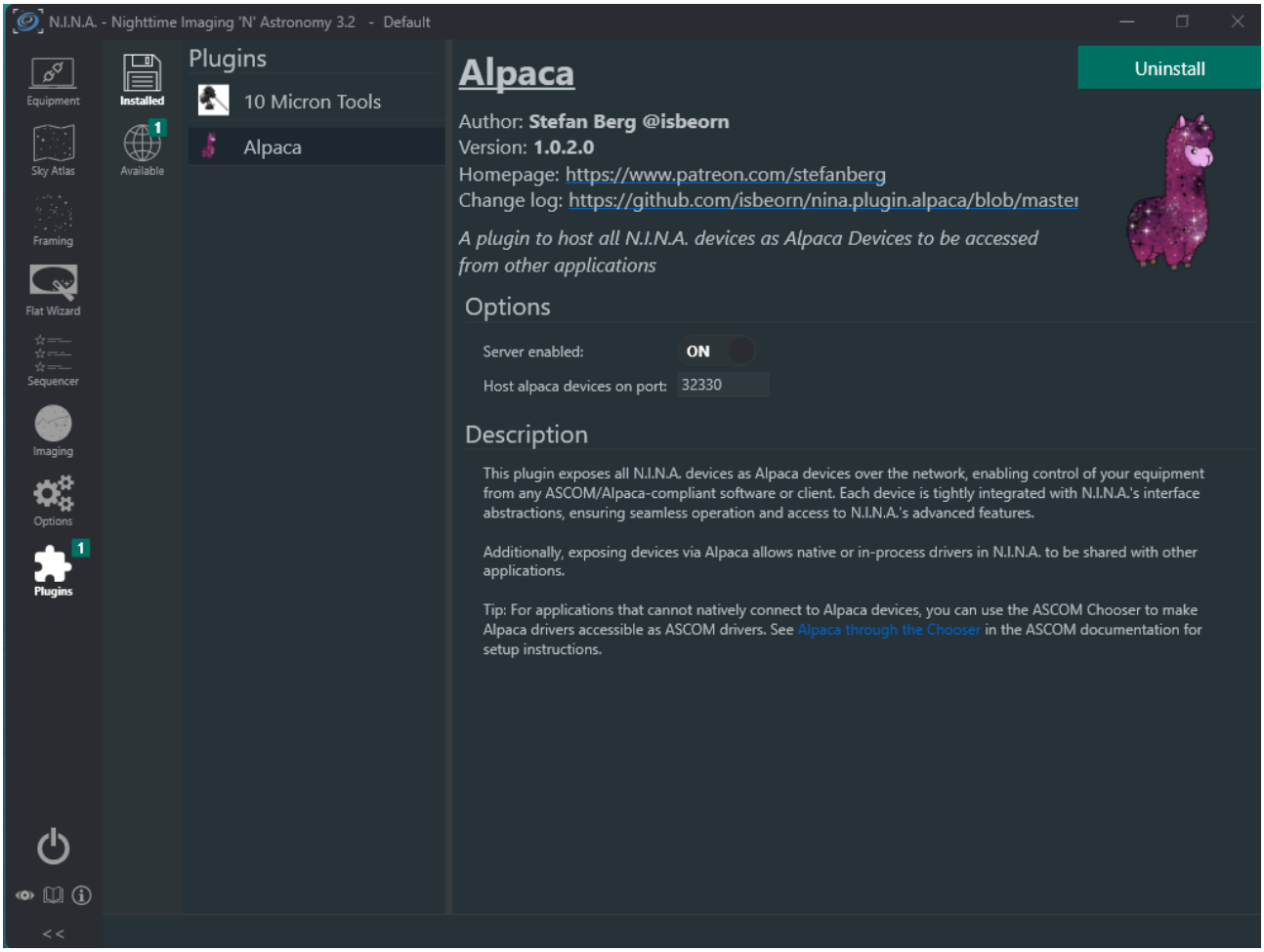
Do the steps 1 - 5 for all devices you need to configure. All configuration are save when leaving MountWizzard4 with Save/Quit button or just when saving the profile. You can add or change any config later on at any time.

## Using N.I.N.A. as device

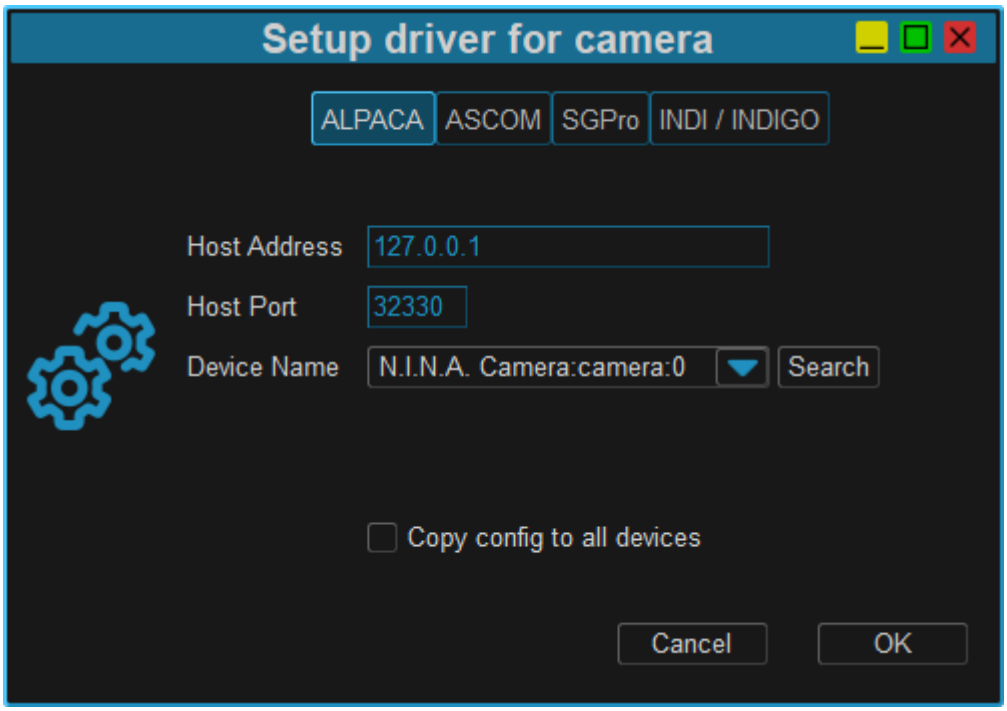
please install the ALPACA plugin for N.I.N.A. and enable it. This will allow you to use N.I.N.A. as a device in MountWizzard4. Beside the camera you could also use other devices than cameras, e.g. filter wheel, focuser, dome as well.



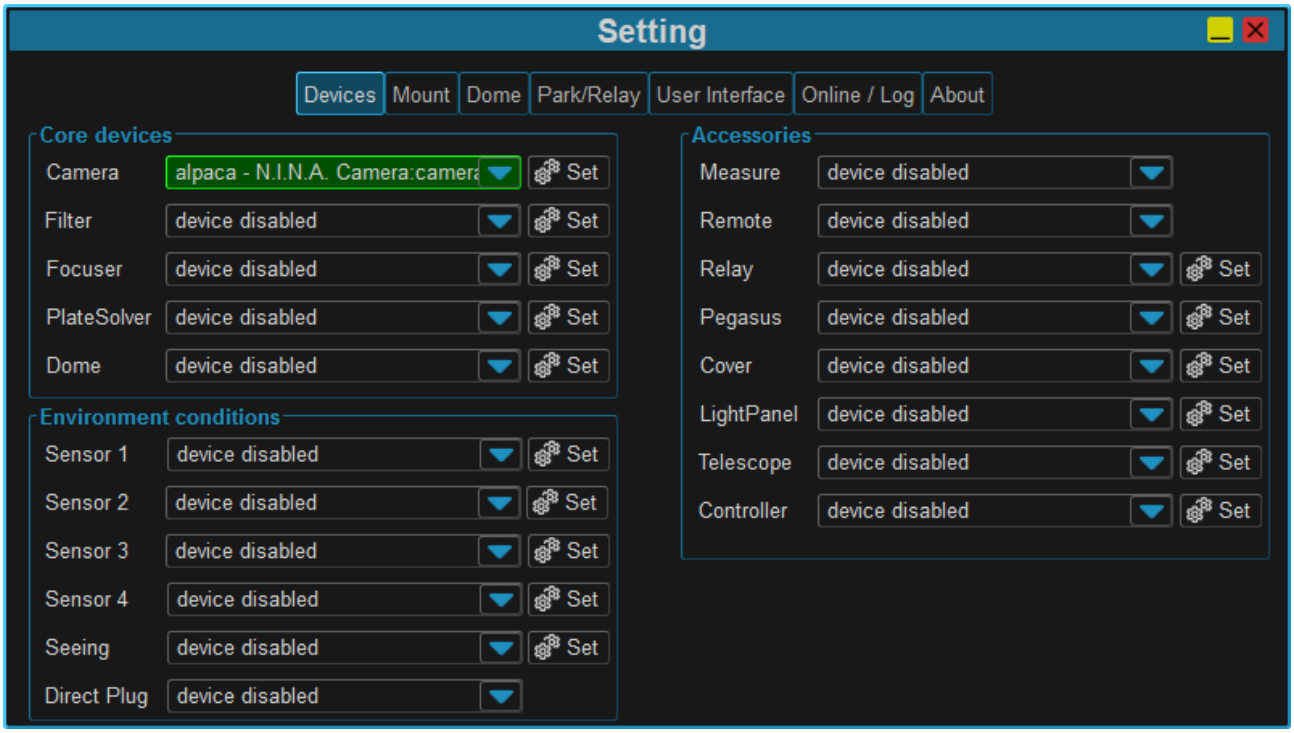

Alpaca plugin for N.I.N.A. is available.




Please enable the plugin and configure the port for the Alpaca server. The default port is 11111. but you could choose it freely. Please note that the port must be the same as in MountWizzard4 an need to be entered in the camera setup as well. The IP address of the computer running N.I.N.A.

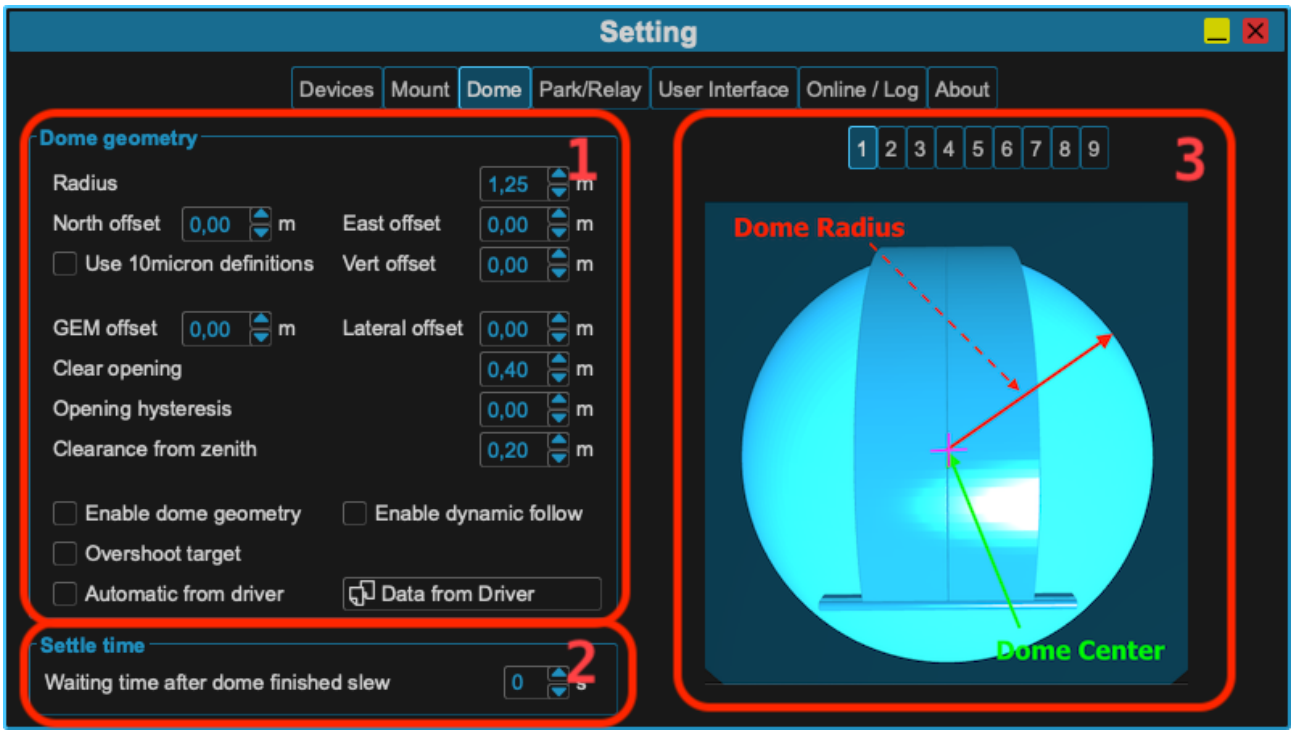



After closing the setup window, the N.I.N.A. device in the camera setup should be available:



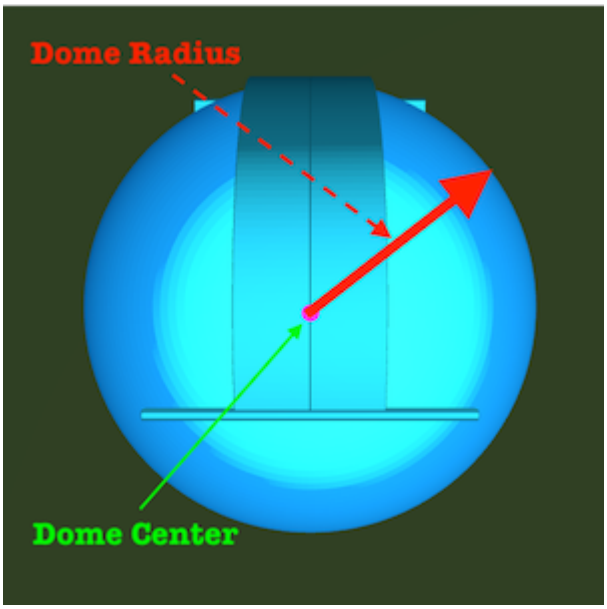

## Dome geometry

If you using dome, you should set up the parameters to get a good performance of the slew calculations. To use the calculations, please check the “Enable dome geometry offset calculations”. If checked, MountWizzard4 will calculate the correct dome azimuth in relation to the desired target of the mount. It take the pierside already into account, so you could slew mount and dome at the same time.

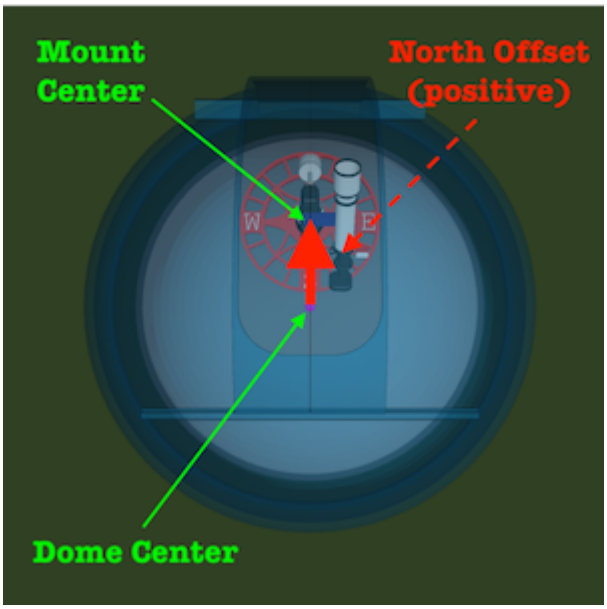


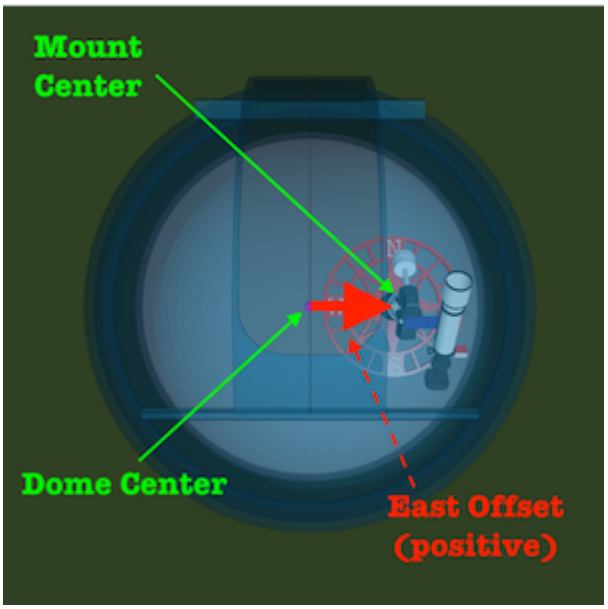
## Area 1: dome geometry parameters

I this section the geometry parameters of the dome and mount in relation to the dome are set. The dome center - the center point of the dome hemisphere - is always the reference point from where the measurements are taken. All measure are in meters. It starts with the radius

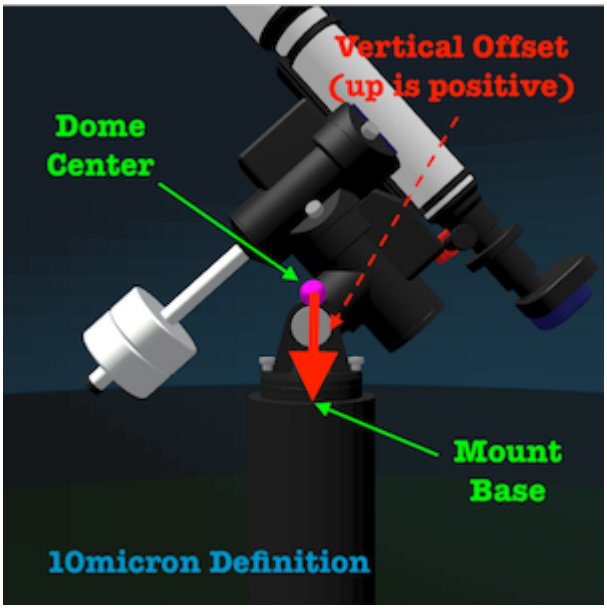


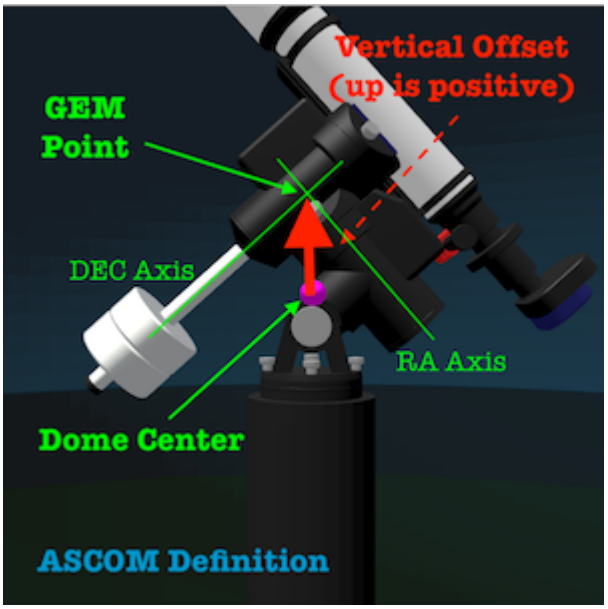

followed by the displacement of the mount in north and east direction. A displacement to the north is treated positive as well as a displacement to the east.



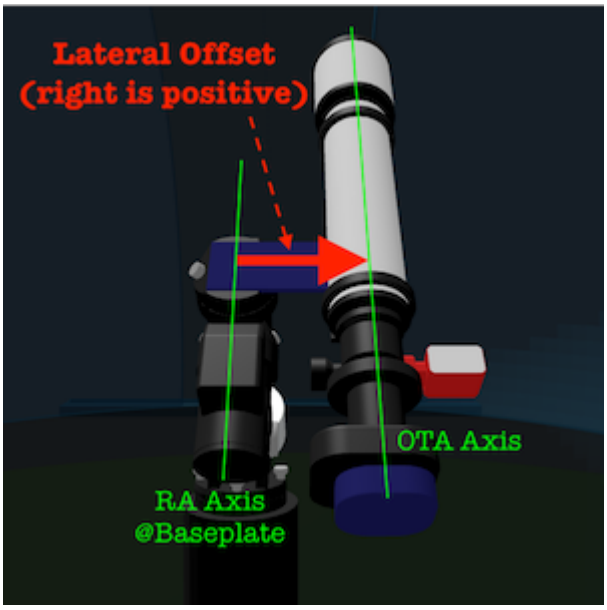




For the vertical displacement you have to take car about different definitions of how to measure this value. A common way many ASCOM drivers do is measuring the distance between the dome center and the GEM point of the mount. The GEM point of the mount is the intersection of the RA and DEC axis. Ideally you will have that point fitting to the dome center. In the 10micron handbook, the measurement is done between the dome center and the base of the mount. This value is mostly negative, because the mount base level is on lower height than the dome center.




If you have mor than one telescope mounted, you will experience a lateral displacement of the OTA you would like to use for reference. The measurement is taken from mw4.base plate of the mount to the OTA axis. Displacement to the right (east) is treated positive.




## Area 2: dome settle time

With the dome settle time you define the waiting time from reaching the target azimuth until the dome emits slew finished. This could be used for avoiding mechanical influence on the mount. This settle time in handled in MountWizzard4 only. If you could enter a settle time in your dome driver etc. the times will add up.

## Area 3: dome geometry help

As geometry parameters are difficult to remember, MountWizzard4 will show you a short explanation picture of the parameter you are currently changing.

## Dome dynamic follow

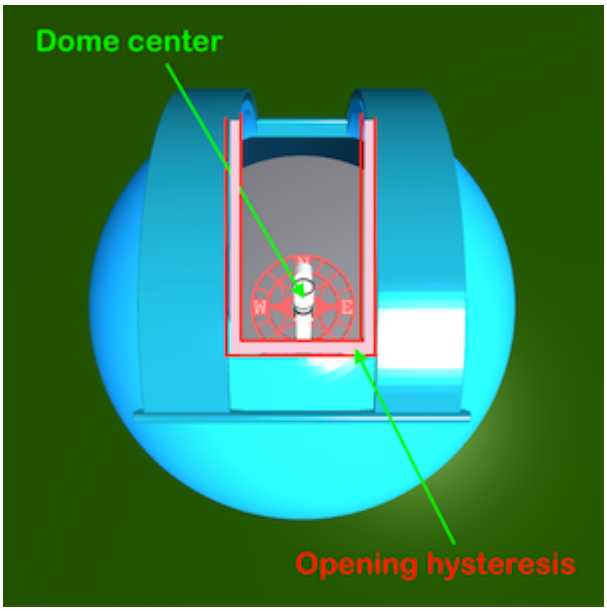
For satellite tracking moving the dome adequately is very important. If you have following on your dome controller, you might have set a hysteresis to avoid continuous dome movements. These values normally are between 1 and 3 degrees. MountWizzard4 could handle this different. If you set your dome geometry correctly, you could add parameters for target shutter distance and shutter zenith distance (better terminus ???). With these parameters MountWizzard4 will calculate if the viewing spot of you telescope moves with the next slew command in the “red zone”. If so, the dome azimuth will be corrected and the dome will slew to it’s new centered position. If there is still a reasonable distance, your telescope still could view your target and no dome movement is necessary.

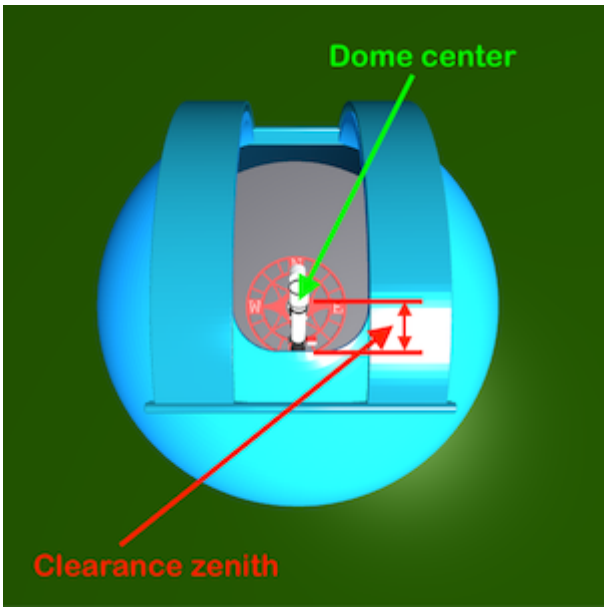
**Hint**

See also on youtube: https://youtu.be/ZmDz-rtvFzc

Top view of your dome setup:





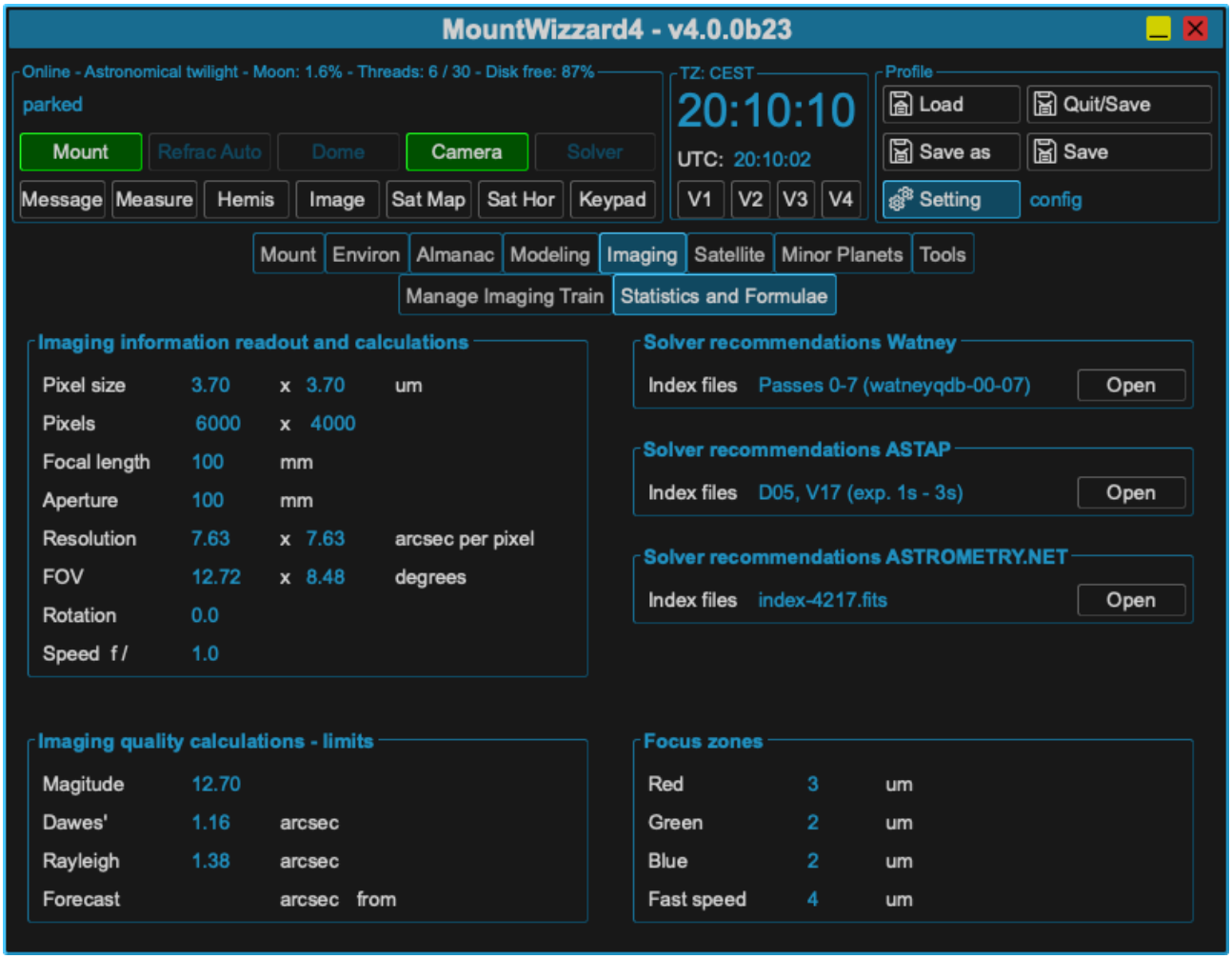

## GUI adaptations

The GUI of MountWizzard4 could be configured to your needs in many ways. You could set your theme (1) or to setup audio signals (2) for some events. There is also a game controller support to move your mount (3) and you write standard setup files for your used OS (4).

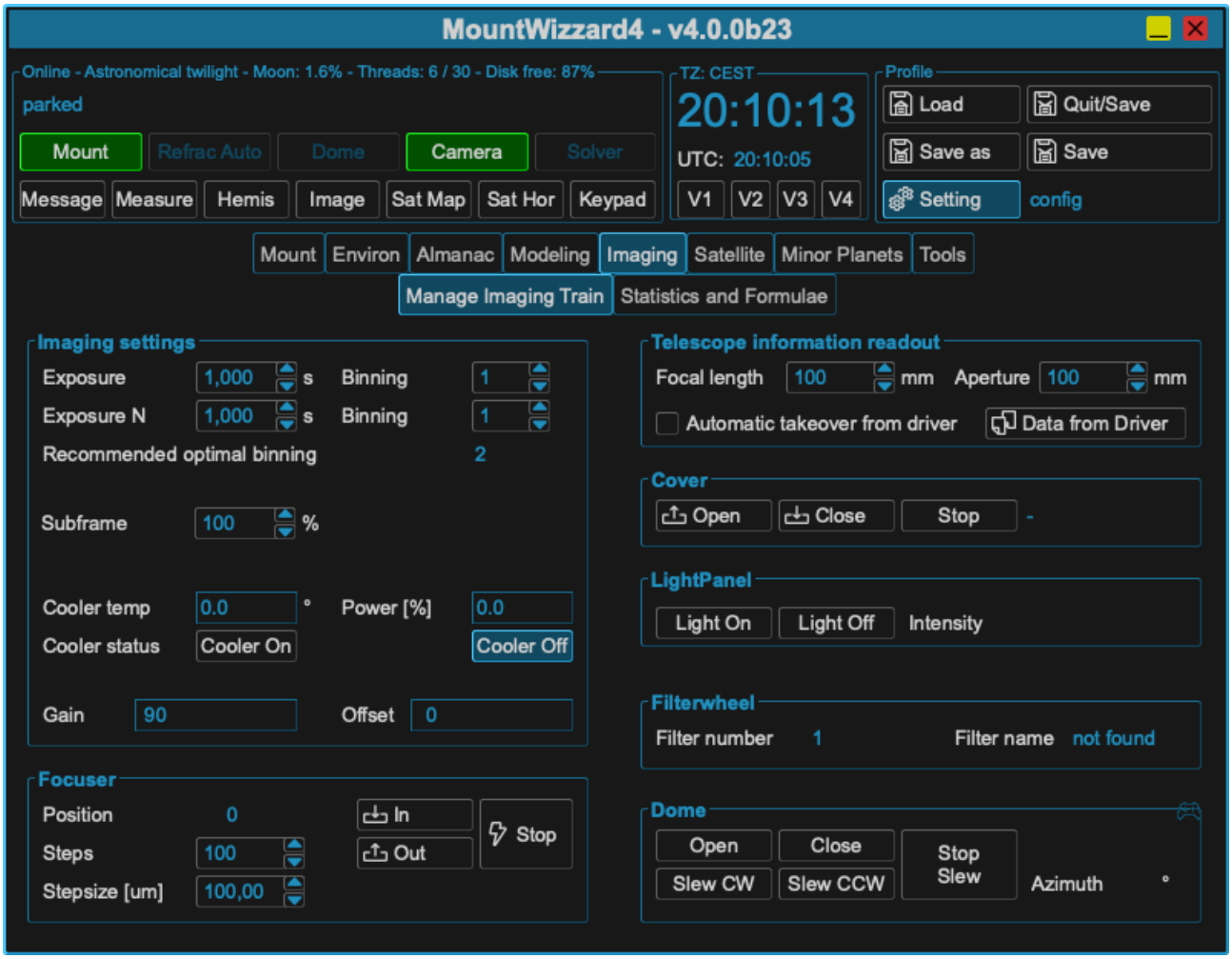



## Imaging setup

For setting up your imaging configuration, you have to enter the following parameters or ensure that these are set correctly: pixel size, pixel count, focal length, aperture. If you have some glitches there, plate solving will not work afterwards in the process. You could check the parameters in the following page:




In addition MountWizzard4 allows you to control some devices related to imaging once they were configured in the device manager:

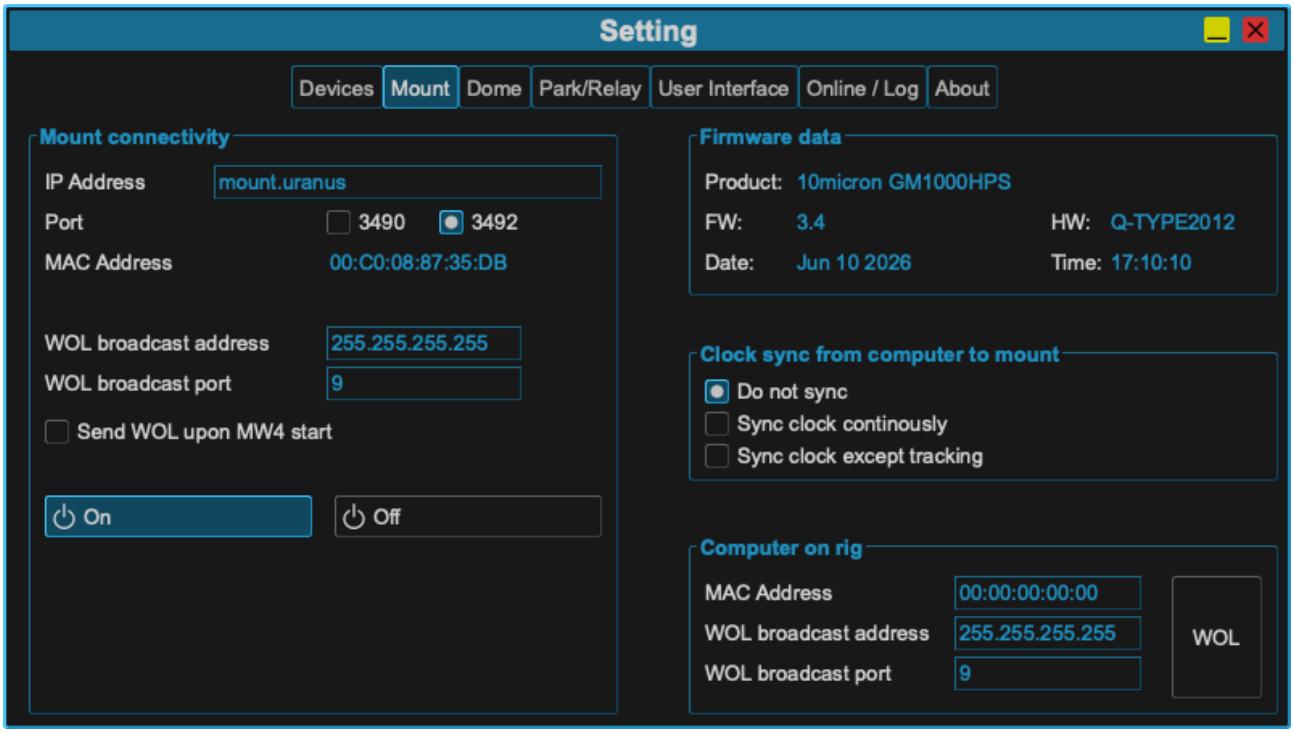



For exposure control you could set the following parameters: exposure time, binning, subframe, gain, offset. Please take into account that the exposure time and binning is split for cyclic exposures (ExposureN) and single ones. The Gain, Offset ans some more control only appear enabled if the driver and device allows accessing them.

# Mount setup

## Basic connectivity

With the first start you have a clean installation. First step should be to enable the mount connectivity. In the settings tab goto Mount and enter under Mount connectivity the IP address of you mount (1). Boot the mount manually and wait until the mount computer is ready.



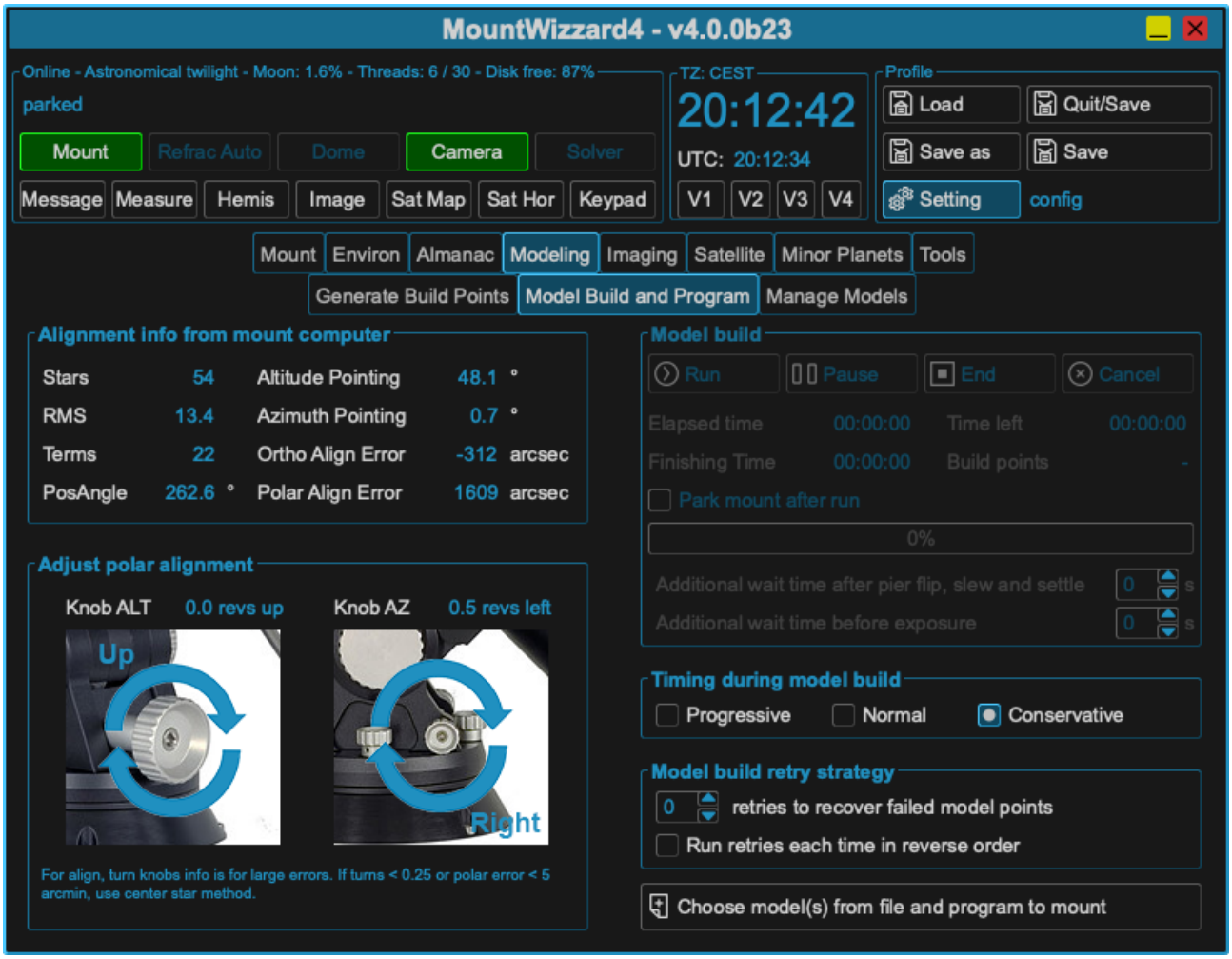

MountWizzard4 will show for Mount connection a green light and enters the MAC Address for remote boot via wake-on-lan (WOL). You should select WakeOnLan on startup in the menu (2). All configuration are save when leaving MountWizzard4 with Save/Quit button or just when saving the profile. You can add or change any config later on at any time.

## Settling Time / Waiting Time

To accommodate several different use cases MountWizzard4 implements additional waiting times to the core settling time, which is implemented and user directly from the mount computer. The following image shows the setting of this parameter, which could be also set and altered through 10micron tools.




This settling time is valid for all slews and movements of your mount once set. Please have a look to the 10micron spec where this behaviour has to be taken into account. Nevertheless for the modeling part MountWizzard4 add two more parameters as the modeling process need heavy movement of the mount. Therefore MountWizzard4 call these parameters not settling time but waiting time. These parameters could be set under the mount parameters:




The working principle is as follows: MountWizzard4 initiates a slew. This command is run by the mount computer and takes the internal settling time into account. This means after the mount came to physical stop, the mount computer will send the signal slew finished after this time period (upper image). This is the case in all used cases and will applied also during modeling process.

For the modeling process MountWizzard4 **adds** a waiting time before moving on after slew, which means waiting the addition set time before starting a next exposure (you know that MountWizzard4 runs asynchron for slew, expose and plate solve to improve speed). The wait is only applied during the modeling process.

Furthermore MountWizzard4 will differentiate if the mount starts and stops on the same pierside or if the was a meridian flip of the mount. For both cases you could set the waiting time.

## Park Positions

MountWizzard4 supports setup and definition of several park positions of the mount. This is useful for mounts which are used in different locations and where the user wants to store the park position for each location.




## Terrain image & horizon mask

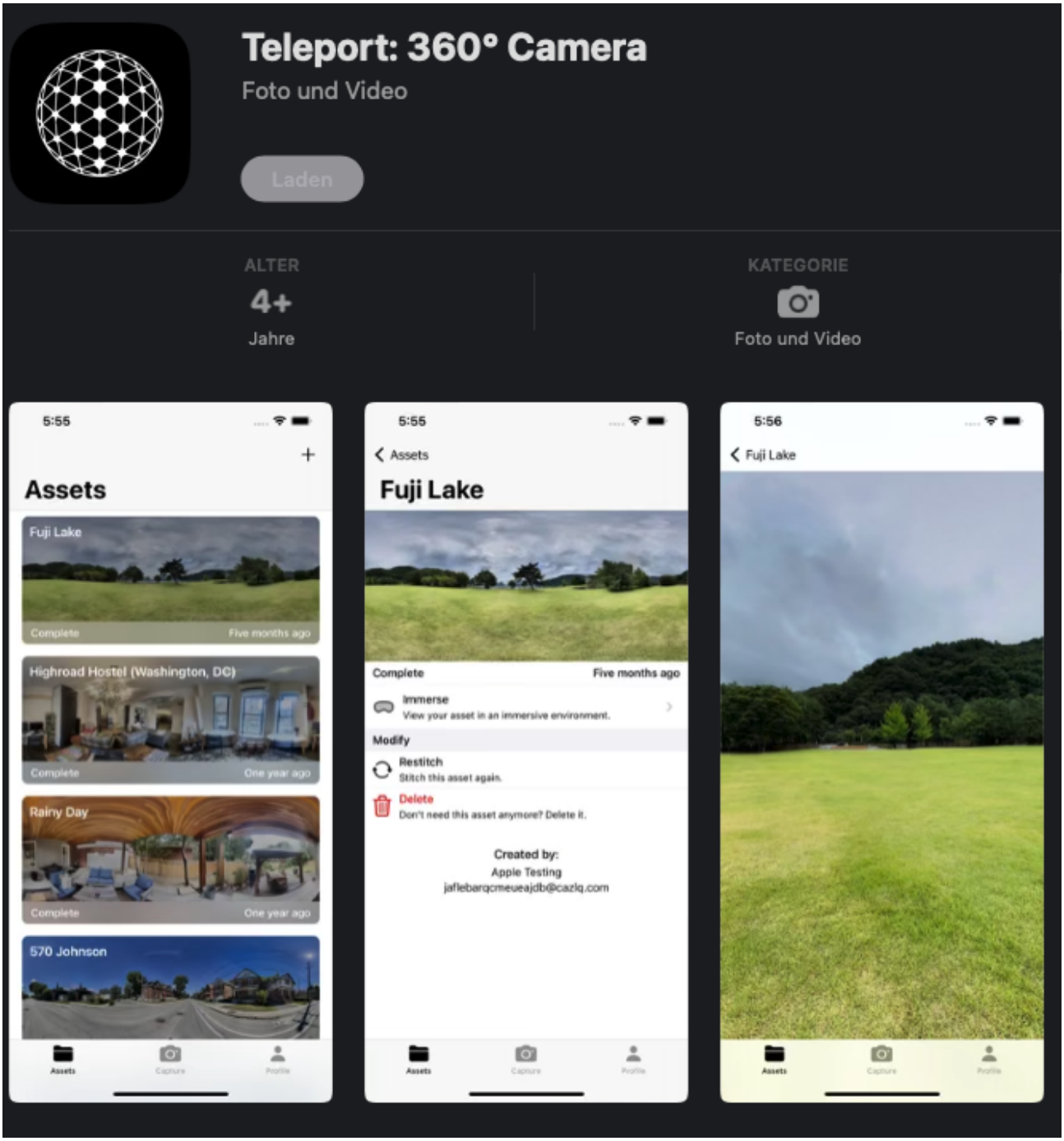
For an optimal use of model points, slew path etc. it is necessary to define a horizon mask for the use in MountWizzard4. There are several way to define and use a mask based on your actual environment

**Hint**

See also on youtube: https://youtu.be/EqDNNMNMS0w

### Using terrain image from Teleport: 360 app

Please use Teleport: 360 app to generate a 360 degrees image around your mount rig. You can save as a .jpeg then rename it to terrain.jpg. I then had to adjust the azimuth paying attention to high points using compass readings.




If you finished the image and transferred it to you computer it might look like:

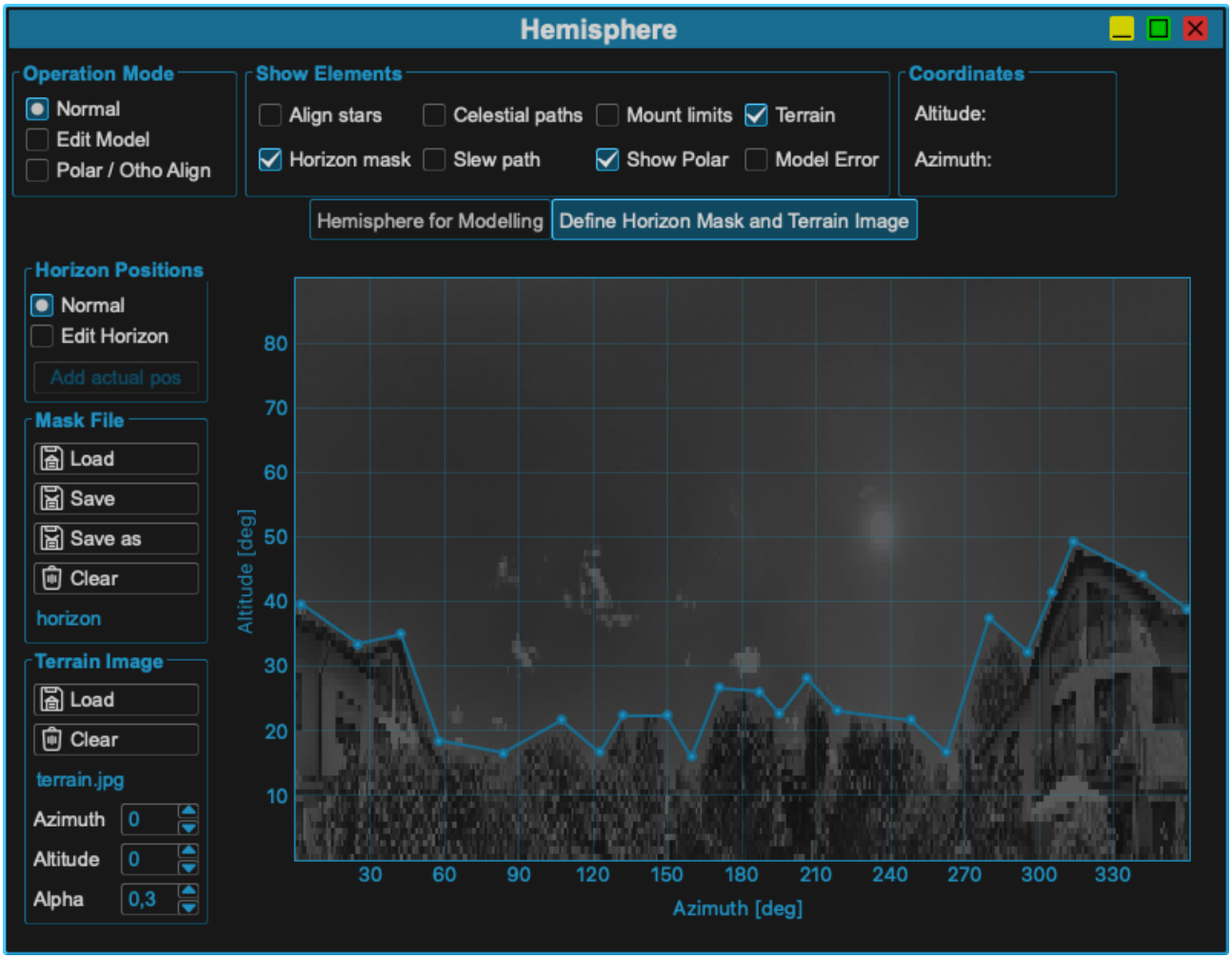


Please put this image to you config dir in your MountWizzard4 working folder. The file format output from streetview is 4096 X 2048 pixel color as JPG. The horizon for altitude 0 degree cuts the image in an upper and lower half. MountWizzard4 uses only the upper half of the image as it expects the image to be taken at the height of the horizon line.

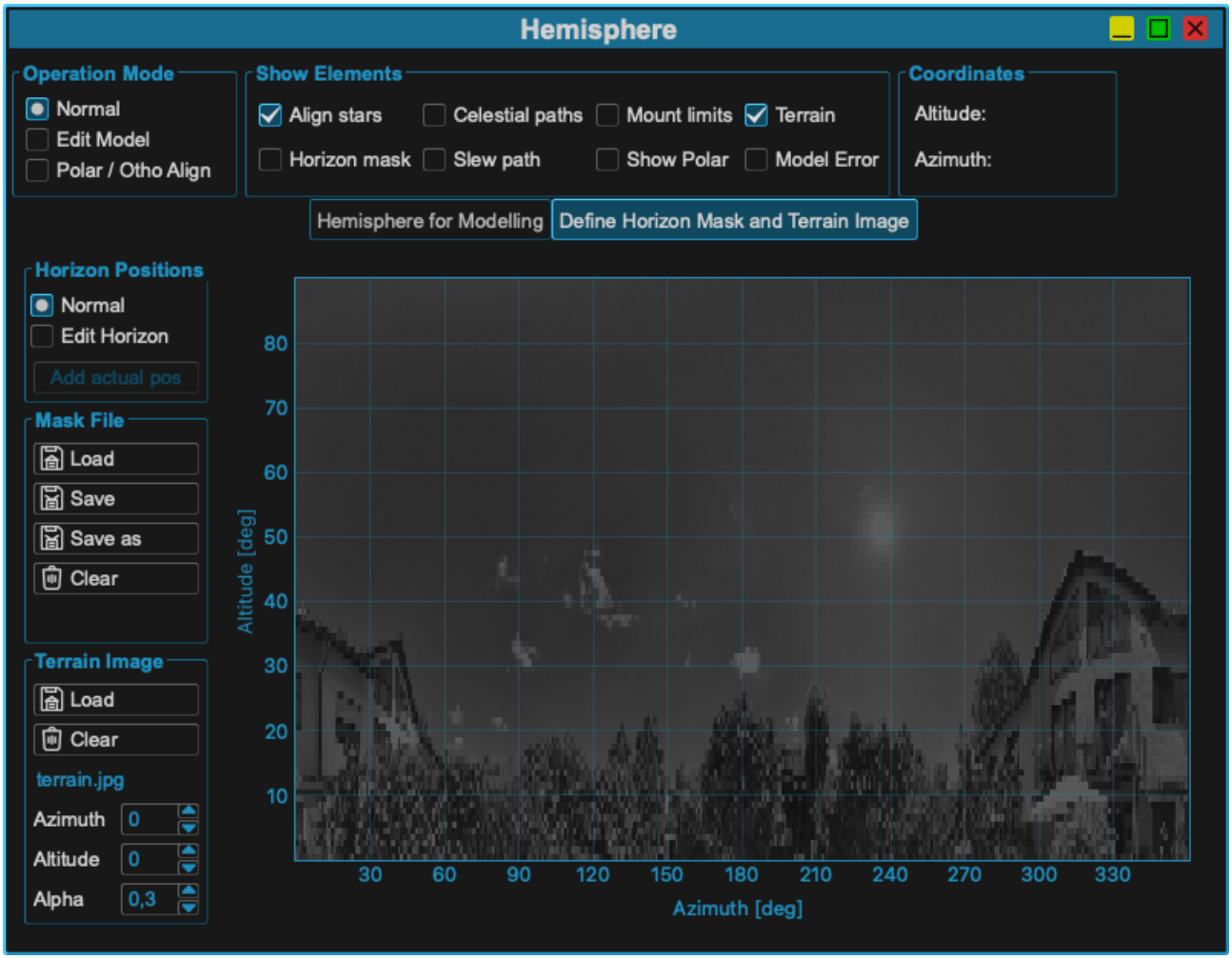
**Note**

The image file has to be named to: **terrain.jpg** and should be in JPG format.

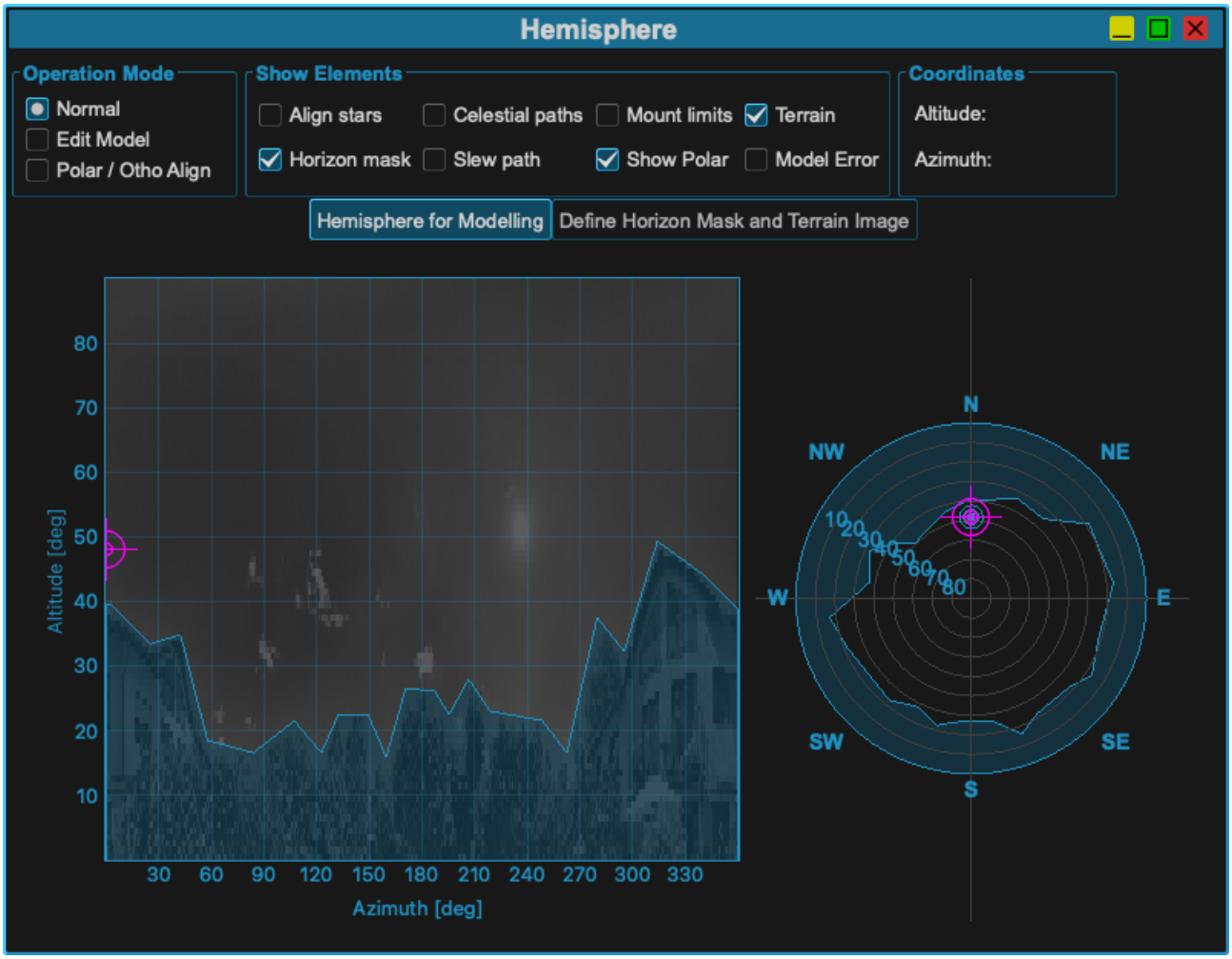
MountWizzard4 will use the image as greyscale image. You could play around how you prepare the image before you copy it to the config directory. Good ideas of improving the image are: make the sky transparent, equalize gamma / lightness settings to avoid highlights in the image, etc. Once you open the hemisphere window you see the setups for the terrain background.




After **use terrain** is checked and a terrain.jpg image is available in config directory, the image will be shown as background of hemisphere.



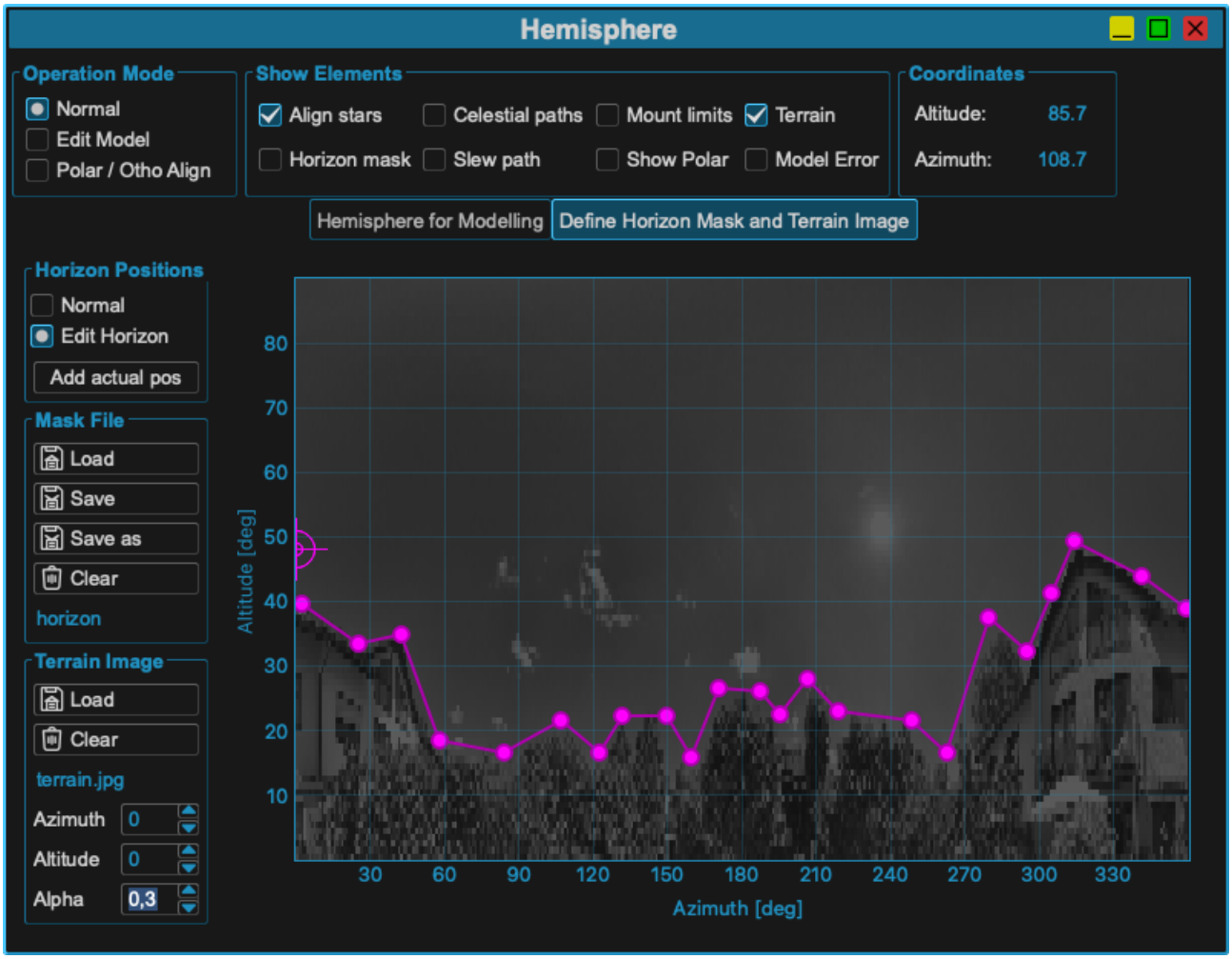

The image will also show up in polar diagram!




Two adjustments could be changed to make the image fit for use: Most important the azimuth adjustment.

- You shift the image by a number of degrees (0 - 359) to get your image fit to the cardinal points of the hemisphere.
- You could change the alpha channel of the image to get a nice view on your screen.

Then you could add the horizon mask quite easily:




**Note**

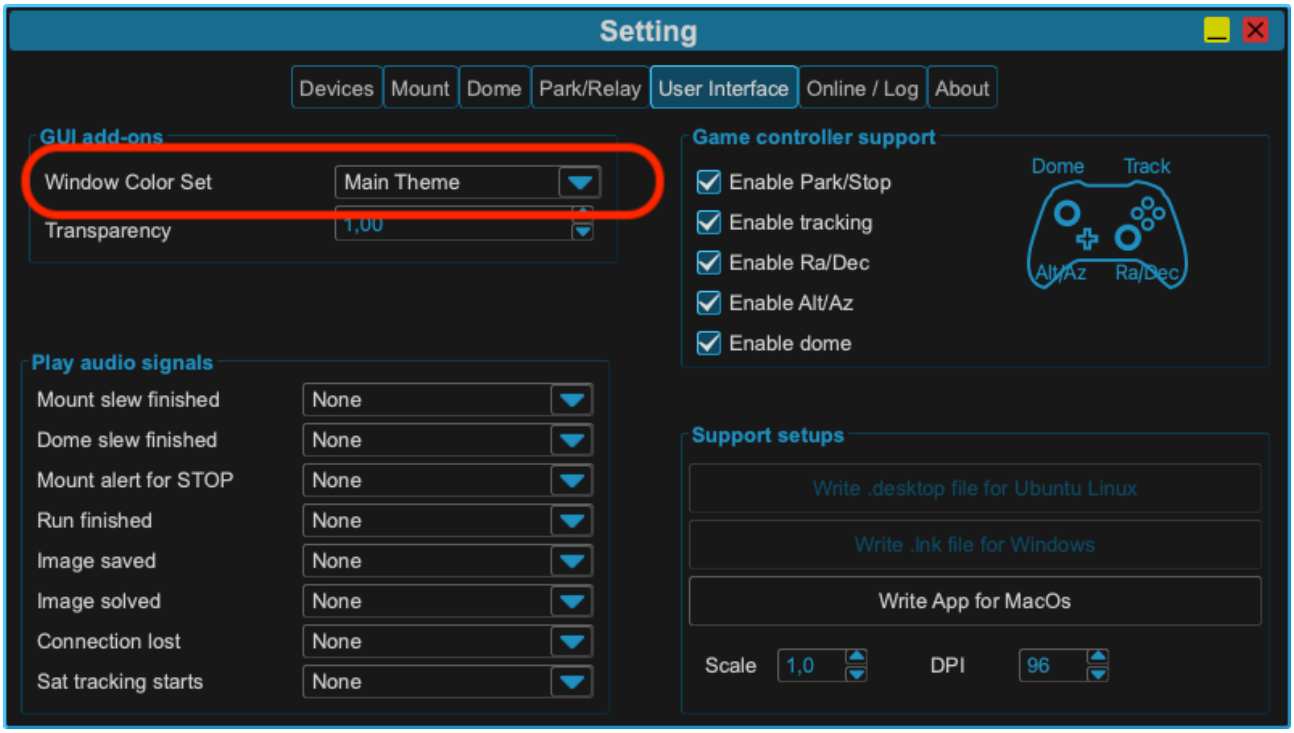
The horizon line (altitude = 0) is set within streetview. Please try to shoot the image at the height of your mount to make it fit. Still there might be some deviations from “real life”.

**Warning**

All calculations which respect a horizon mask reference to a given mask not the image itself. So even if you have an image set, you need to define horizon mask points!

## Themes selection

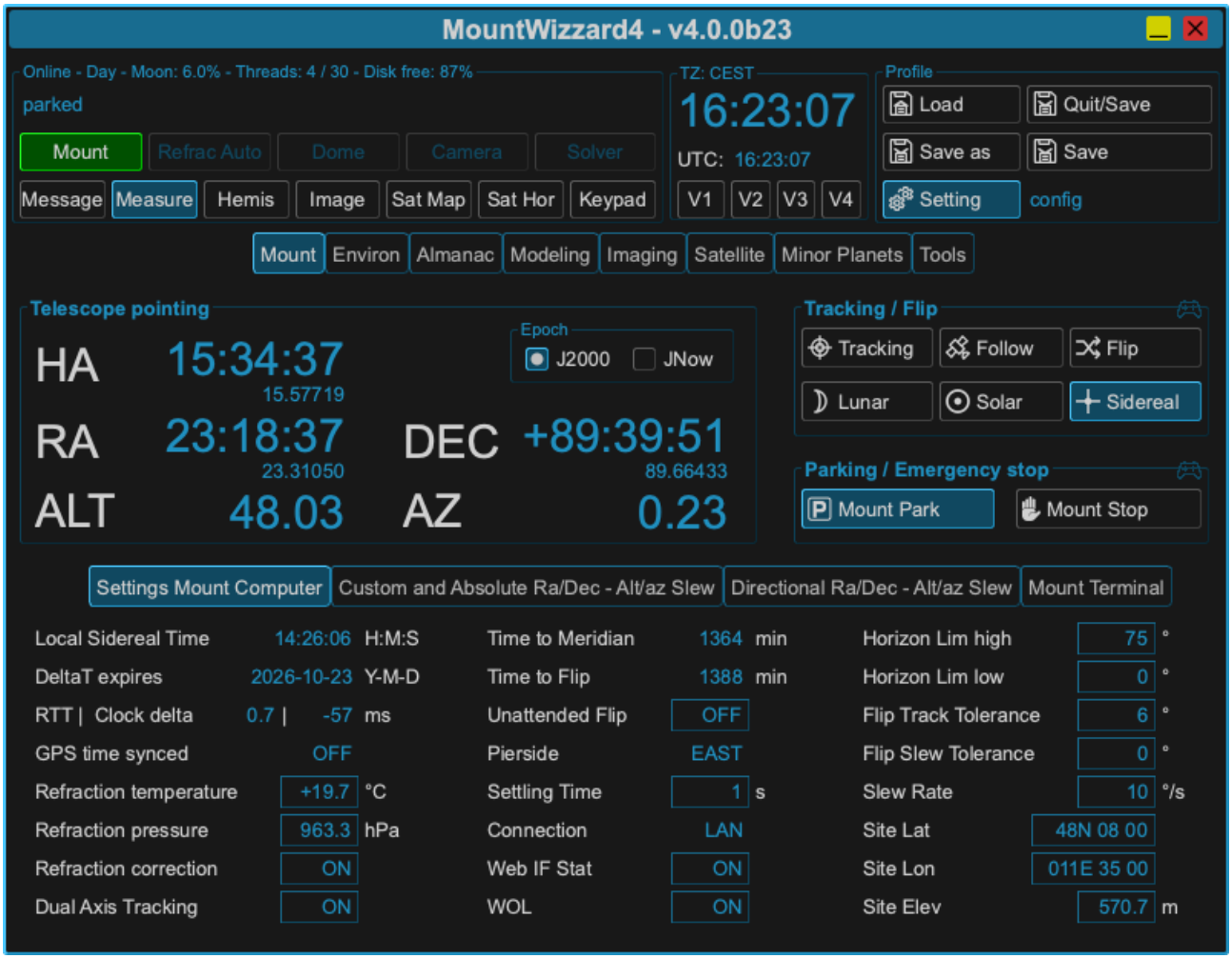
MountWizzard4 supports 3 different color sets. These could be selected in settings:




and will be persisted after closing the application. They could be changed during runtime.

## Main theme

It the theme which was present from the beginning on.




## Red night theme

Getting a nor dim color set for those, who use the application in the field.



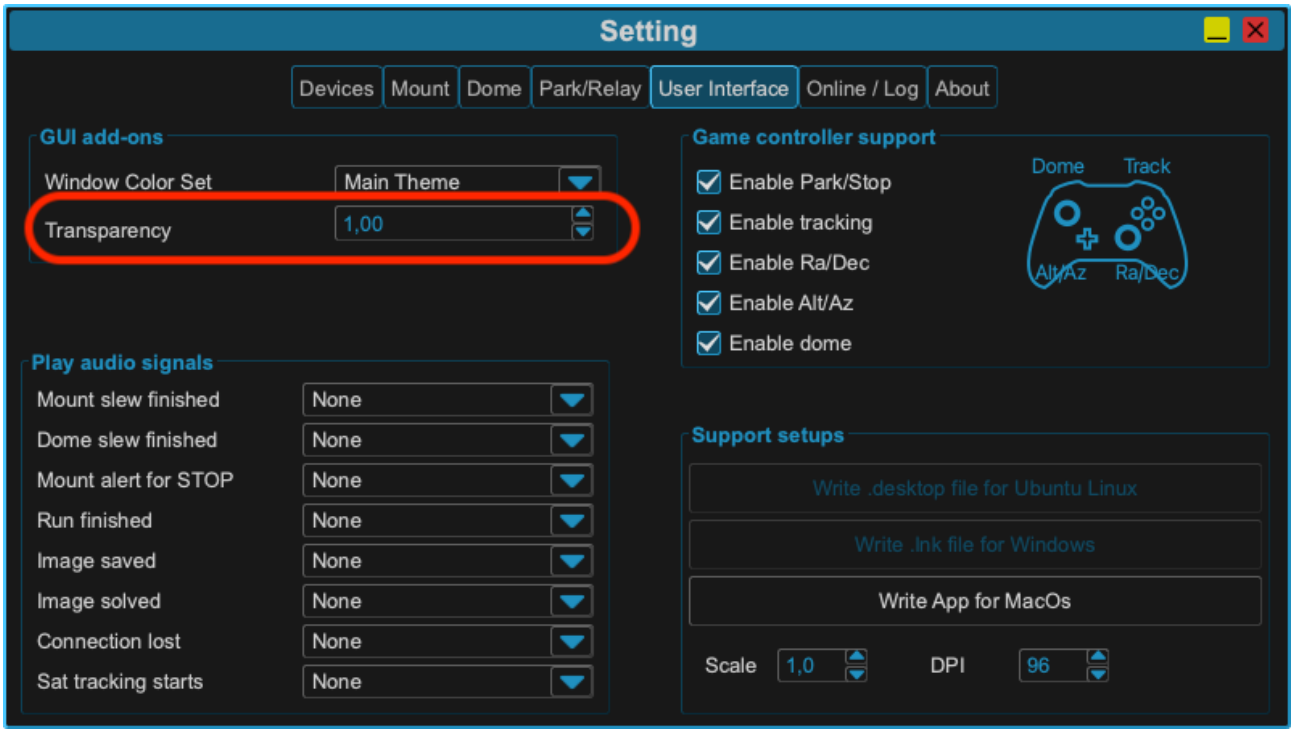

## Light day theme

For those ones who sit more inside and like lighter themes as well, but need not to take care about brighter coloring.



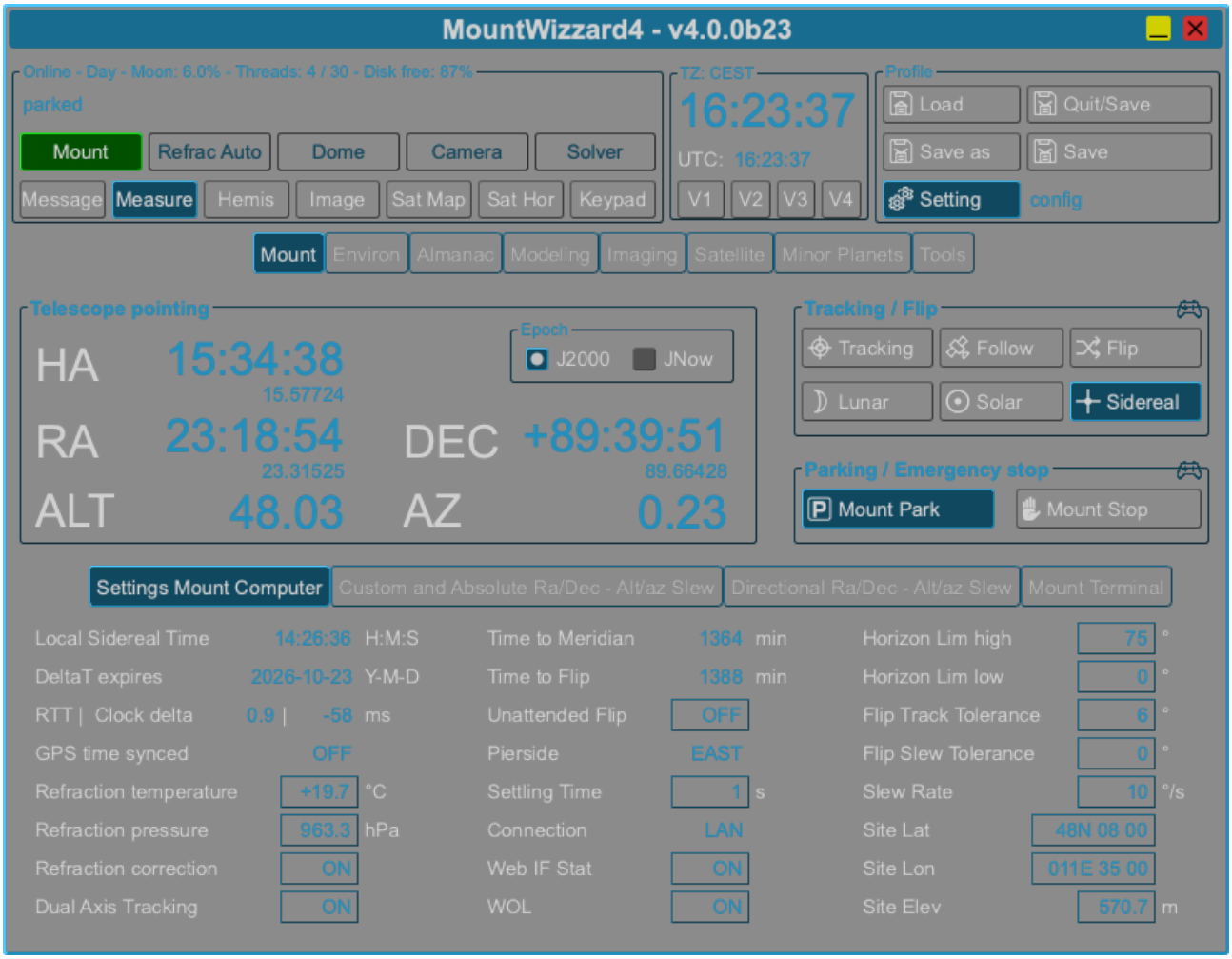

## Transparent theme

For those ones who like to see the background of the application.




This adds-on to any existing theme and could be combined with any of the above mentioned themes. It will be also stored persisted after closing the application and could be changed during runtime.

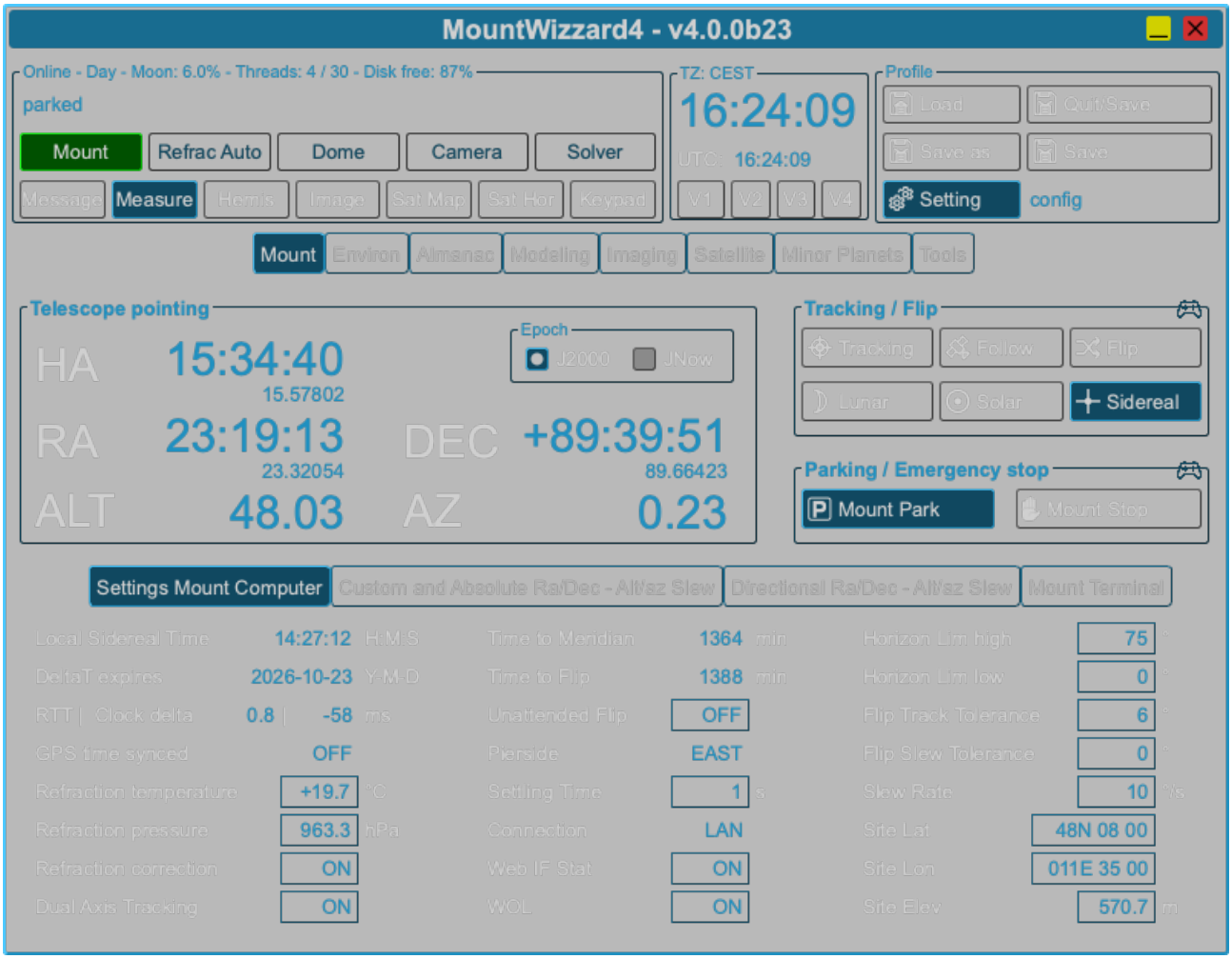
The following image shows the themes with the transparent add-on and set alpha to 50%:


MountWizzard4 - v4.0.0b23
Online - Day - Moon: 6.0% - Threads: 4 / 30 - Disk free: 87%
parked
Mount
Refrac Auto
Dome
Camera
Solver
Message
Measure
Hemis
Image
Sat Map
Sat Hor
Keypad
TZ: CEST
16:23:37
UTC: 16:23:37
V1
V2
V3
V4
Profile
Load
Quit/Save
Save as
Save
Setting
config
Mount
Environ
Almanac
Modeling
Imaging
Satellite
Minor Planets
Tools
Telescope pointing
HA
15:34:38
15.57724
Epoch
J2000
JNow
RA
23:18:54
23.31525
DEC
+89:39:51
89.66428
ALT
48.03
AZ
0.23
Tracking / Flip
Tracking
Follow
Flip
Lunar
Solar
Sidereal
Parking / Emergency stop
Mount Park
Mount Stop
Settings Mount Computer
Custom and Absolute Ra/Dec - Alt/az Slew
Directional Ra/Dec - Alt/az Slew
Mount Terminal
Local Sidereal Time
14:26:36
H:M:S
DeltaT expires
2026-10-23
Y-M-D
RTT | Clock delta
0.9 |
-58
ms
GPS time synced
OFF
Refraction temperature
+19.7
°C
Refraction pressure
963.3
hPa
Refraction correction
ON
Dual Axis Tracking
ON
Time to Meridian
1364
min
Time to Flip
1388
min
Unattended Flip
OFF
Pierside
EAST
Settling Time
1
s
Connection
LAN
Web IF Stat
ON
WOL
ON
Horizon Lim high
75
°
Horizon Lim low
0
°
Flip Track Tolerance
6
°
Flip Slew Tolerance
0
°
Slew Rate
10
°/s
Site Lat
48N 08 00
Site Lon
011E 35 00
Site Elev
570.7
m


MountWizzard4 - v4.0.0b23
Online - Day - Moon: 6.0% - Threads: 4 / 30 - Disk free: 87%
parked
Mount
Refrac Auto
Dome
Camera
Solver
Message
Measure
Hemis
Image
Sat Map
Sat Hor
Keypad
TZ: CEST
16:23:43
UTC: 16:23:43
V1
V2
V3
V4
Profile
Load
Quit/Save
Save as
Save
Setting
config
Mount
Environ
Almanac
Modeling
Imaging
Satellite
Minor Planets
Tools
Telescope pointing
Epoch
J2000
JNow
HA
15:34:37
15.57711
RA
23:18:58
23.31627
DEC
+89:39:51
89.66427
ALT
48.03
AZ
0.23
Tracking / Flip
Tracking
Follow
Flip
Lunar
Solar
Sidereal
Parking / Emergency stop
Mount Park
Mount Stop
Settings Mount Computer
Custom and Absolute Ra/Dec - Alt/az Slew
Directional Ra/Dec - Alt/az Slew
Mount Terminal
Local Sidereal Time
14:26:42
H:M:S
DeltaT expires
2026-10-23
Y-M-D
RTT | Clock delta
1.0 |
-58
ms
GPS time synced
OFF
Refraction temperature
+19.7
°C
Refraction pressure
963.3
hPa
Refraction correction
ON
Dual Axis Tracking
ON
Time to Meridian
1365
min
Time to Flip
1389
min
Unattended Flip
OFF
Pierside
EAST
Settling Time
1
s
Connection
LAN
Web IF Stat
ON
WOL
ON
Horizon Lim high
75
°
Horizon Lim low
0
°
Flip Track Tolerance
6
°
Flip Slew Tolerance
0
°
Slew Rate
10
°/s
Site Lat
48N 08 00
Site Lon
011E 35 00
Site Elev
570.7
m




and the following image shows the main theme with the transparent add-on and set alpha to 30%:

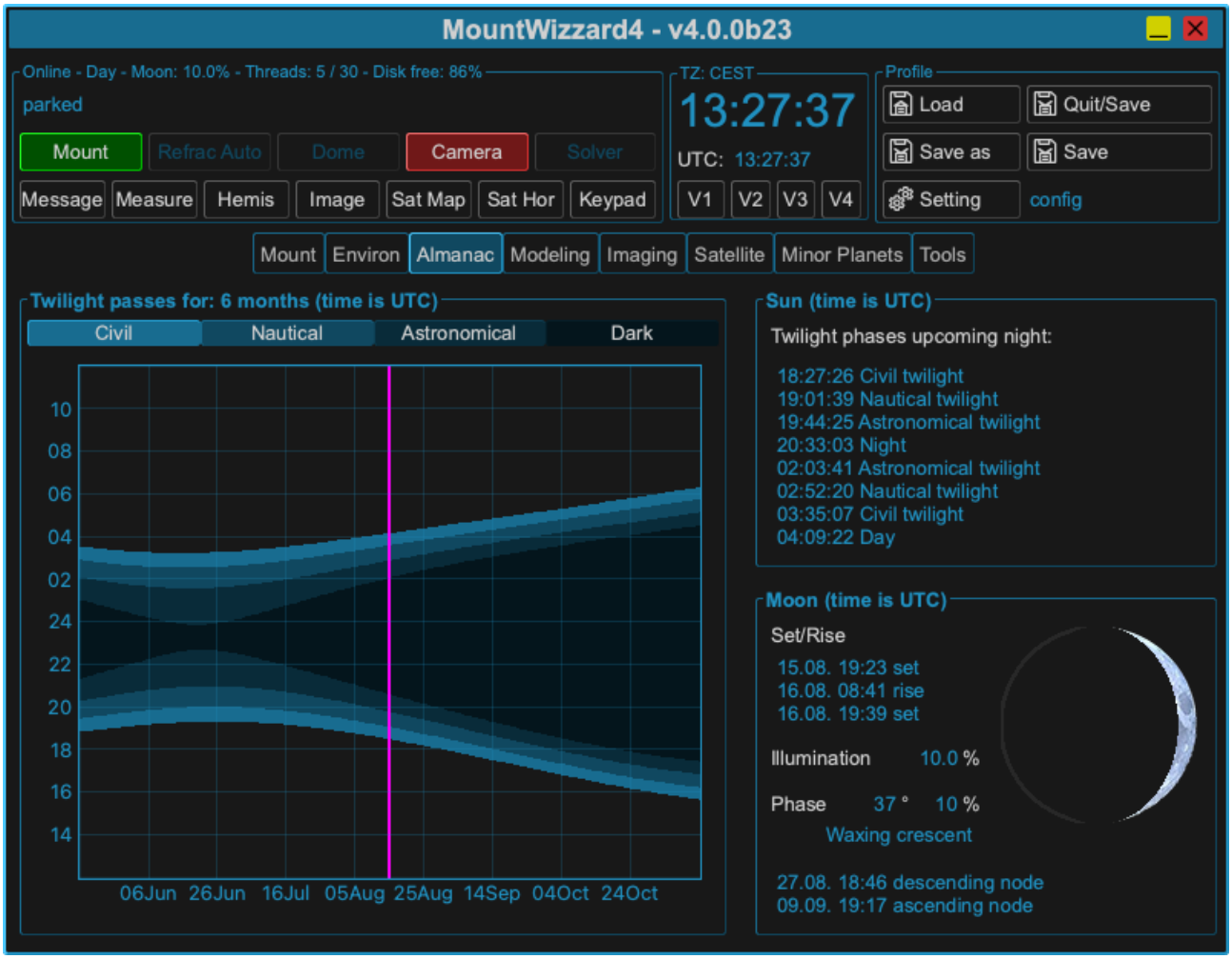



# Workflows and functions

The following part should describe and explain the workflows and functions of the different parts of the software. Often some hints and explanations are given to understand and they are based of the experience of the author or other users.

## Almanac

MountWizzard4 provides some basic information about twilight (1), (2) and moon (3) phases.



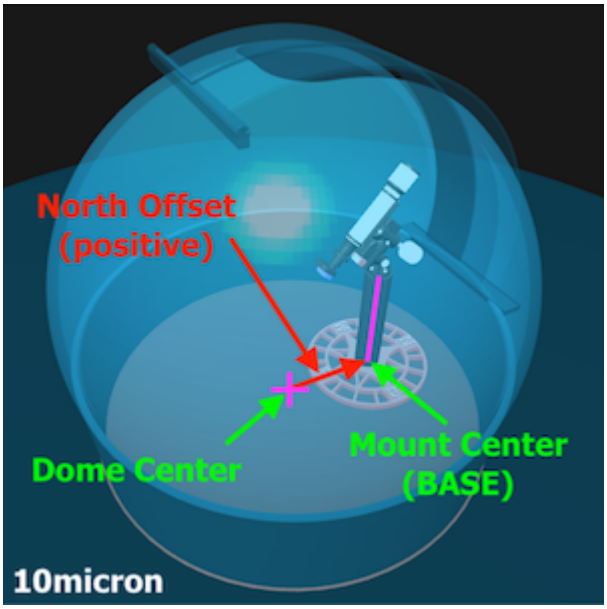

You could always select an area for zoom in or just use the mouse wheel to zoom.




## Dome

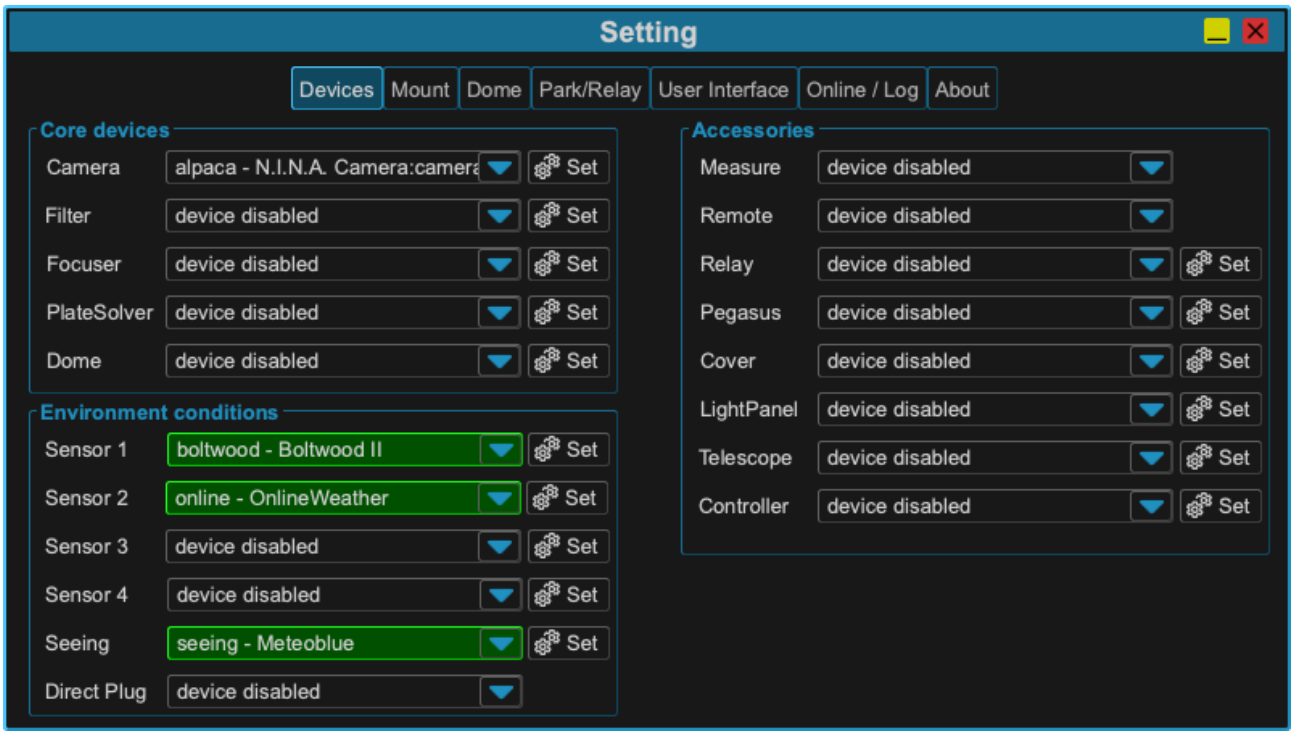
MountWizzard4 starts slewing the dome in parallel to the mount as it knows where the mount will land. Normally it should be able to detect how long the dome slews. The waiting time for the dome should be only relevant if dome movements influence mount by vibrations. It could be the case that the slewing signal has some specialities.

Sorting there are some optimizations to be set in MountWizzard4. Default ist without dome and it sorts for minimum mount slews distance.

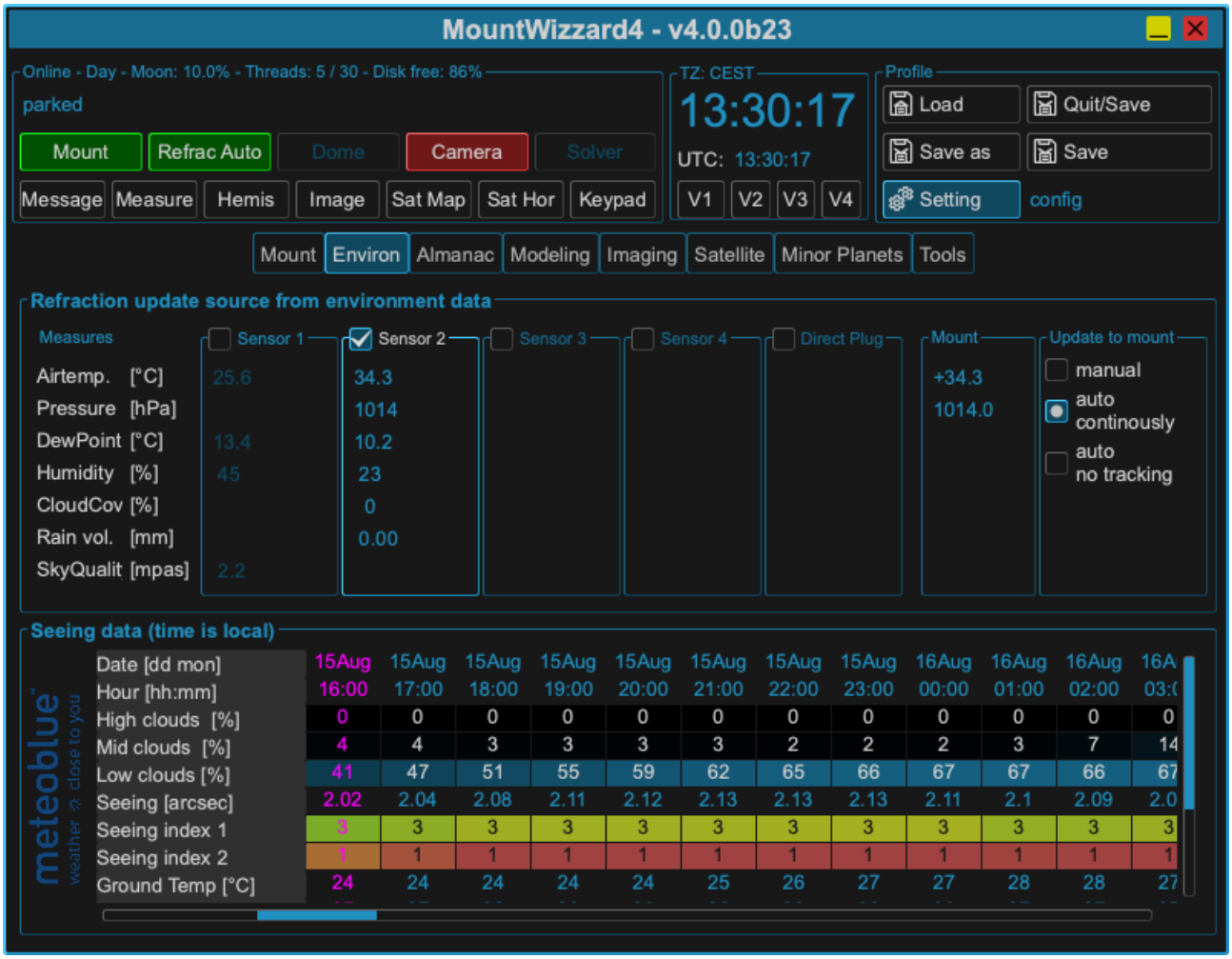



## Environment

Setup the environment:

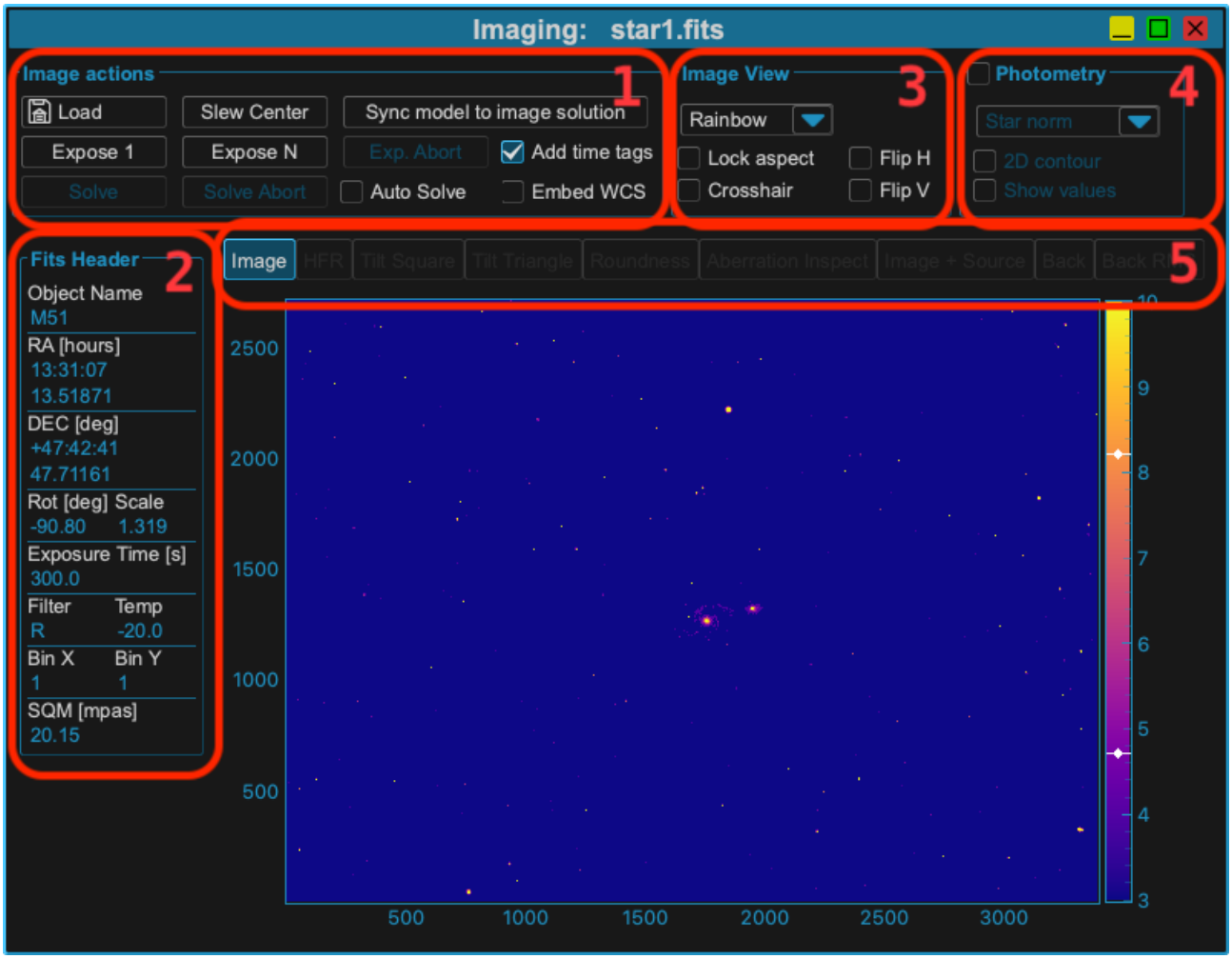



Showing:




## Imaging

The imaging window shows FITS files loaded from disk or images exposed manually or during model build. It is split in different areas to work with.




## Area 1: Image exposing and solving

MountWizzard4 supports single (expose 1) and multiple (expose N) exposures. Continuous imaging could be stopped with abort. You also could explicitly load a fits file (extension .fit or .fits). If you have a plate solver (e.g. ASTAP) installed, you could solve the actual displayed image. The solved results are shown in message window. If you would like to add the results to the image, please check “embed data”. This will make MountWizzard4 to write the plate solving results in the fits header of the file.

**⚠ Warning**

Please be aware that MountWizzard4 will write the data to be embedded directly in the FITS file without making a copy of the file!

When “auto solve” is checked, MountWizzard4 will automatically plate solve every new picture and show or embed the results in the message window or fits header.

A simple stacking method is available when mount is in tracking and keeps point accurate. When “stacking” is checked, MountWizzard4 will add all exposed images (expose N running) and calculate the mean of the image.

## Area 2: FITS Header entries

Some of the FITS header entry of the actual image are shown.

## Area 3: Image attributes

MountWizzard4 calculates and extracts some image attributes. For example if a WCS header information is available, the distortion parameters are present or the actual image is flipped with regard to real position in sky.

## Area 4: Image display

Show the image and it’s different view selected in area 5. For standard view the scale is pixel with 0/0 to be the center of the image. There will be a colorbar in each view with the values of the image.

## Area 5: View options

For the image you have different tabs to analyse the image. The first tab is the standard view of the image.

## Some examples for the windows

View Image with sources:

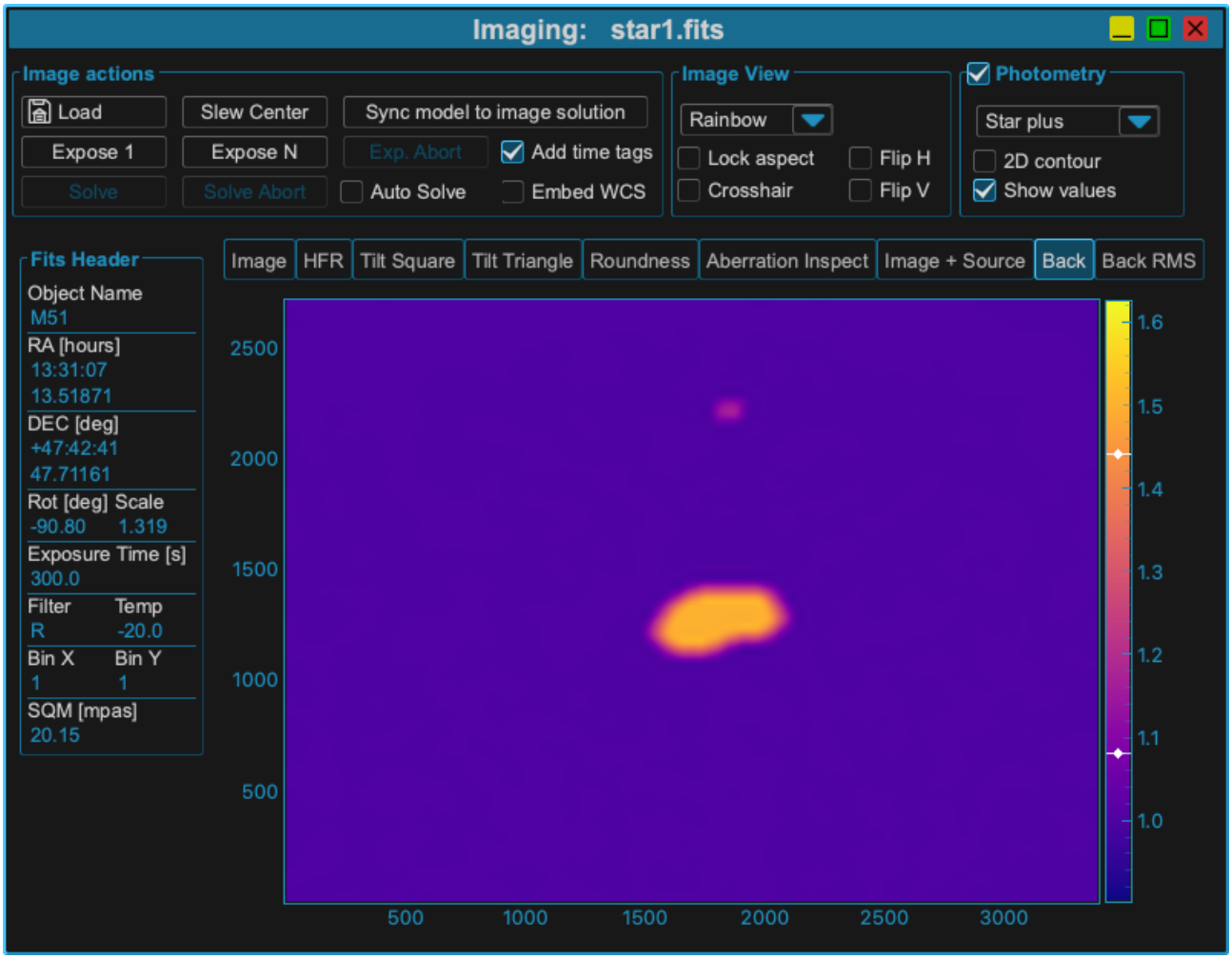



View image with HFR values

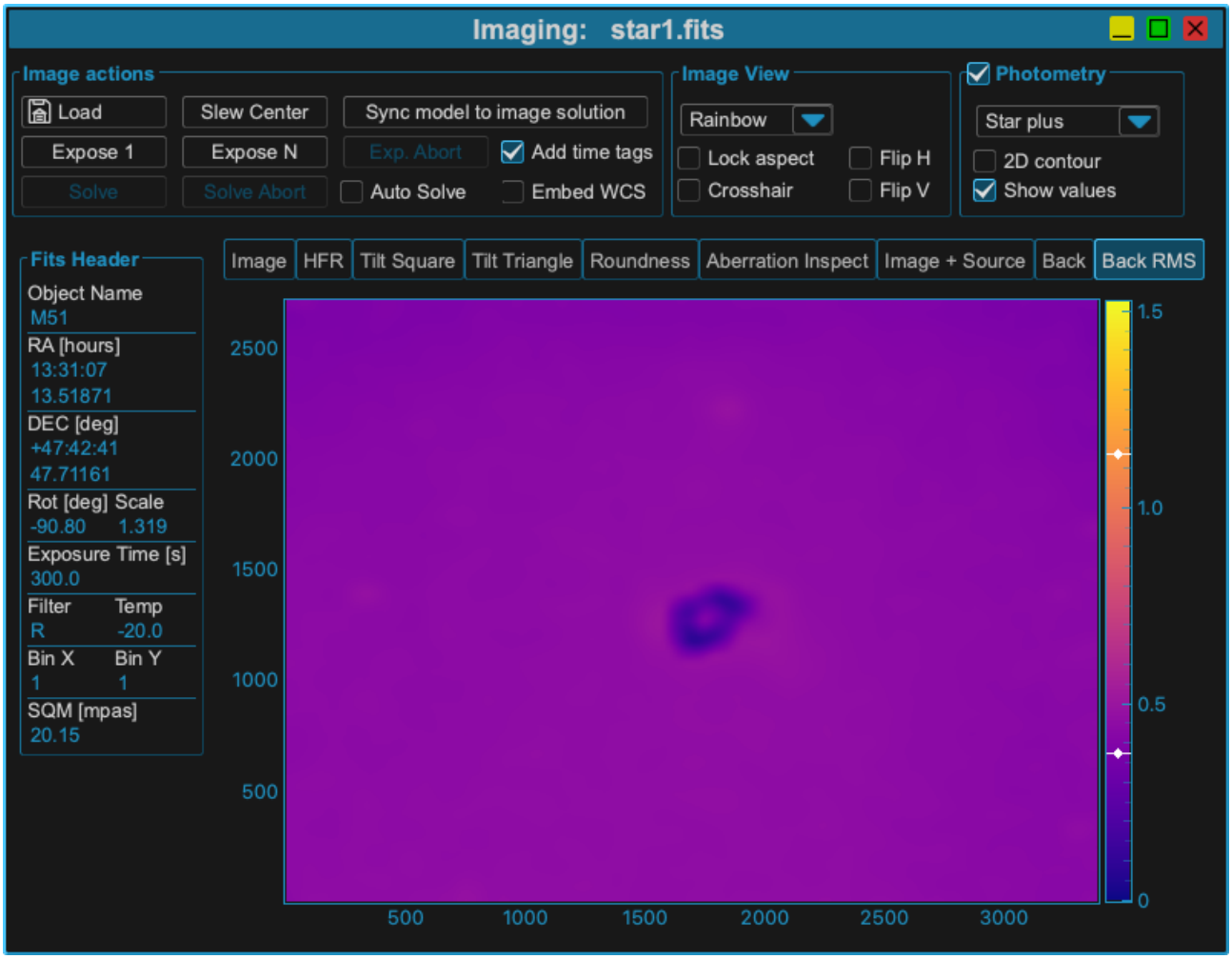
workflows/imaging/image/show_hfr.png
System Message: WARNING/2 (/home/runner/work/MountWizzard4/MountWizzard4/doc/workflows/imaging/index.rst, line 66)

Cannot scale image! Could not get size from "workflows/imaging/image/show_hfr.png": [Errno 2] No such file or directory: 'workflows/imaging/image/show_hfr.png'

View image with background level

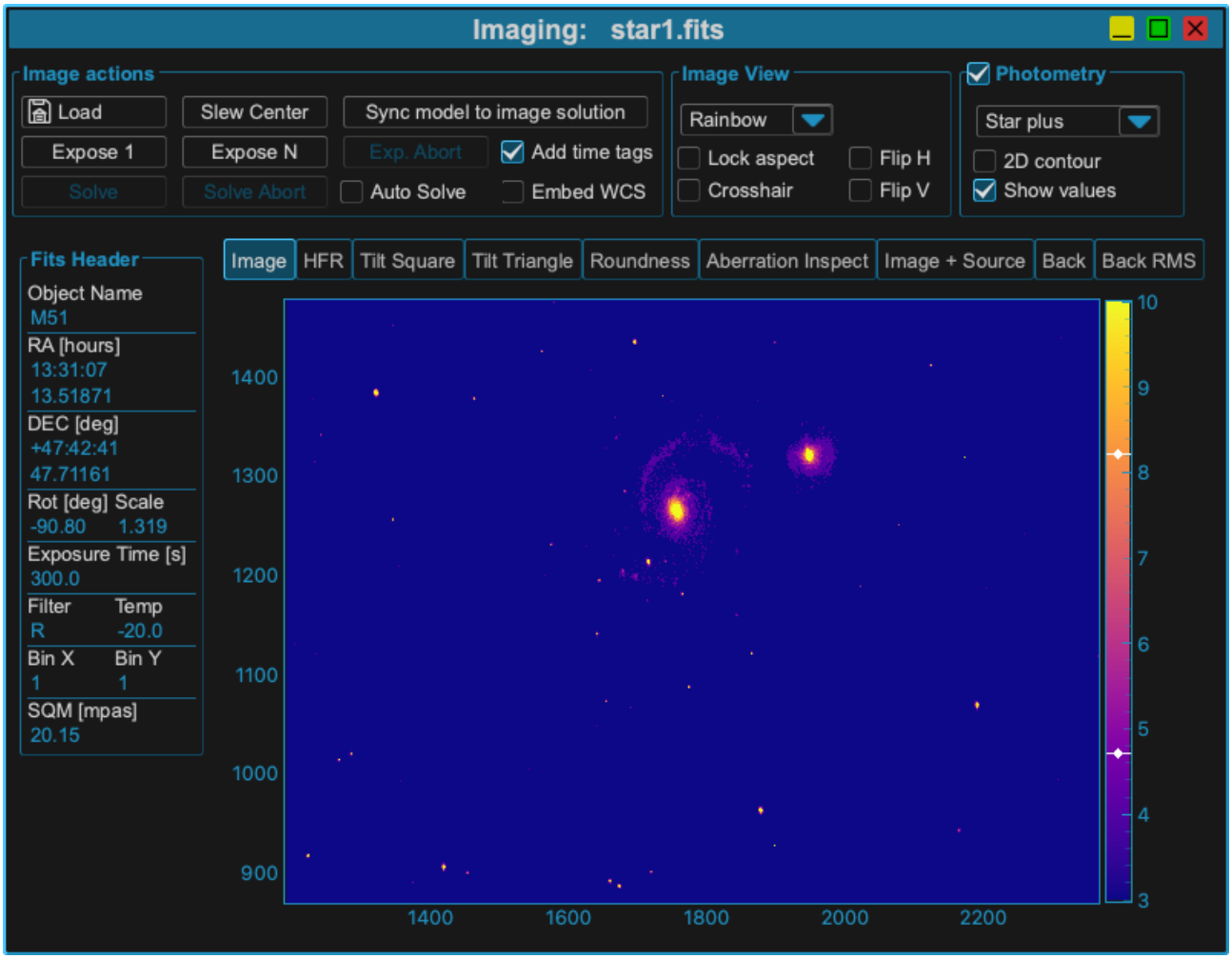



View image with background noise

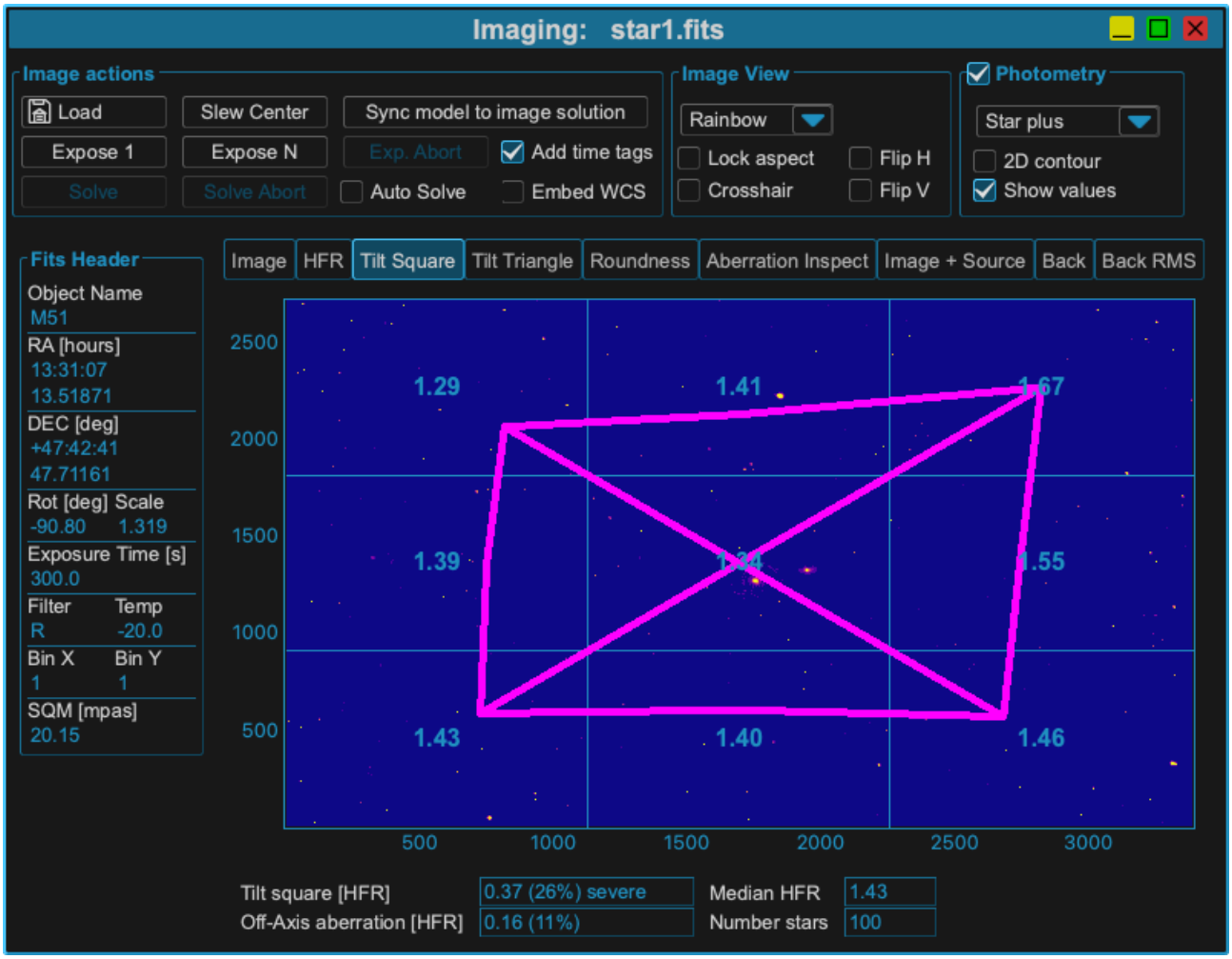



View image with photometry flux

View image with different zoom



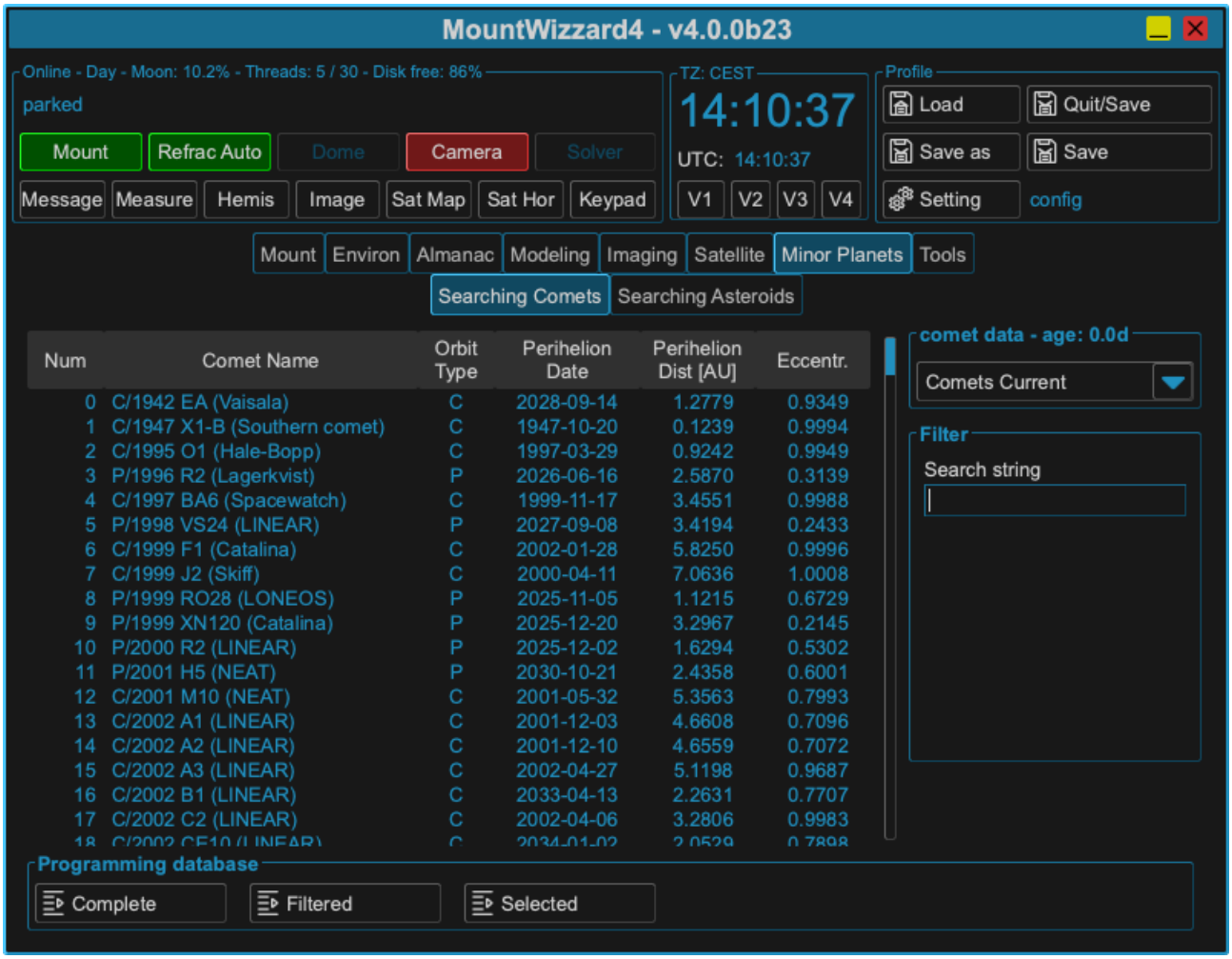

View image tilt



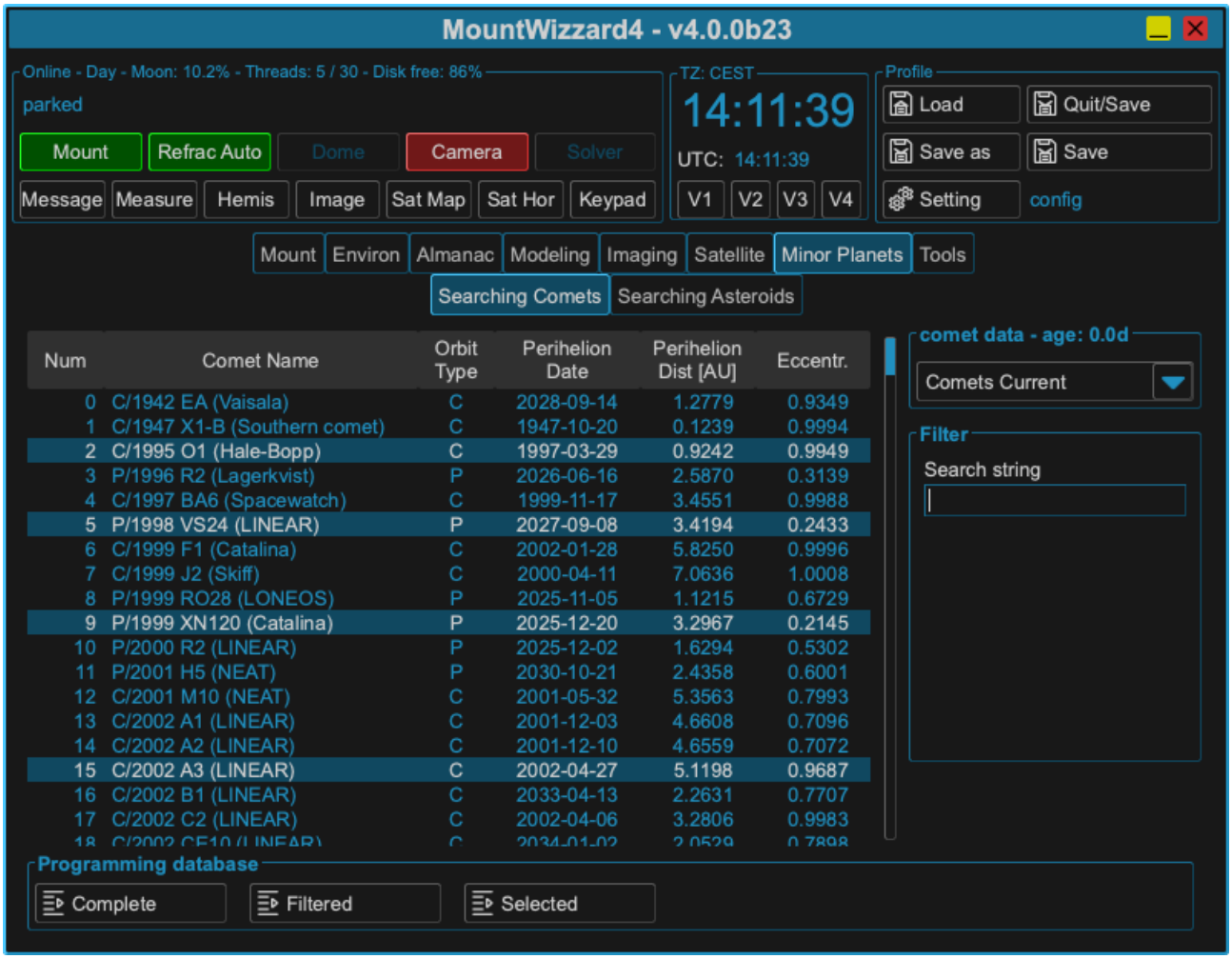

## Minor Planets

Under module *Minor Planets* you can download and visualize the data of comets and asteroids. The data is downloaded from the Minor Planet Center.

The data is stored in a local database and can be used for various purposes. The comets you will be able to select under the comets tab. By clicking ont the header of each type of data, you will be able to sort the list by the selected column.




Once downloaded, you can select a comet by text search or by clicking in the list, also selecting multiple comets is possible:

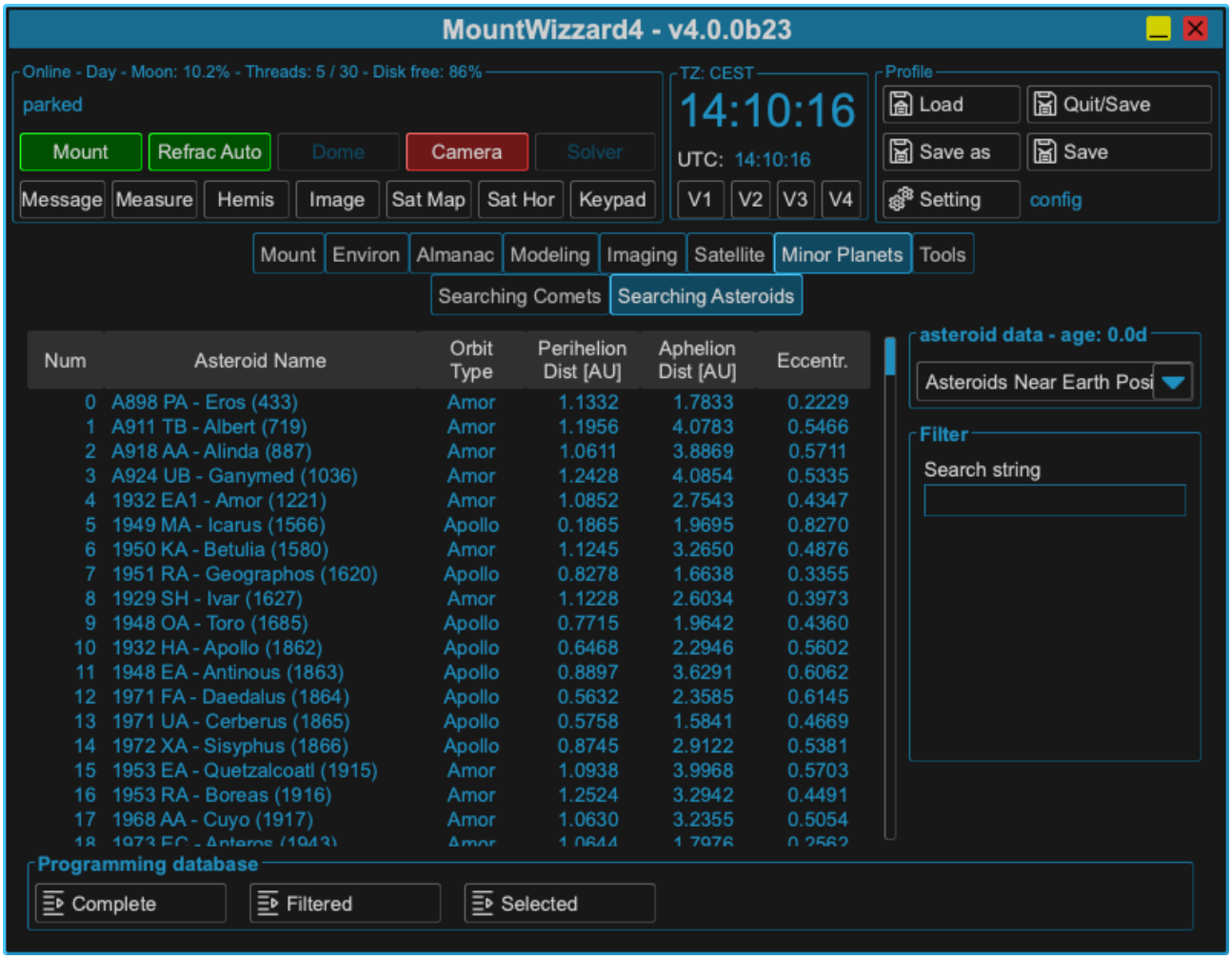



You also could filter the list by text search or even combine methods:




After selection, you could upload the selected / filtered or complete database to the mount computer. The mount computer will use the data for tracking. Programming will be only possible if the mount is connected to the computer.

The same is valid for asteroids:




Within asteroids, you could select different databases to download the data from via the drop down selector.

In both cases, comets and asteroids, the local databases will be updated every day to offer actual data.

## Modeling

I only focus on model building with tool support, so no words about manual model building. I don't know the internal algorithms of the 10micron mount how the calculate their corrections. So many of the hints derive from pure logical or mathematical approaches and even there I personally might have some misconceptions or make some errors.

So my goal in model building is quite simple: I'm lazy in doing setups, so I want a solutions which gives me a correction model most accurate in minimum of time automatically. I rely heavily on the corrections capability on the 10micron mounts, so I use them always with dual tracking on. For doing a setup there are many things to think of beside the model (leveling, rigidity etc.). Keep them perfect, but I don't talk about them. So this results in two tasks I have to do to get a model

to do unguided images: Polar alignment and the model for correction itself. I refer to the Blog Filippo Riccio from 10micron:

https://www.10micron.eu/forum/viewtopic.php?f=16&t=846

All the hints you get from the mount (how to turn knobs, alignment star) improve the alignment. As the model is only an approximation for the error correction, it will be not an one step approach. If you aim for the best result, please think of 2-3 iterations of the whole procedure. In my setup I normally need 2 iterations for doing an alignment which is good for 20-30 min exposures and have round stars.

As Overview: MountWizzard4 has a straight forward approach for building models. The following chart shows each basic step.

workflows/modeling/image/workflow.drawio.svg
System Message: WARNING/2 (/home/runner/work/MountWizzard4/MountWizzard4/doc/workflows/modeling/index.rst, line 30)

Cannot scale image! Could not get size from "workflows/modeling/image/workflow.drawio.svg": [Errno 2] No such file or directory: 'workflows/modeling/image/workflow.drawio.svg'

The following sections describe each step in detail:

## Step 1: Polar alignment

All credits for explaining the following section go to 10micron and many experts using the mount.

### Initial alignment

Initial three star alignment insufficient to get a good polar alignment error (PAE) value. A two star alignment can only give a poor PAE estimate as the cone error correction is missing, can be many arc minutes off true value Minimum that can be used is 3 stars but they recommend at least 5 and ideally 10 points (ideally based on experiments say that 12 to 20 points have better convergence without spending too much effort).

**Note**

20 points in MountWizzard4 using the fast ASTAP plate solver will do 20 points in around 7 minutes - not too onerous for a good PAE measurement.

Pay attention also that the stars are not near to the same maximum circle (i.e. not all near the same right ascension, or all near the same azimuth, etc.). When using few stars, how the stars are chosen is critical.

**Note**

MountWizzard4 helps here by automating the spread of points to avoid the problems listed through its various point layout patterns.




The hemisphere window is showing different colors and sizes for selecting the right stars once the Polar/Ortho mode is selected. You need a valid first model in the mount to use this feature!

As you add stars, more and more model terms will be added um of either (# points - 1) or 11 terms for a single all sky model - 11 terms meaning all required terms have a solution. Note that this may be split into two models for east side and west side in which case the number of terms shown by the mount is doubled.

A polar alignment error less than maybe one arcminute will be possible. If you are using Dual Axis Tracking it sorts all this out for you. (A 10 minute exposure with a 10 arc minute (1/6 degree) PAE will not show field rotation on a large full frame sensor up to latitude 60 north or south)

## How polar align works

The polar align function works in a really easy way. It removes the polar axis error from the model before pointing at the star, so when you center the star with the azimuth/altitude movements the right ascension axis will be automatically aligned with the celestial pole.

Choosing location in the sky of the star to use to do the polar alignment correction is critical. The choice of the star is made so you avoid the zones where the manual movements have no effect. Avoid the zenith (this is where the azimuth knob has no effect). Avoid the East-West axis (this is where the altitude adjustment knob has no effect)

**Note**

MountWizzard4 will help you selecting the right location by marking the right regions and tight stars. In addition you don't need to use the handpad for selecting the stars: in MountWizzard4 you could select the reference star with a mouse click and start the polar alignment.

In MountWizzard4 the polar alignment works by changing the mode to polar/ortho alignment. In this mode, you could select a star in the hemisphere window with a mouse click and start slew the mount to the reference star. All you need is to center the selected reference star in you image.

**Note**

For centering a star you could use the image window, select for continuous exposure and enable the crosshair. You could also magnify the region around the crosshair center to improve the precision.

## Should I deactivate the dual axis tracking?

If you have the dual axis tracking active, the mount will track using the model built in the initial phase (the initial 3 points generates a minimum model with 2 terms - its not a good model).

You will not be getting the correct tracking rate while you adjust the alt/az knobs. So if you wait too much before centering the star and giving confirmation to the mount controller, or if the initial error is very large, there will be a drift. Therefore you must turn off dual tracking while performing the PA procedure.



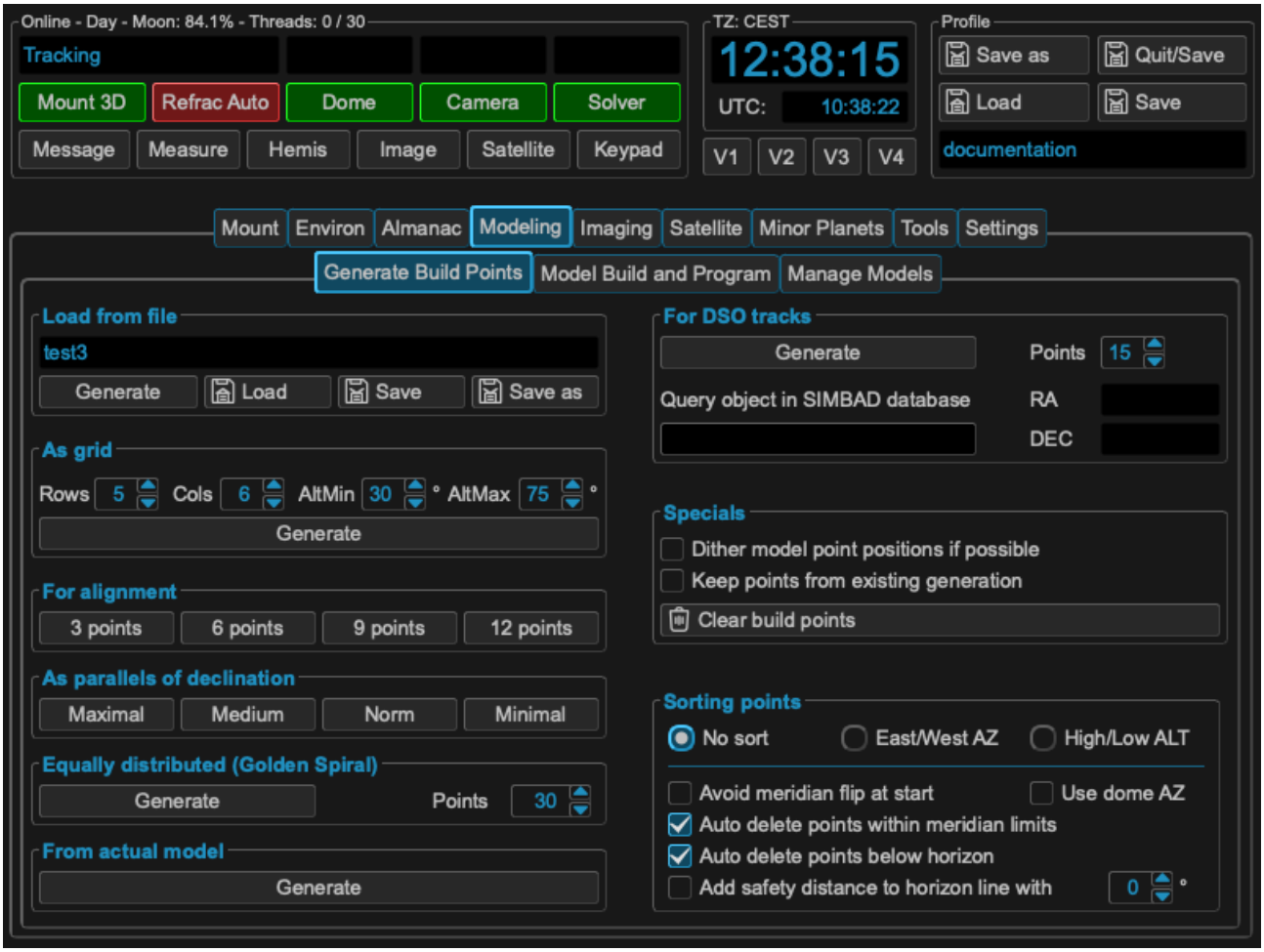

## Effects of polar misalignment

Even if your mount is wildly away from a correct polar alignment, when you make a model the correction of the polar misalignment will be performed exactly when Dual Axis Tracking is activated. This means that the pointing error and the tracking error will be the same - if you make a model with the same number of stars and the same accuracy - whether the mount is aligned with the celestial pole or not. When dual axis tracking is deactivated, the mount will track like a normal equatorial mount, by moving the right ascension axis at a constant sidereal speed.

The only important thing that will change with increasing PAE is the field rotation.

## Step 2: Selecting build points

The first step for building a model is the selection of the desired model points. Handling you horizon is an important step as MountWizzard4 helps you could in finding points, which were not covered through obstacles in your location defined by the horizon line.




There were multiple ways of setting your model points within MountWizzard4:

From file

In a generated alt/az grid

First align points on defined alt

Points along the celestial slew paths

Equally distributed over the sphere (golden spiral)

From the actual model (replicating tha actual one)

Based on a DSO track of an object with define Ra/Dec

Manually set point (in hemisphere window)

There are multiple parameters or selections available to refine your point selection. the best way to find out is simplest to test them.

**Note**

Please think that you distribute the model points at the end over a sphere! Using alt/az is convenient, but might be misleading. You could enable the polar view in hemisphere window to get an image how the points are spread in a polar diagram.

It makes a lot of sense if you have discovered you favourite model points setup to store them in a file and reload them whenever you want. You could store as many files as you want.

### Sorting build points

MountWizzard4 does a default sorting for you. If the default does not fit to your setup, you could tweak the order and additional constraints like you want. These settings are stored persistent in your profile.

No sort point -> default

East / West: points from East coordinate to west coordinate (az)

High / Low: point from high altitude to low altitude (good for domes)

Avoid meridian flip: modeling start at the actual pier side

Auto delete points within meridian limits

Auto delete points below horizon

Add safety distance: adds selected degrees safety margin to horizon

**Warning**

Regardless what you select, the 10micron mount computer could only handle a maximum of 100 points

If you are using a dome, you could select the option to sort the points to optimize the slewing of the dome. This will happen the following way:

MountWizzard4 simulates a full model run, calculates the mount positions, calcs from mount the resulting dome azimuth with all geometry corrections and sorts afterwards the build points that you get ascending azimuth for the dome.

### Build points specials

In addition you could also mix different types of model point generation by checking

**keep points from existing generation**

MountWizzard4 will not cleanup the point map before adding new point, but add them.

## Step 3: Running the model build

After you have selected you build points, MountWizzard4 could automatically run through the points and generate the data to generate a model. These are steps you could basically all manual as the final model calculation itself is done in the mount computer! There are two different ways offered from 10micron to build a model:

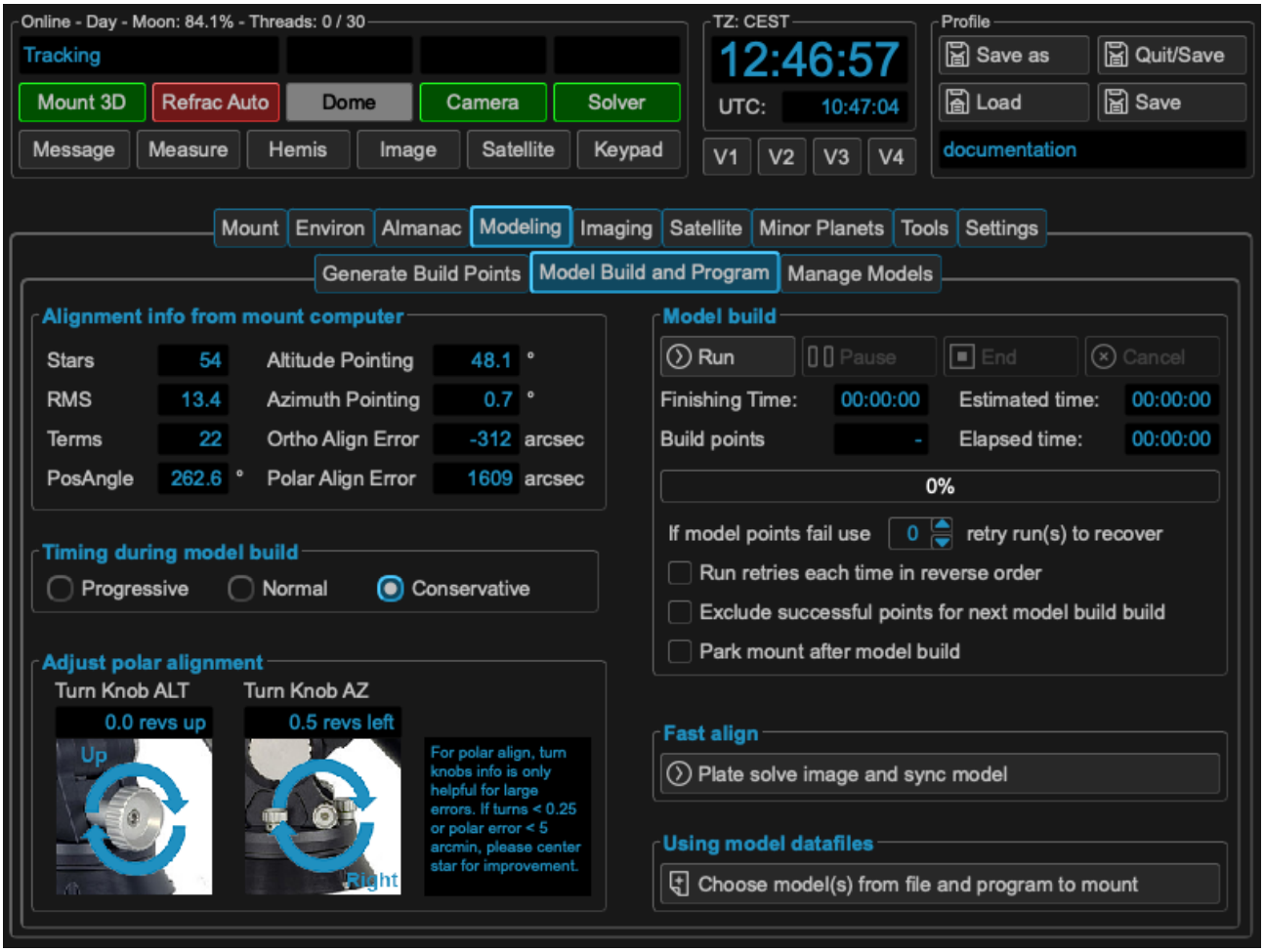
Incremental

Batch

In incremental model build, each new point will start a recalculation of the model in the mount. This behavior is also called sync refine, as with each point the model is processed.

In batch model build, a complete set of point and their data are transferred to the mount computer, which calculates from this data a new model.

In result both variant will produce the same model with the same quality. The difference is the way and the context you are working in. MountWizzard4 uses the batch model build as there are advantages (not in the model quality) in the automatic handling of point data.




In the “model build and program” tab, that actual model spec from the mount computer is shown. If you are in an early stage of your setup, you might use some of the hints given to refine and tune your mechanical setup before building your final model for imaging.

**Note**

All values which are show in these graphs were calculated by the mount computer itself and just read out and displayed by MountWizzard4. As the 10micron algorithm of the model optimizer is not know. The given hints are observations.

**Warning**

Any changes in your mechanical setup invalidates the model!

### Explanation model build process:

The model correct for error. Some could be removed exactly, some not. The way is a mathematically optimization method. In max the mount could calculate 22 terms (which means two models of a set of 11 terms, one for WEST and one for EAST side). If you need 22 parameters for the model (for whatever reason the mount thinks), you have to have at minimum 22 alignment stars or more. Otherwise this will not result in 22 parameters. Again like in polar alignment: think of what is the goal of this task. For sure you would like to remove as much of the alignment error to be able to get unguided images.

## Modeling automatics

The automation process is explained by the next drawing. It is quite simple, but there are some important topic, which you should care about.

workflows/modeling/image/model_process.drawio.svg

MountWizzard4 tries to do as much steps at the same time as possible to reach the minimum model build time. So **“Take an image”** means that the signal to do the next slew to the point is already started when the camera sends image integration finished. During image download and image saving, slewing is already on the way. The same happens to image solve. MountWizzard4 does not wait until an image is plate solved, but manages as much images in the shortest time. If your plate solver is fast, your don’t see much of this asynchronous behavior, if you use a slower one or if you are running MountWizzard4 on a slower machine, you might experience a stack of plate solving tasks when the mount slewing and imaging is already finished.

**Note**

For the plate solver a starting Ra/Dec and scale hint is necessary for a successful plate solve. Please check your environment if INDI or INDIGO embeds this data in the FITS header of the image file.

You could **“Pause”** the model build at any time, but MountWizzard4 will finish after a sequence which allows safely to pause. It will not happen immediately.

You also could **“End”** the model build process at any time. In this case MountWizzard4 will take all already existing data and tries to program a model to the mount.

In addition you could **“Cancel”** a model build run. In this case all data is lost.

MountWizzard4 offers a disabling of dual tracking during the model build run and restores this state afterwards or a mount park after a model build is finished.

During a model build it might happen due to clouds or other events, that some of the images from model point could not be solved an therefore used for model build. For this situation you

could enable a retry mechanism for the **“missed”** points. After running through all points, MountWizzard4 will retry the missed ones N times and will add their data if possible afterwards.

### Using model datafiles

MountWizzard4 stores for each model build run all data (and some more for analyse) to build a mount model. With this data you could rebuild at any time you mount model from scratch if you for example deleted to much points during optimisation or other reasons.

In addition you have the chance to combine multiple model runs to a single mount model! Please think of the maximum of 100 point the mount computer will handle to calculate a mount model.

The warning about the invalidation of the model when mechanical changes are made are true for this step as well. Combining model data from different mechanical setups lead into an invalid model. Rebuilding a model from old data when mechanical changes were made result also in an incorrect model.

### Fast Align

There is a way to adjust an existing model to a certain sphere position. If you choose fast align, MountWizzard4 will do for the actual position an image, plate solve it an align the model to the solved coordinates. This means the whole model is move in a way, that the actual pointing coordinates and solved coordinates are equal. But this changes model pointing for all other position in the sky as well without knowing if this shift really fits there.

**Note**

Fast align is a step to adjust quickly a reasonable model to an actual pointing position. This action is not a model build process! You most probably loose the pointing accuracy of you model!

## Step 4: Managing mount models

Managing your models is a central task especially if you image in a mobile environment. MountWizzard4 supports this in the Manage Model Tab in the main window. All functions there are related to the actual mount model or the models stored in the mount.

Three graphs are available for checking the model:

Model Positions

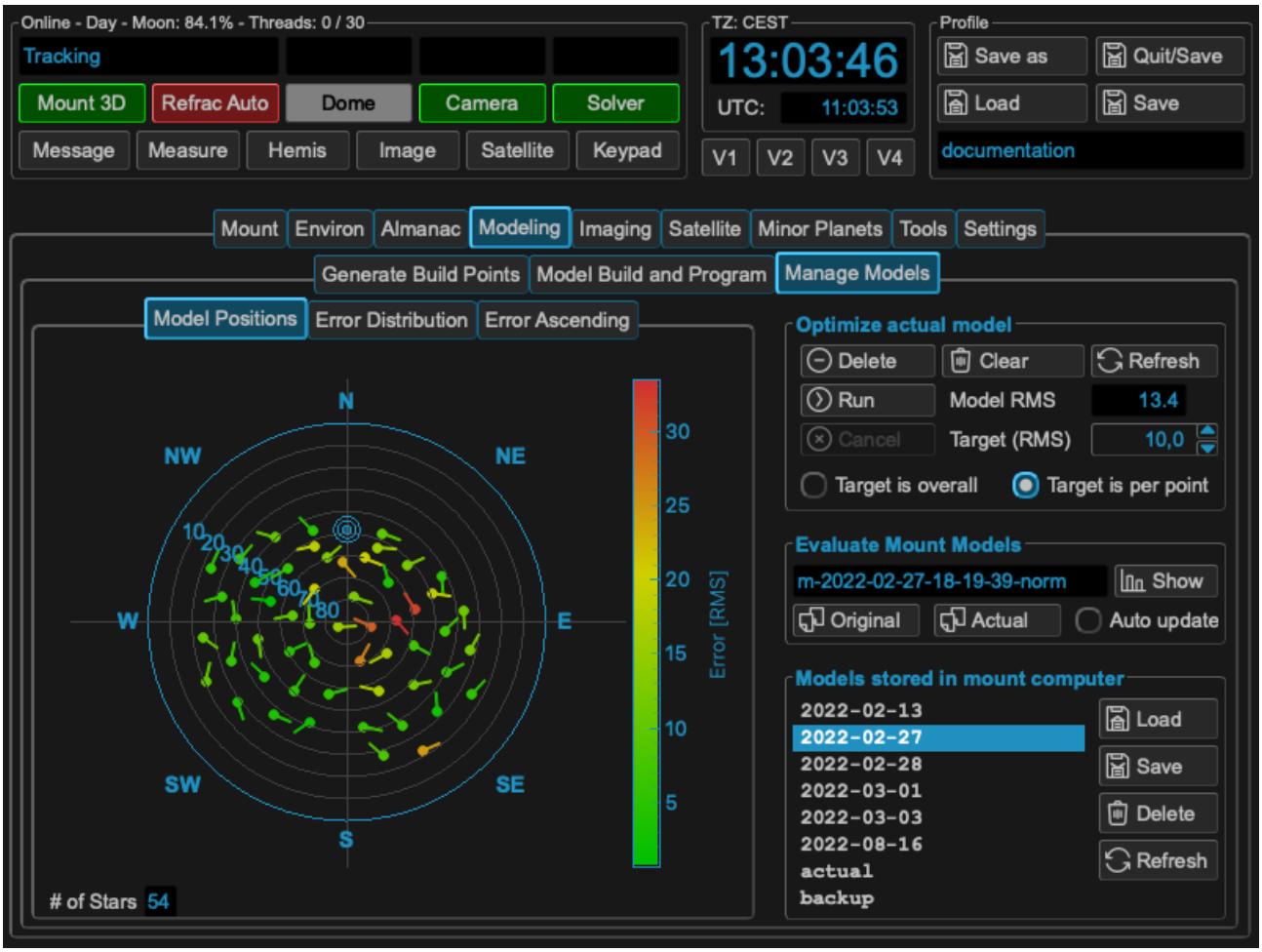
Error ascending

Error Distribution

**Note**

All values which are show in these graphs were calculated by the mount computer itself and just read out and displayed by MountWizzard4. As the 10micron algorithm of the model optimizer is not know. The given hints are observations.

The Model Positions graph shows in a polar plot how the model point were distributed over the hemisphere and how large the errors of each point is in a color scheme. If necessary you could check “Show error values” and numerical values will label each point.

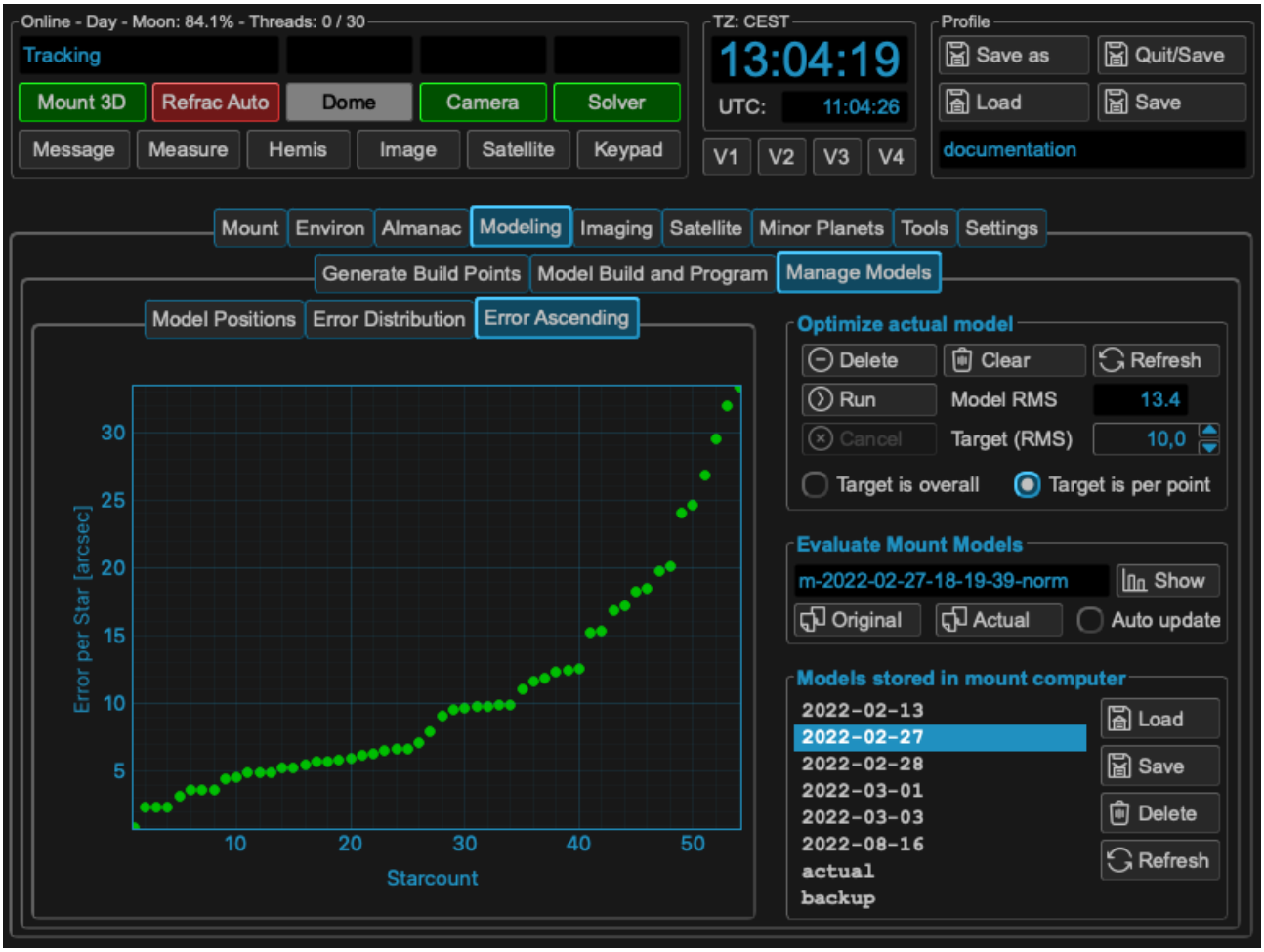



In the give example you see the western point are in general better than the eastern ones, so in the setup there might be differences in setup depending on the pier side.

The Error Ascending graph show all model points sorted by ascending error values. You might see some outside points which differ a lot from other ones or recognize an elbow type of curve which also might lead to some issues with you setup.



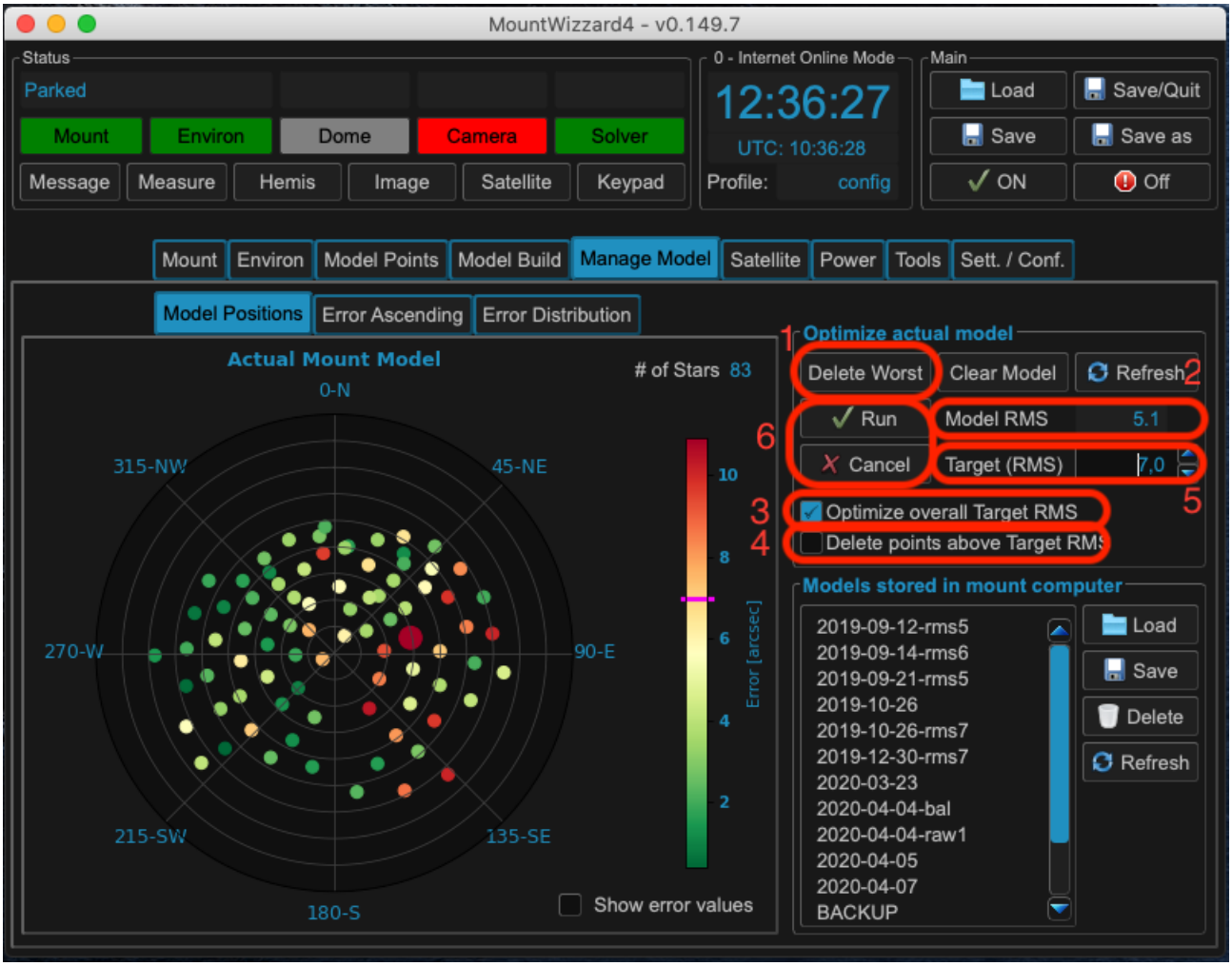

The Error Distribution graph shows in a polar plot how the error values (higher values on outer rings) and the polar angle of the error value.



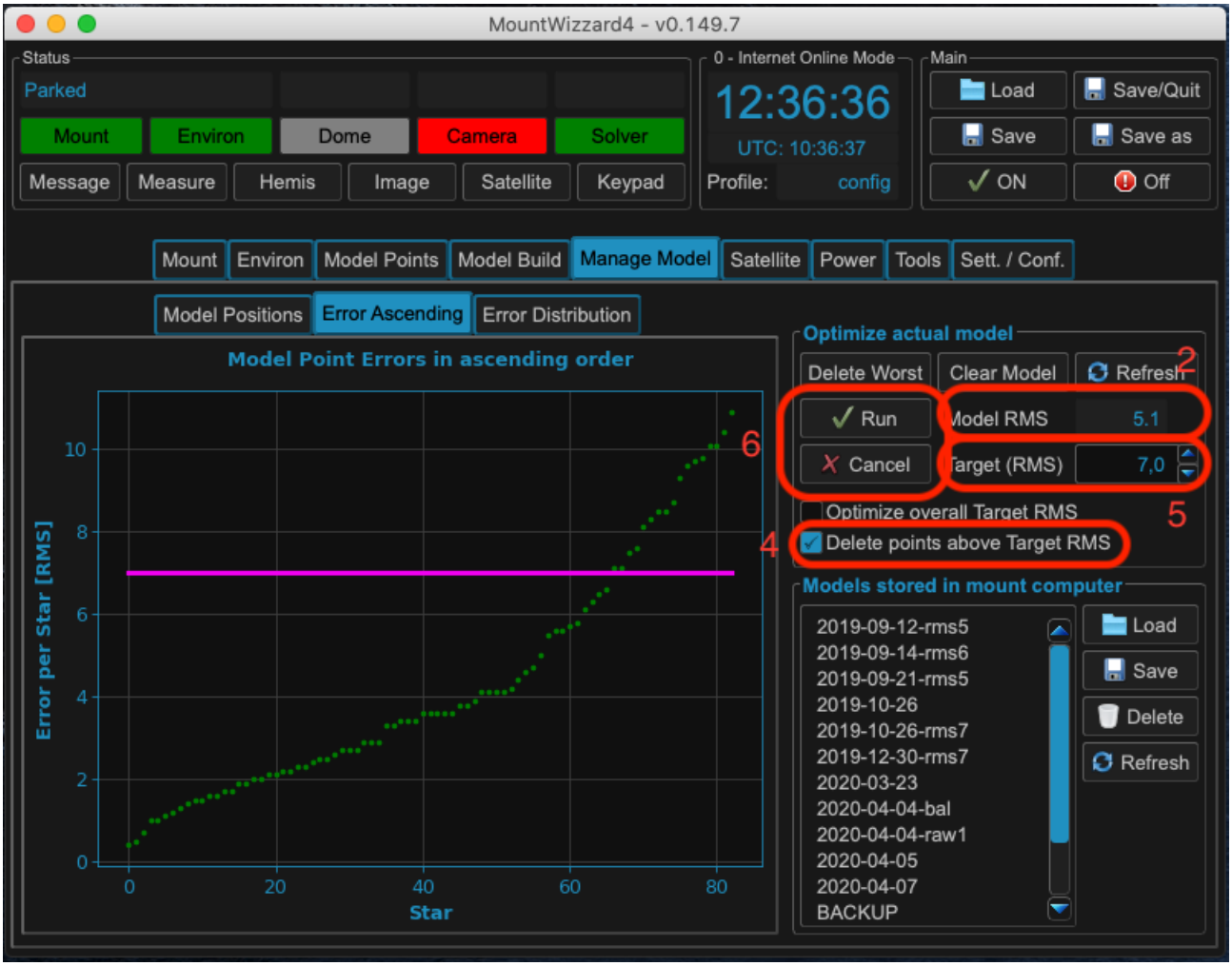

In all graphs you will see the pink lines which show the value of the Target (RMS) setting in the “Optimize actual model” box. This helps in adjusting a cut-off value for automatic model point removal.

## Optimizing the model

I some cases you will discover that some model points and their errors are outside a reasonable area. In that case three different model optimization scenarios come into place. The first and the obvious one is to remove manually the worst point of the model (1).




If you have more than a single point to remove, the second and third come into place, an automation process within MountWizzard4. With setting the checkbox “Optimize overall Target RMS” (3) and starting the optimization run (6), MountWizzard4 will delete as many point until the Model RMS (2) is below the selected threshold (5).




Checking “Delete point above Target RMS” (4) and run (6) the optimization, MountWizzard4 will delete all points where their individual Error is above the selected threshold (5). Please note that this will lead to a different Model RMS (2).

## Explanation of optimization of process:

Yes you could remove bad points from the model. But does it help ? Again from mathematics: you bend an error curve like a metal plate over a rough surface to equalize it. If the is a single stone under this plate -> approximation might be bad. So removing this stone might help in getting a better approximation for the rest of the surface. But it is not good to remove the gravel under the plate to improve just numbers in RMS! If I see large outliers in alignment errors within an area which shows good numbers around, I remove that single point. But not to much. In average I remove 2-5 points from a 60 point model max. Yes if you remove a point the over RMS could rise ! That’s because the whole model is newly calculated and that’s no subtraction of a bad point.

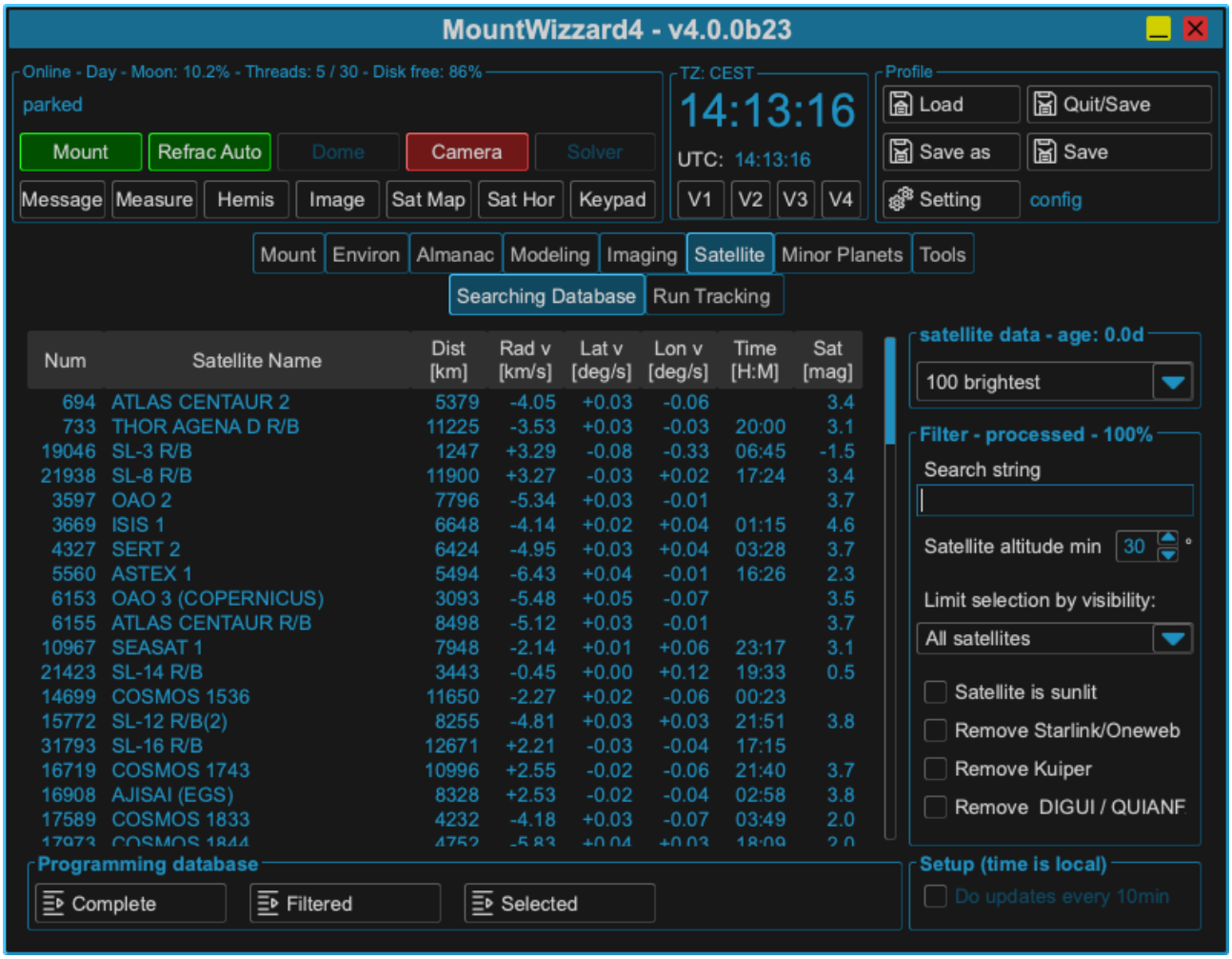
# Satellite tracking

MountWizzard4 supports the mount capability of tracking satellites. The planning is based on Two Line Elements (TLE), which are provided by organisations (e.g. http://www.celestrak.com) and deliver a set of orbital parameters for calculating the satellite track with regard to your location in Alt / Az coordinates. Since the introduction, 10micron does the calculation internally. Therefore it is needed to upload the satellite TLE data to the mount and let it calculate. Before FW 3.x this is the only possibility (using the internal satellite database works similar) to do tracking.

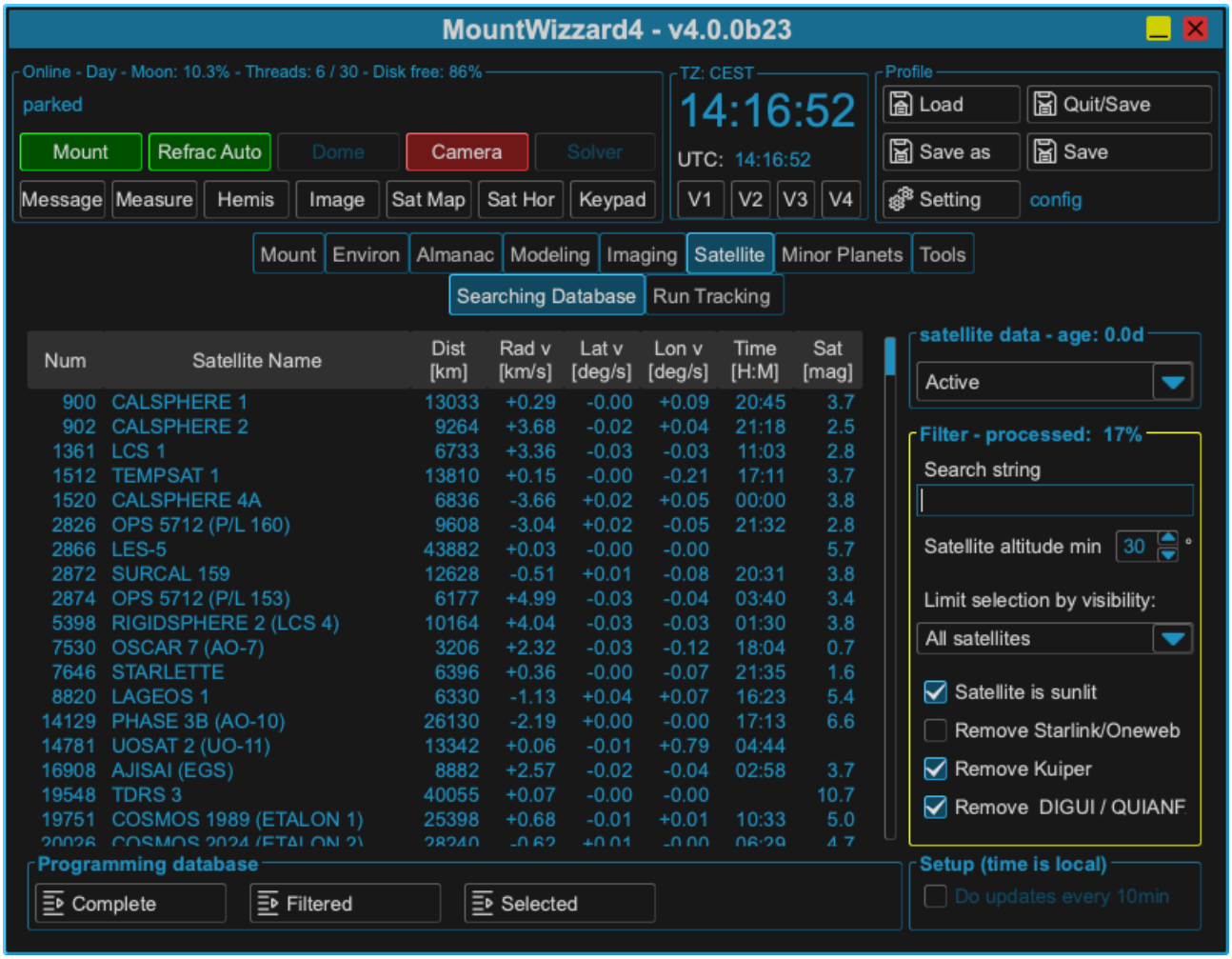
**Note**

As the orbital elements of a satellite change over time, it is necessary to get most actual data for get a good tracking. Data which is older than 10 days is definitely outdated. MountWizzard4 marks them red. Older than 3 days may work, MountWizzard4 marks them yellow. Ideal is to get them right in time.

You could select different databases on the right upper part of “Searching Database” tab {1}. Once selected and online enabled, MountWizzard4 will download the newest data and offers the list of included satellites. For finding the satellite of your choice you could use the search field (2) to reduce the list. The string is *not* case sensitive and will be stored persistent.



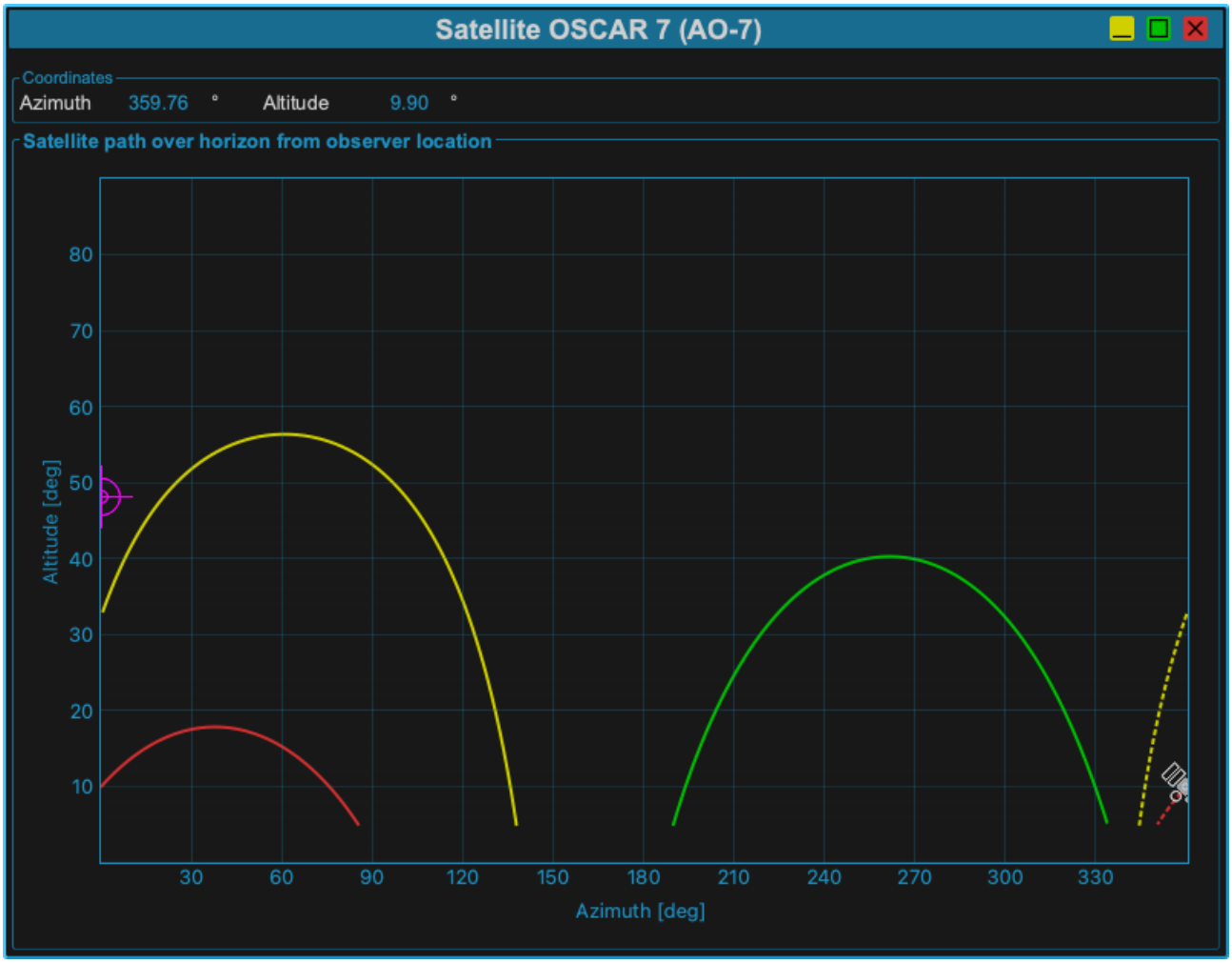

There are many more selection criteria available to filter the list of satellites with real-time calculations of some properties like velocity, distance and a glimpse of the apparent magnitude. This calculation takes some time and you will see a yellow frame and progress counter in the filter area.

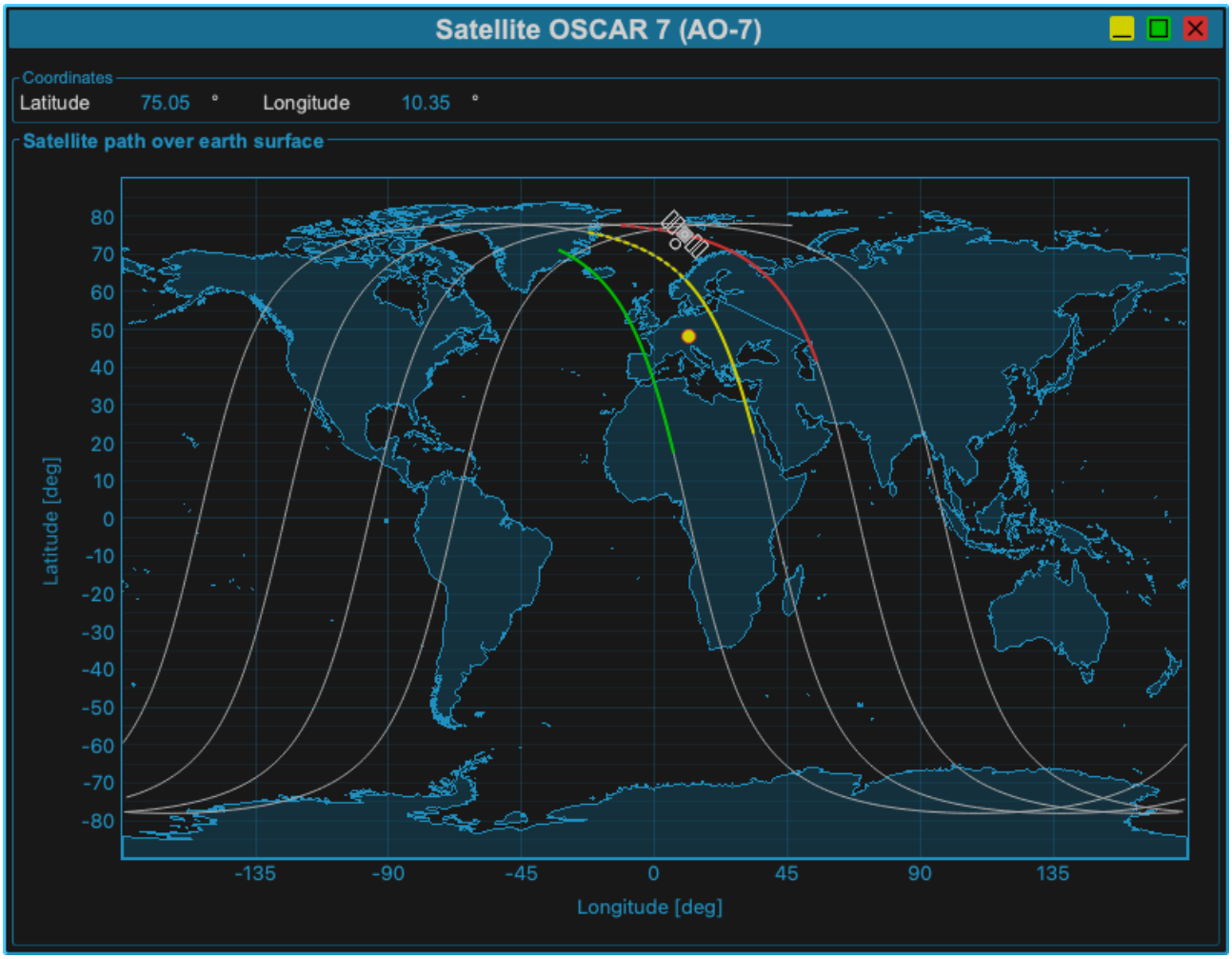



Once you choose a satellite with double click, data is programmed to mount controller, parameters are displayed, MountWizzard4 calculated the next 3 orbits of the satellite with rise / culminate / settle and if it occurs the flip time when crossing the meridian.

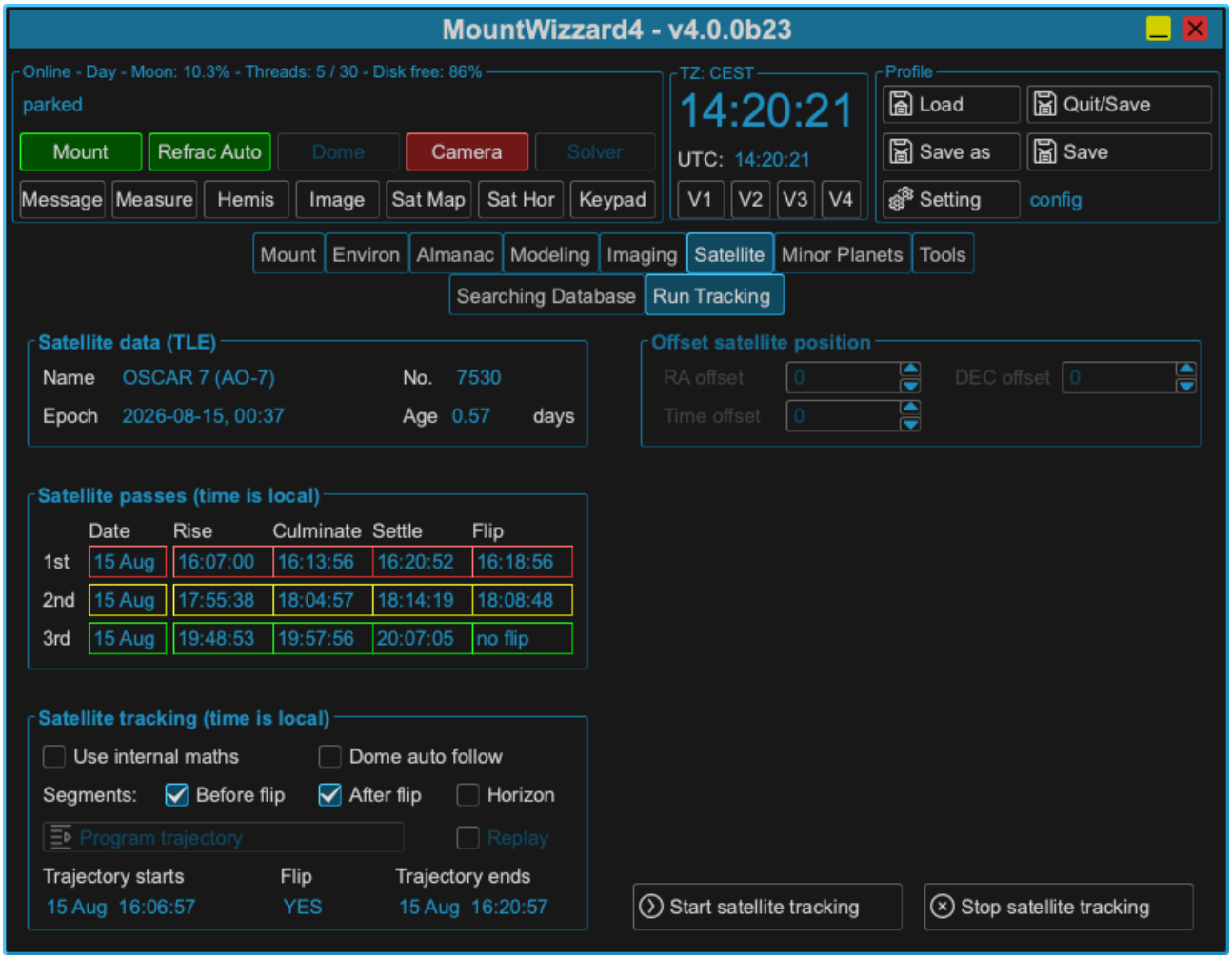



The selected satellite will also be shown in the **Satellite** windows:


Satellite OSCAR 7 (AO-7)
Coordinates
Azimuth 359.76 °
Altitude 9.90 °
Satellite path over horizon from observer location
Altitude [deg]
80
70
60
50
40
30
20
10
30
60
90
120
150
180
210
240
270
300
330
Azimuth [deg]

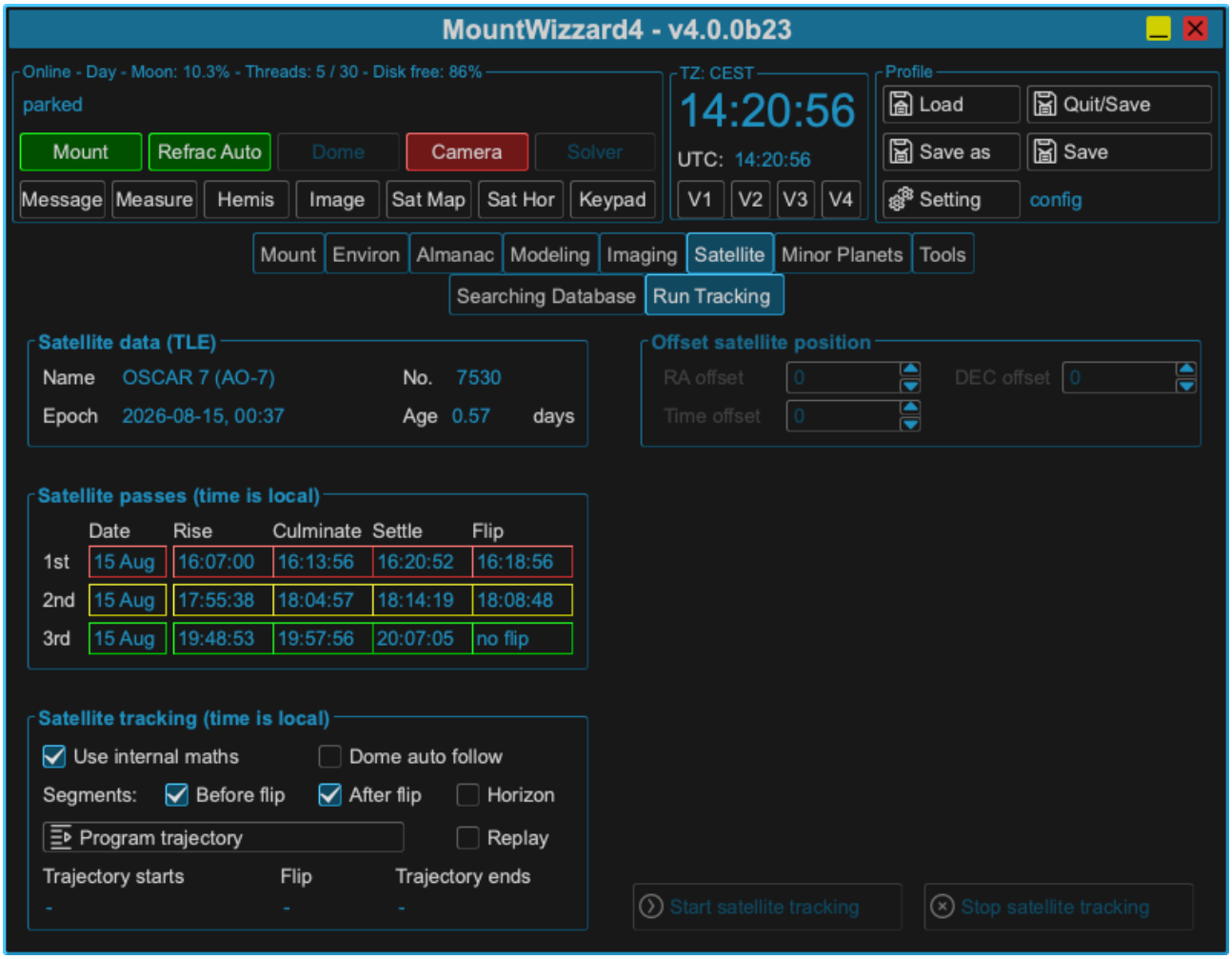



After the mount has done it's calculations as well, the result will be shown in the **Trajectory starts** and **Trajectory ends** fields and a possible flip will be announced on the "Run Tracking" tab, MountWizzard4 offers now (need Firmware 3.x) some adjustments for centering the satellites in you imaging FOV.




As soon as a valid tracking path is present in the mount, the **Start satellite tracking** and **Stop satellite tracking** button are enabled. Once started, the mount will slew to the begin of the tracking path and wait for the satellite to rise. Selecting partial tracks and respecting constraints is not possible.

Since FW 3.x the command protocol offers the programming of a custom satellite track. This offers the capability of takings care of avoiding flips, respect horizon and other constraints. The operation is different to the classic approach: Instead of programming TLE data to the mount, MountWizzard4 programs Alt / Az coordinates in a one second interval to the mount (max. 900s) which the mount after start tracking will follow. You could enable this feature with checking **Use internal maths** if the firmware is 3.x or higher. After enabling, additional elements will be enabled.



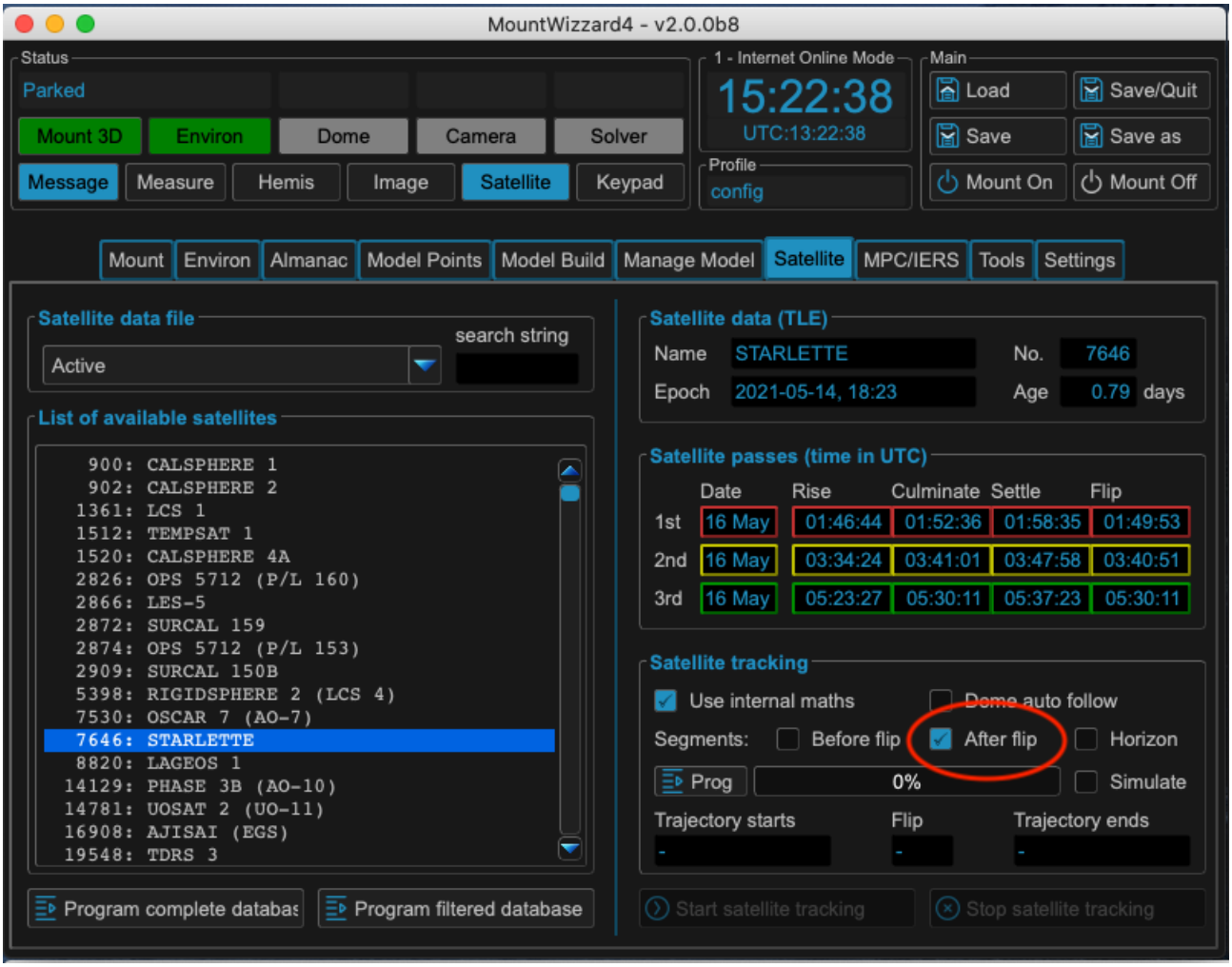

As the calculation now happens outside the mount, we could take a look to the difference between tracks calculated by the 10micron mount and MountWizzard4 based on the same satellite TLE data! You will find some comparison under the architecture / math section: precision of internal calculations.

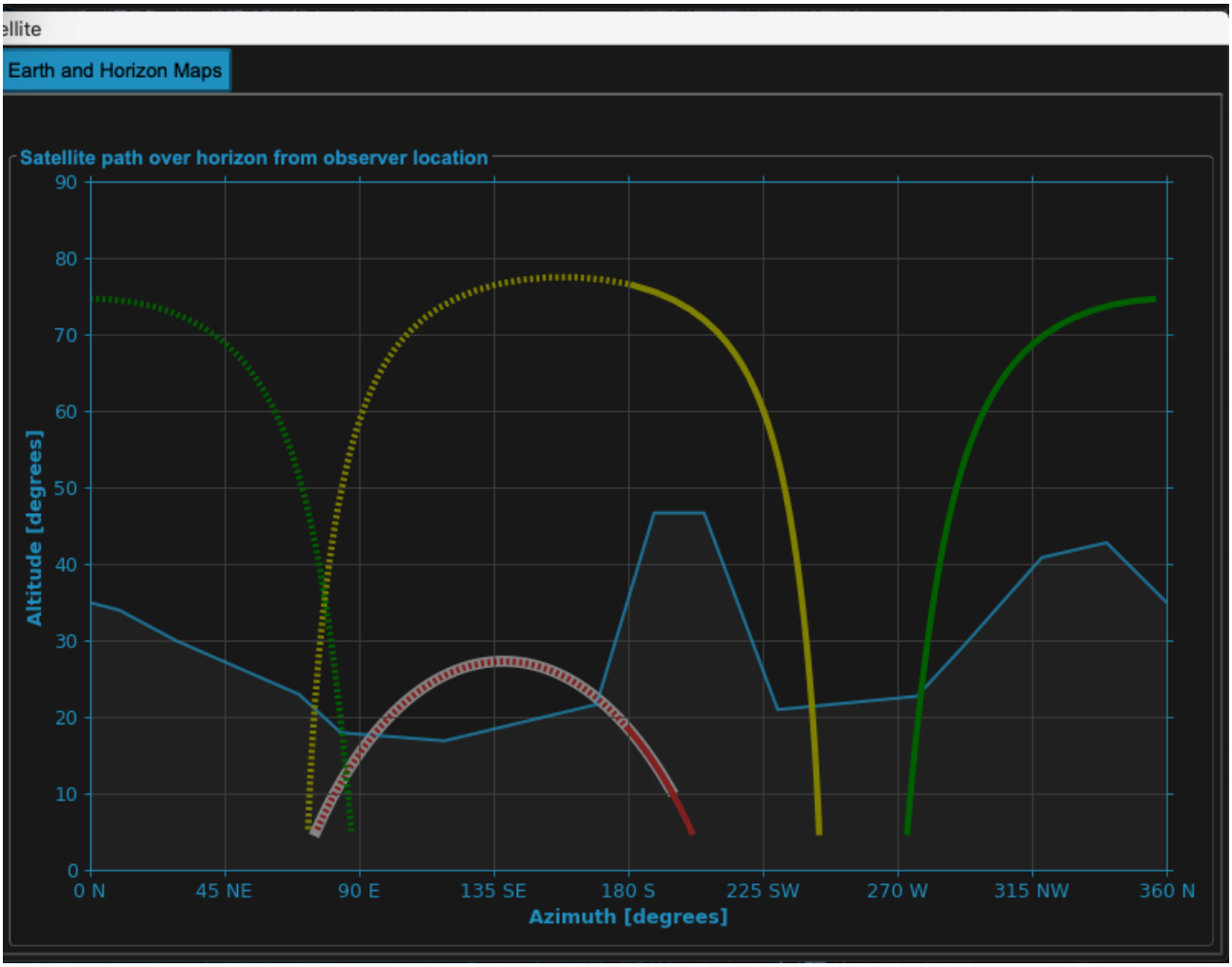
If you are using the internal math as well as classic mode, you could open the satellite window. There you could see the next three orbits and for internal math the resulting satellite track with an white underlay. If you change any setup, MountWizzard4 will recalculate all data and updates the plots accordingly.

**Note**

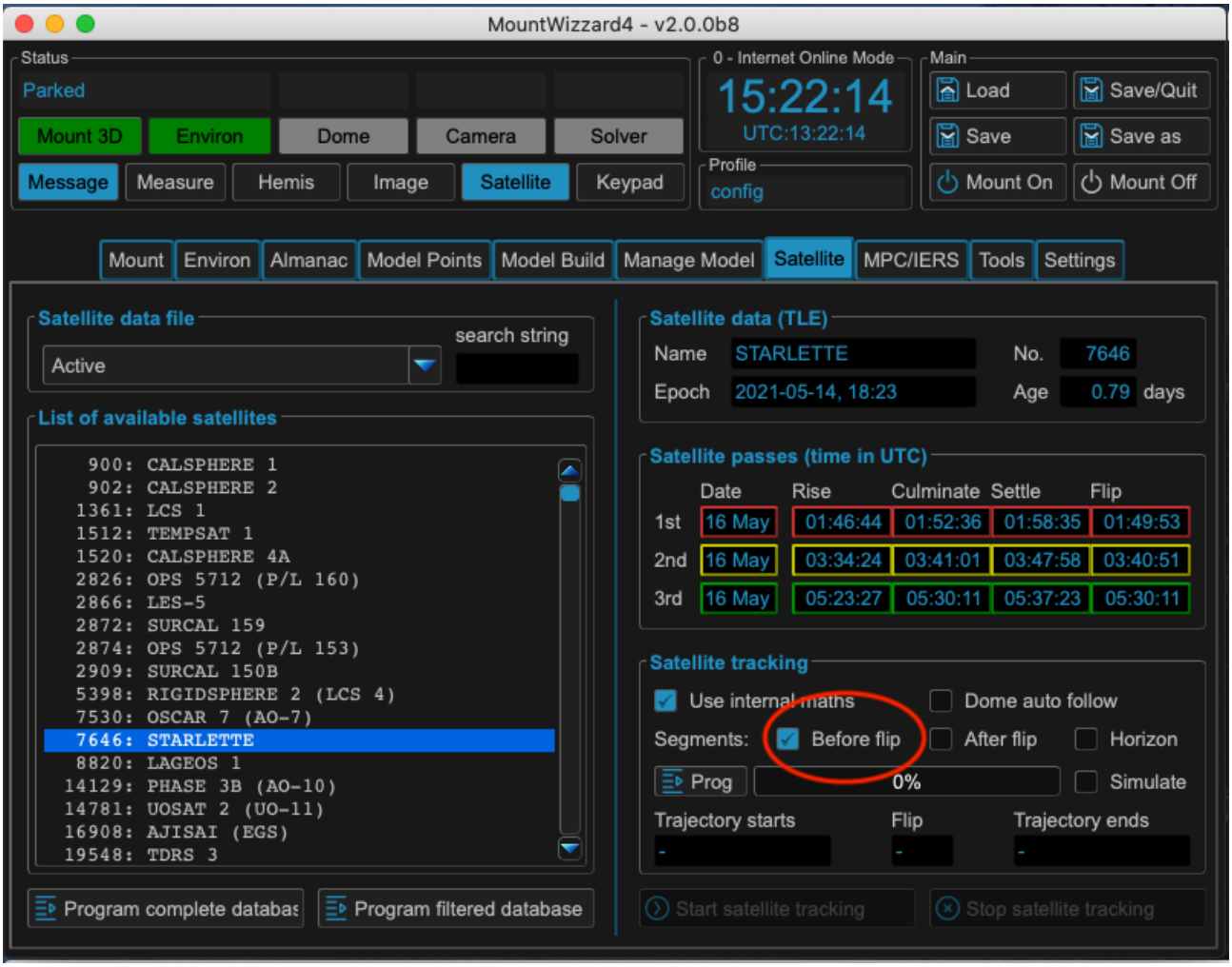
The solid line shows the track before meridian transit, the dotted line the track after meridian transit. The resulting white background shows the resulting track, which takes flip track tolerance into account. If you change settings in satellite (or even limits), MountWizzard4 will update the tracks path in plot accordingly and reset the prog state.

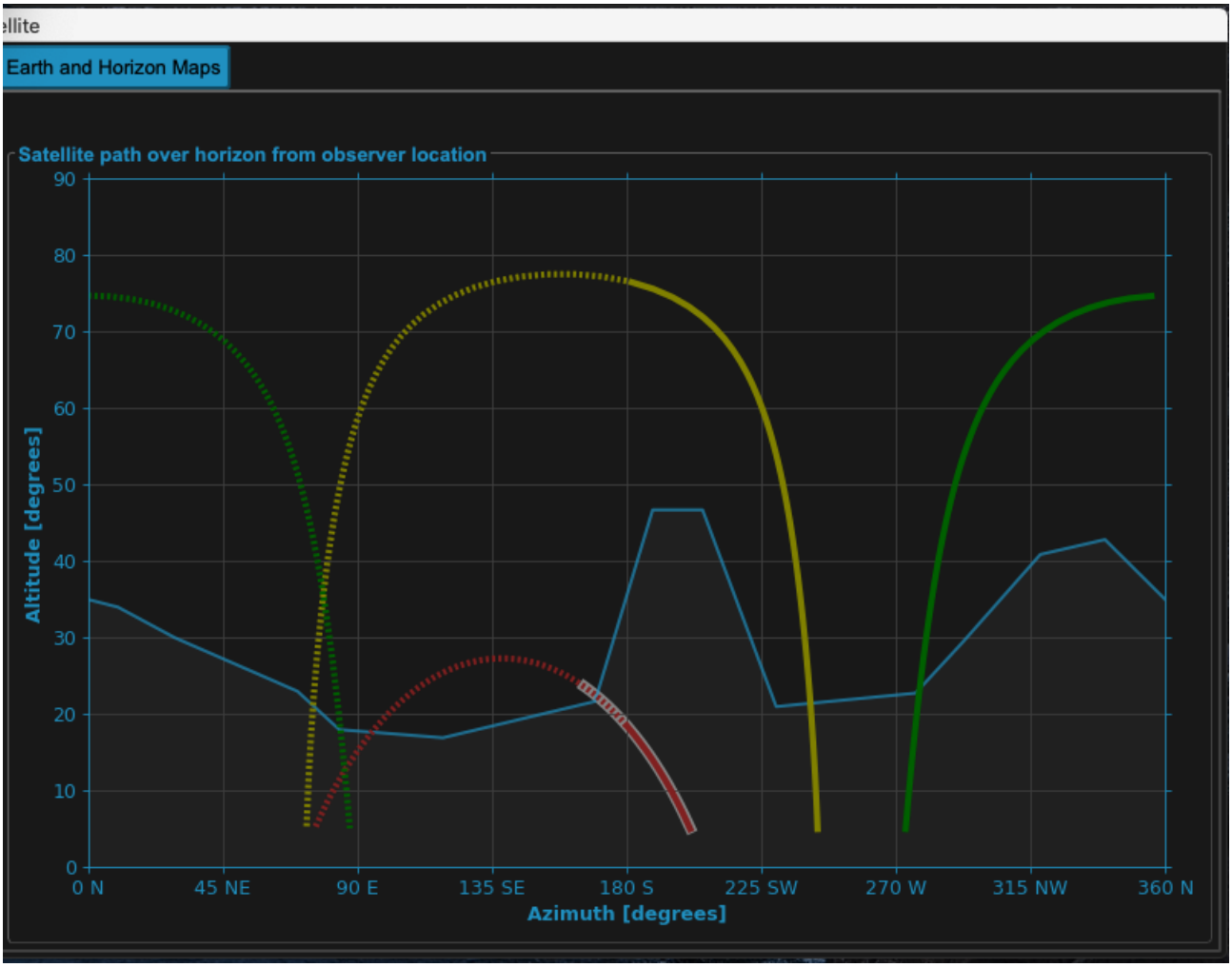
Select only a segment before a meridian transit (and therefore avoiding a flip during tracking):


MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
1 - Internet Online Mode
15:22:38
UTC:13:22:38
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Satellite data file
search string
Active
List of available satellites
900: CALSPHERE 1
902: CALSPHERE 2
1361: LCS 1
1512: TEMPSAT 1
1520: CALSPHERE 4A
2826: OPS 5712 (P/L 160)
2866: LES-5
2872: SURCAL 159
2874: OPS 5712 (P/L 153)
2909: SURCAL 150B
5398: RIGIDSPHERE 2 (LCS 4)
7530: OSCAR 7 (AO-7)
7646: STARLETTE
8820: LAGEOS 1
14129: PHASE 3B (AO-10)
14781: UOSAT 2 (UO-11)
16908: AJISAI (EGS)
19548: TDRS 3
Program complete databas
Program filtered database
Satellite data (TLE)
Name STARLETTE
No. 7646
Epoch 2021-05-14, 18:23
Age 0.79 days
Satellite passes (time in UTC)
Date Rise Culminate Settle Flip
1st 16 May 01:46:44 01:52:36 01:58:35 01:49:53
2nd 16 May 03:34:24 03:41:01 03:47:58 03:40:51
3rd 16 May 05:23:27 05:30:11 05:37:23 05:30:11
Satellite tracking
Use internal maths
Dome auto follow
Segments:
Before flip
After flip
Horizon
Prog
0%
Simulate
Trajectory starts
Flip
Trajectory ends
Start satellite tracking
Stop satellite tracking

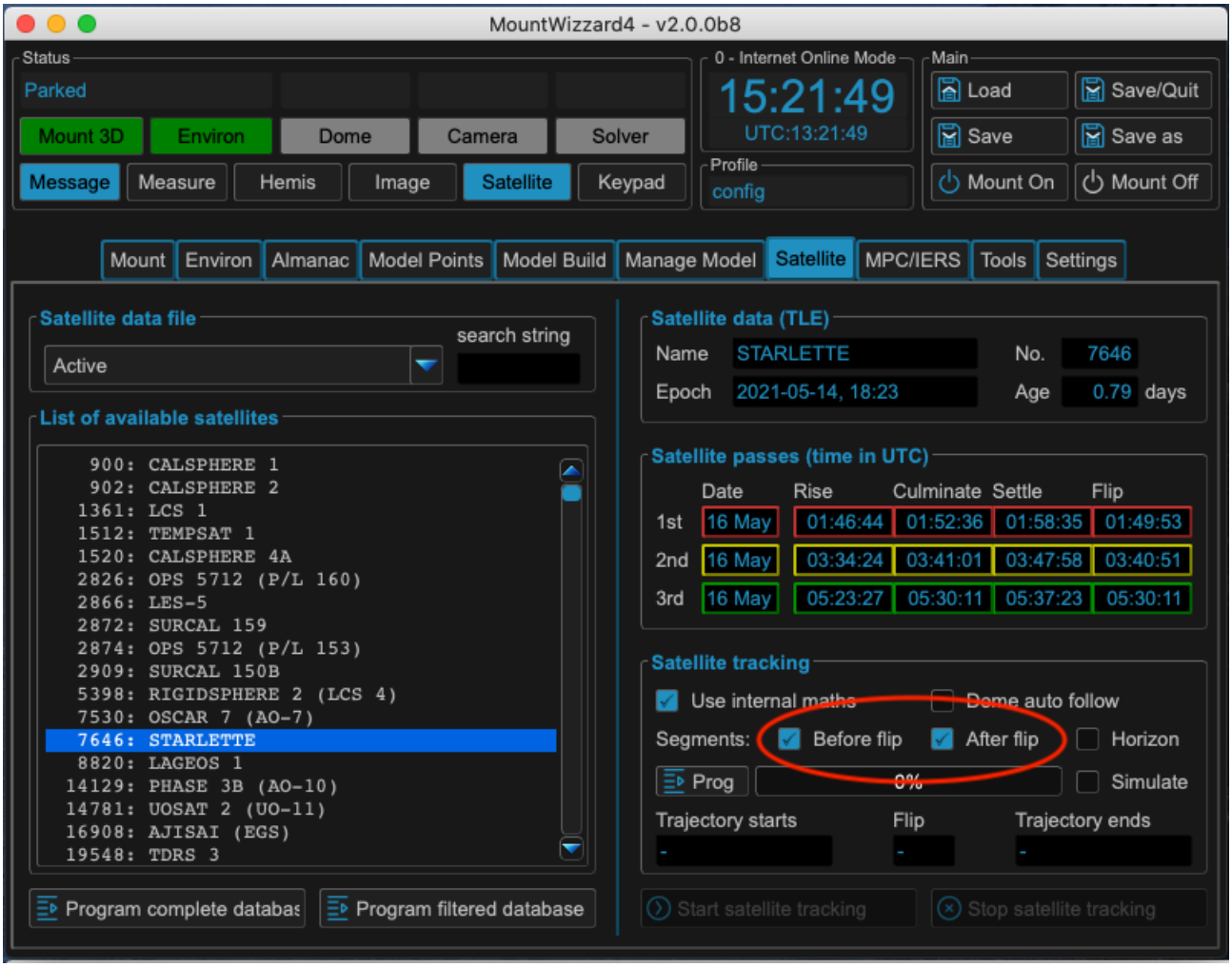


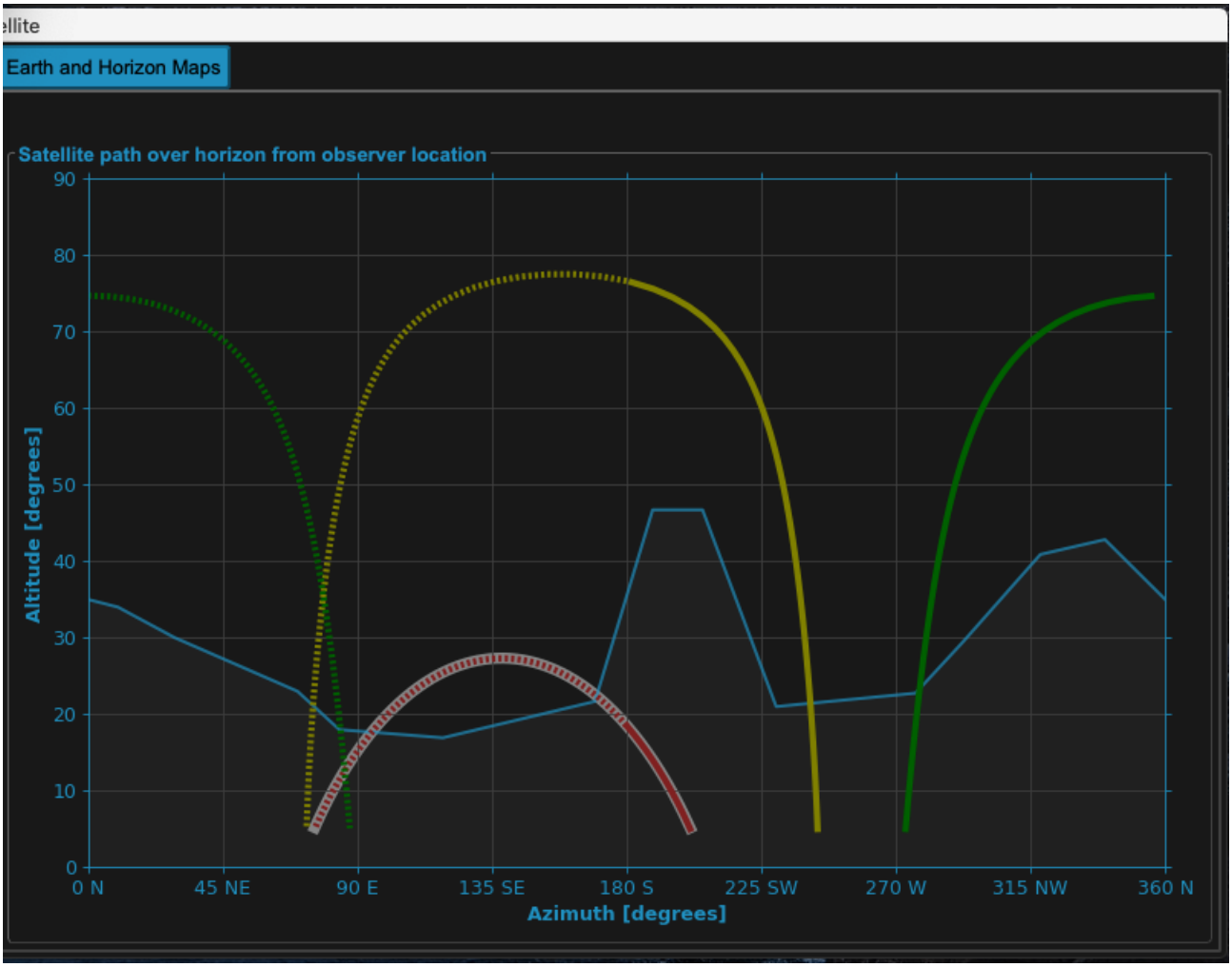

Select only a segment after a meridian transit (and therefore avoiding a flip during tracking):


MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
0 - Internet Online Mode
15:22:14
UTC:13:22:14
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Satellite data file
search string
Active
List of available satellites
900: CALSPHERE 1
902: CALSPHERE 2
1361: LCS 1
1512: TEMPSAT 1
1520: CALSPHERE 4A
2826: OPS 5712 (P/L 160)
2866: LES-5
2872: SURCAL 159
2874: OPS 5712 (P/L 153)
2909: SURCAL 150B
5398: RIGIDSPHERE 2 (LCS 4)
7530: OSCAR 7 (AO-7)
7646: STARLETTE
8820: LAGEOS 1
14129: PHASE 3B (AO-10)
14781: UOSAT 2 (UO-11)
16908: AJISAI (EGS)
19548: TDRS 3
Program complete databas
Program filtered database
Satellite data (TLE)
Name STARLETTE
No. 7646
Epoch 2021-05-14, 18:23
Age 0.79 days
Satellite passes (time in UTC)
Date Rise Culminate Settle Flip
1st 16 May 01:46:44 01:52:36 01:58:35 01:49:53
2nd 16 May 03:34:24 03:41:01 03:47:58 03:40:51
3rd 16 May 05:23:27 05:30:11 05:37:23 05:30:11
Satellite tracking
Use internal maths
Dome auto follow
Segments:
Before flip
After flip
Horizon
Prog
0%
Simulate
Trajectory starts
Flip
Trajectory ends
Start satellite tracking
Stop satellite tracking

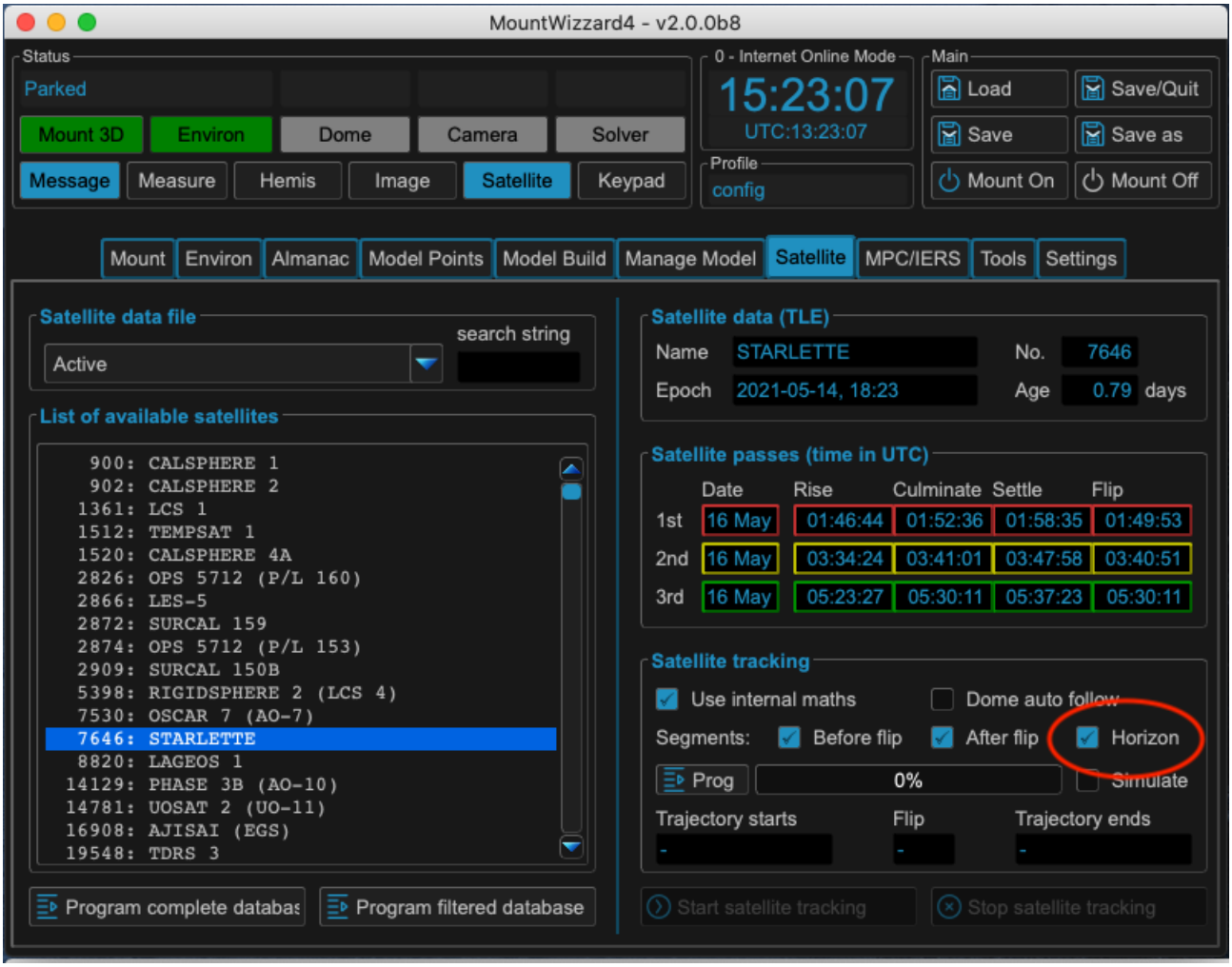


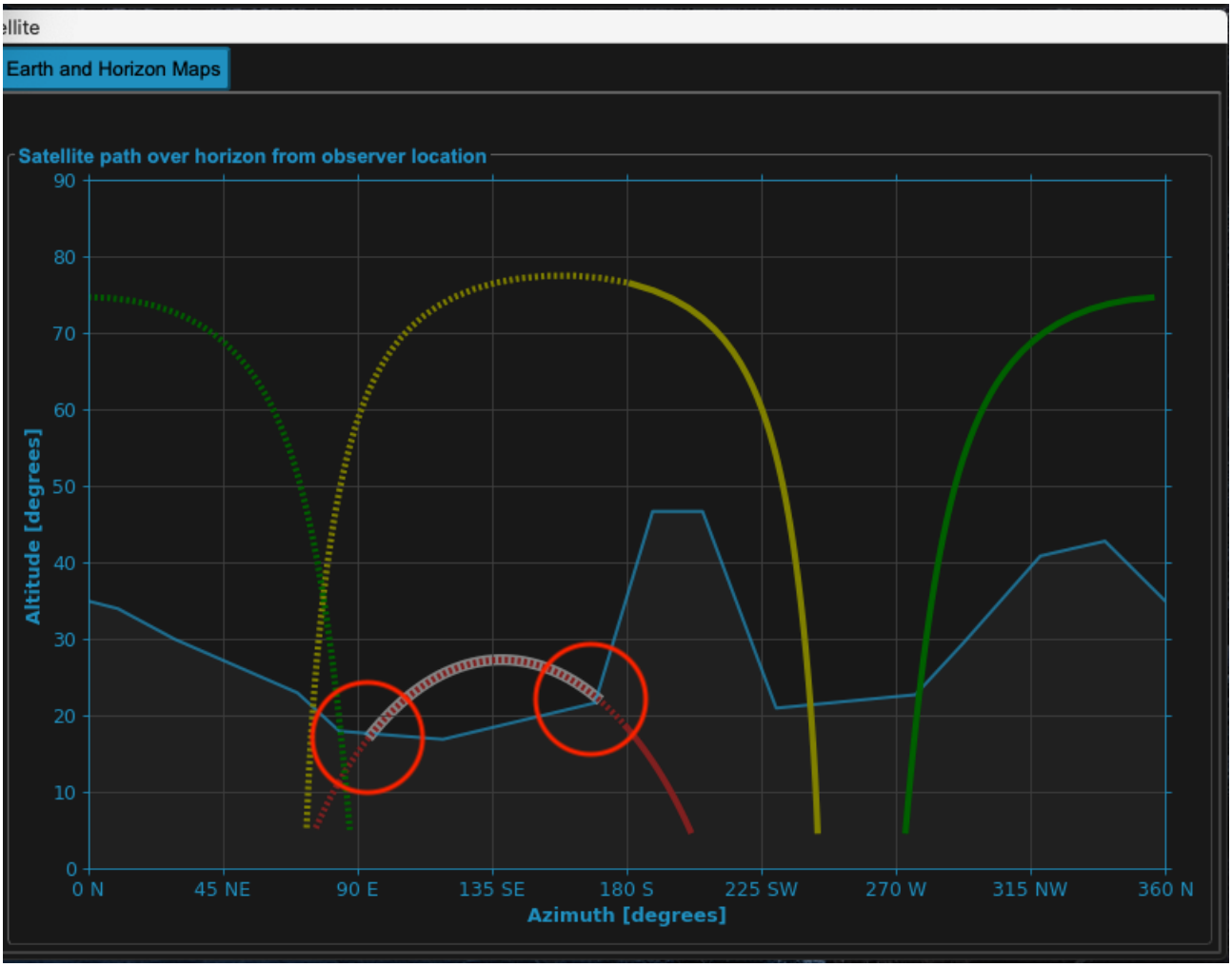

Select both segments of the meridian (this might cause a flip during tracking):


MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
0 - Internet Online Mode
15:21:49
UTC:13:21:49
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Satellite data file
search string
Active
List of available satellites
900: CALSPHERE 1
902: CALSPHERE 2
1361: LCS 1
1512: TEMPSAT 1
1520: CALSPHERE 4A
2826: OPS 5712 (P/L 160)
2866: LES-5
2872: SURCAL 159
2874: OPS 5712 (P/L 153)
2909: SURCAL 150B
5398: RIGIDSPHERE 2 (LCS 4)
7530: OSCAR 7 (AO-7)
7646: STARLETTE
8820: LAGEOS 1
14129: PHASE 3B (AO-10)
14781: UOSAT 2 (UO-11)
16908: AJISAI (EGS)
19548: TDRS 3
Program complete databas
Program filtered database
Satellite data (TLE)
Name
STARLETTE
No.
7646
Epoch
2021-05-14, 18:23
Age
0.79
days
Satellite passes (time in UTC)
Date
Rise
Culminate
Settle
Flip
1st
16 May
01:46:44
01:52:36
01:58:35
01:49:53
2nd
16 May
03:34:24
03:41:01
03:47:58
03:40:51
3rd
16 May
05:23:27
05:30:11
05:37:23
05:30:11
Satellite tracking
Use internal maths
Dome auto follow
Segments:
Before flip
After flip
Horizon
Prog
0%
Simulate
Trajectory starts
Flip
Trajectory ends
Start satellite tracking
Stop satellite tracking

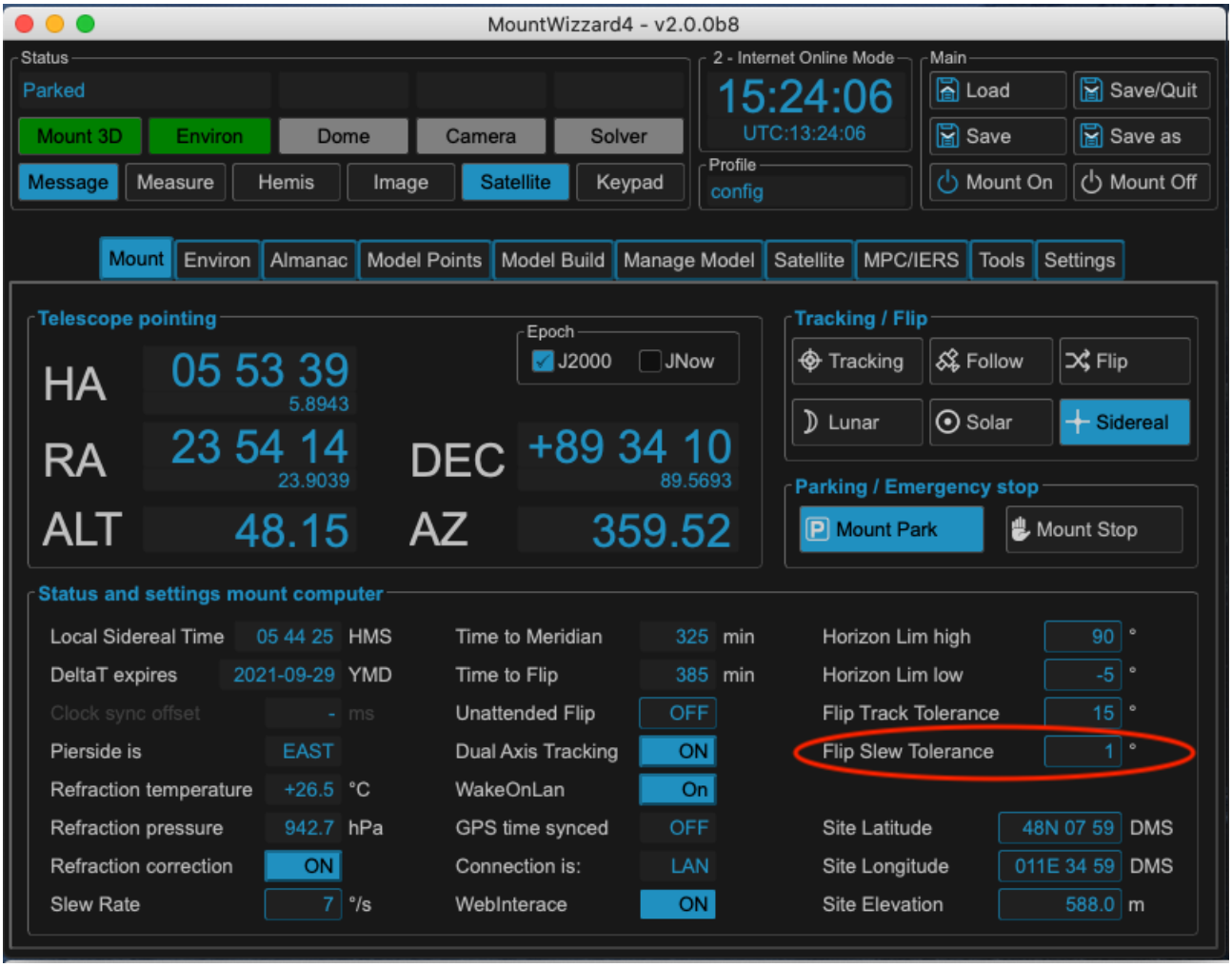


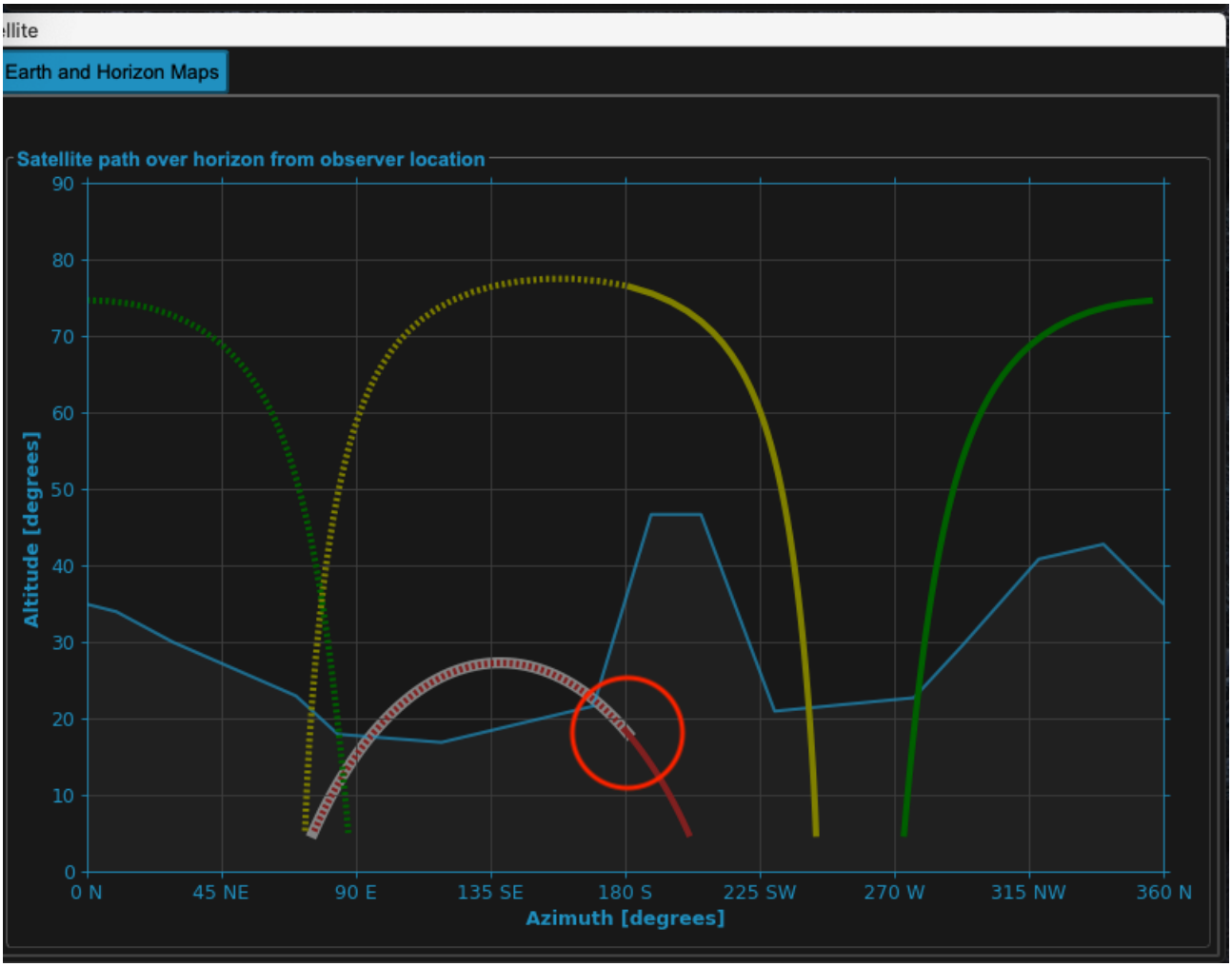

Select respecting the horizon line. This filters out additional all segments, which are below the given horizon mask.


MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
0 - Internet Online Mode
15:23:07
UTC:13:23:07
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Satellite data file
search string
Active
List of available satellites
900: CALSPHERE 1
902: CALSPHERE 2
1361: LCS 1
1512: TEMPSAT 1
1520: CALSPHERE 4A
2826: OPS 5712 (P/L 160)
2866: LES-5
2872: SURCAL 159
2874: OPS 5712 (P/L 153)
2909: SURCAL 150B
5398: RIGIDSPHERE 2 (LCS 4)
7530: OSCAR 7 (AO-7)
7646: STARLETTE
8820: LAGEOS 1
14129: PHASE 3B (AO-10)
14781: UOSAT 2 (UO-11)
16908: AJISAI (EGS)
19548: TDRS 3
Program complete databas
Program filtered database
Satellite data (TLE)
Name
STARLETTE
No.
7646
Epoch
2021-05-14, 18:23
Age
0.79
days
Satellite passes (time in UTC)
Date
Rise
Culminate
Settle
Flip
1st
16 May
01:46:44
01:52:36
01:58:35
01:49:53
2nd
16 May
03:34:24
03:41:01
03:47:58
03:40:51
3rd
16 May
05:23:27
05:30:11
05:37:23
05:30:11
Satellite tracking
Use internal maths
Dome auto follow
Segments:
Before flip
After flip
Horizon
Prog
0%
Simulate
Trajectory starts
Flip
Trajectory ends
Start satellite tracking
Stop satellite tracking

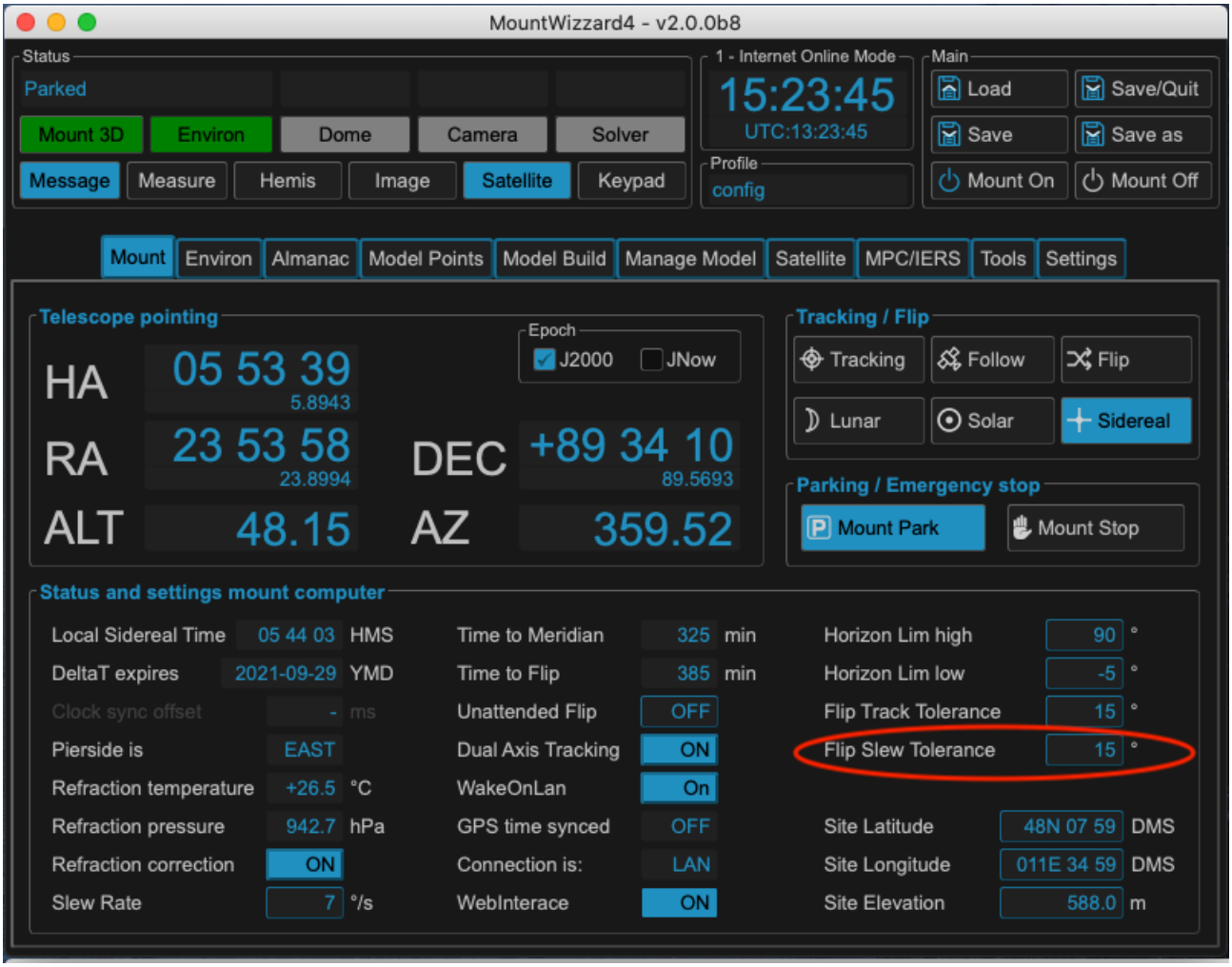


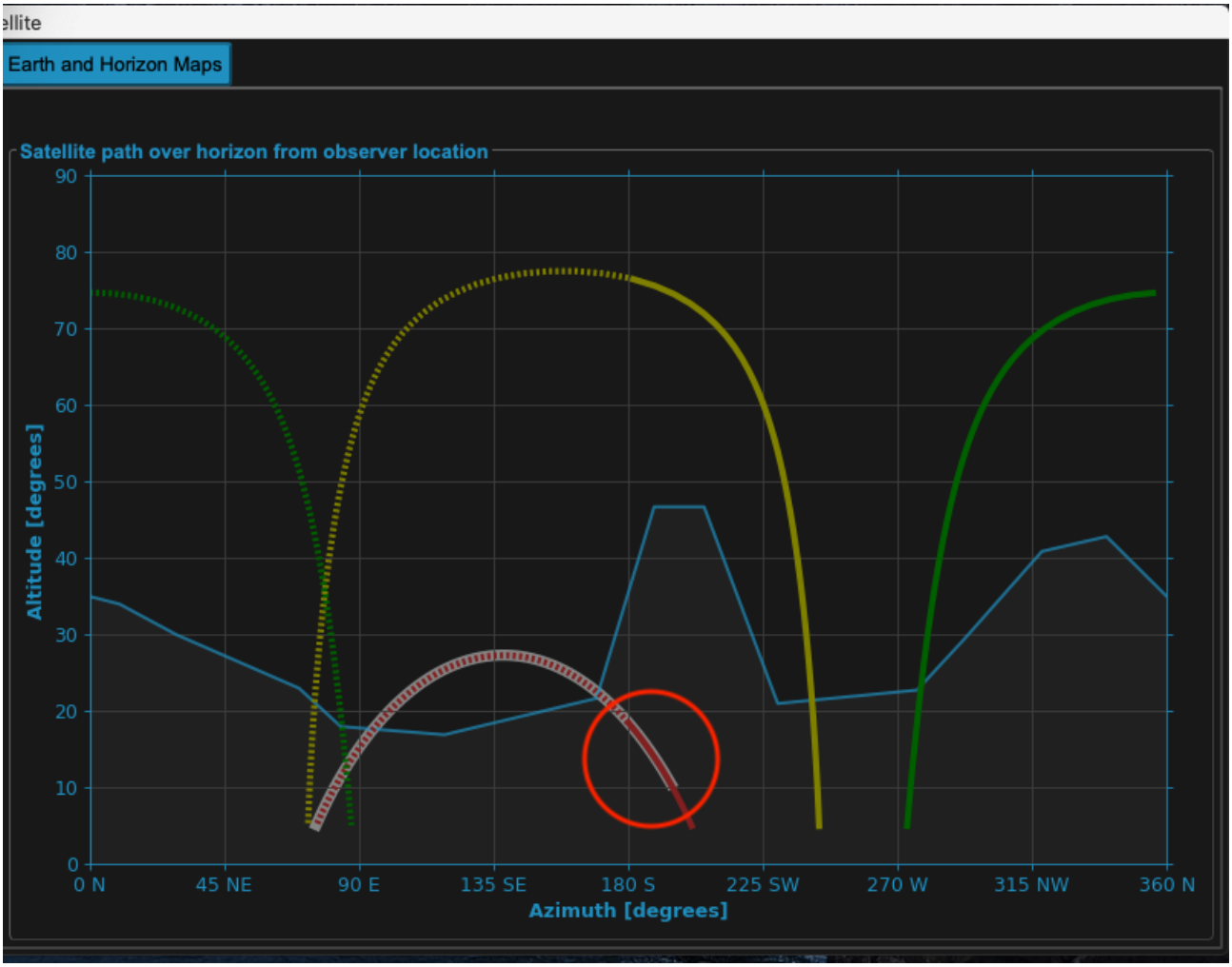

MountWizzard4 will take into account the meridian track limits of your mount. Here set to 1 degree (which is close to meridian)


MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
2 - Internet Online Mode
15:24:06
UTC:13:24:06
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Telescope pointing
Epoch
J2000
JNow
HA
05 53 39
5.8943
RA
23 54 14
23.9039
DEC
+89 34 10
89.5693
ALT
48.15
AZ
359.52
Tracking / Flip
Tracking
Follow
Flip
Lunar
Solar
Sidereal
Parking / Emergency stop
Mount Park
Mount Stop
Status and settings mount computer
Local Sidereal Time
05 44 25
HMS
DeltaT expires
2021-09-29
YMD
Clock sync offset
-
ms
Pierside is
EAST
Refraction temperature
+26.5
°C
Refraction pressure
942.7
hPa
Refraction correction
ON
Slew Rate
7
°/s
Time to Meridian
325
min
Time to Flip
385
min
Unattended Flip
OFF
Dual Axis Tracking
ON
WakeOnLan
On
GPS time synced
OFF
Connection is:
LAN
WebInterace
ON
Horizon Lim high
90
°
Horizon Lim low
-5
°
Flip Track Tolerance
15
°
Flip Slew Tolerance
1
°
Site Latitude
48N 07 59
DMS
Site Longitude
011E 34 59
DMS
Site Elevation
588.0
m




MountWizzard4 will take into account the meridian track limits of your mount. Here set to 15 degrees (which could avoid a meridian flip or at least extend the tracking time)

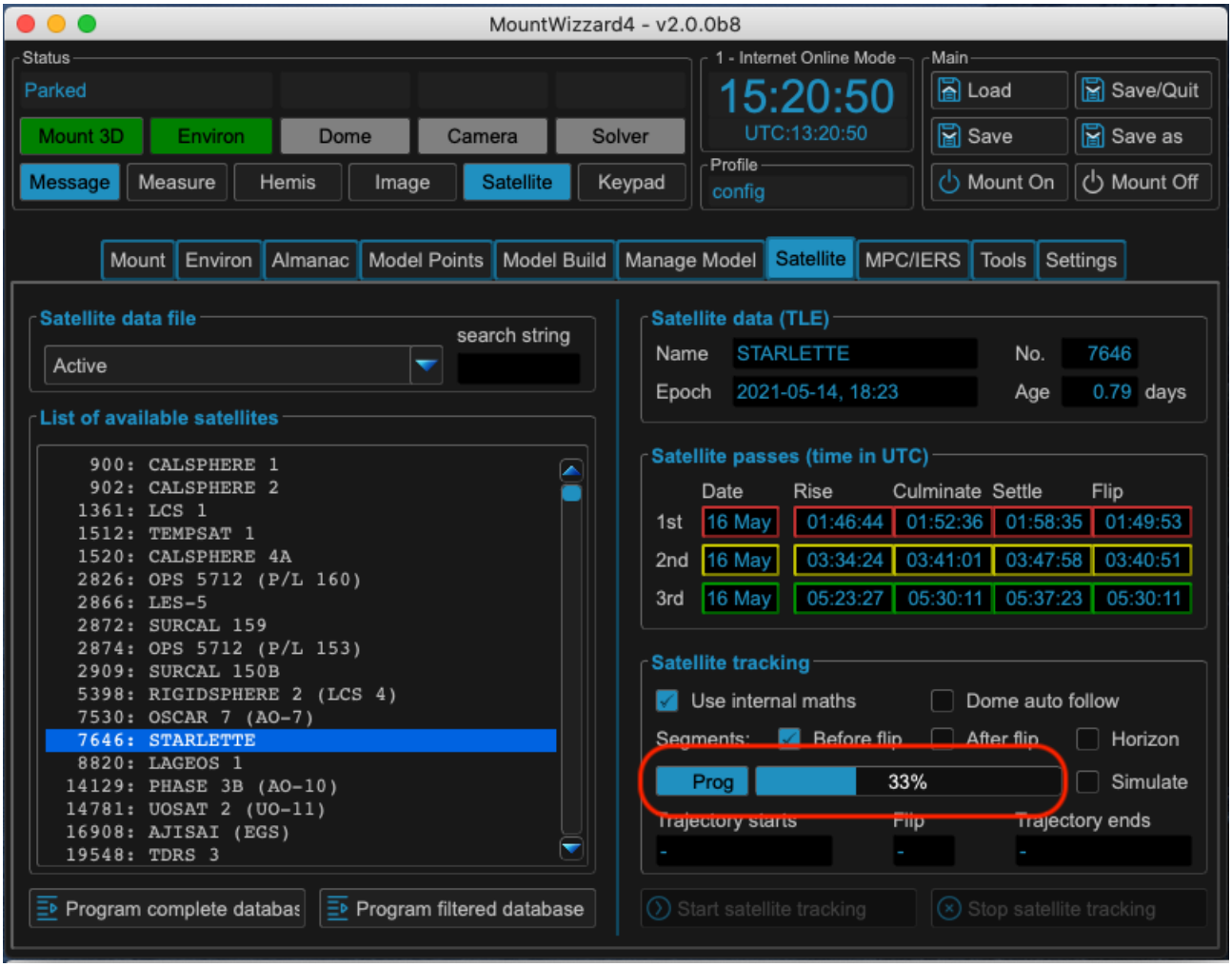

MountWizzard4 - v2.0.0b8
Status
Parked
Mount 3D
Environ
Dome
Camera
Solver
Message
Measure
Hemis
Image
Satellite
Keypad
1 - Internet Online Mode
15:23:45
UTC:13:23:45
Profile
config
Main
Load
Save/Quit
Save
Save as
Mount On
Mount Off
Mount
Environ
Almanac
Model Points
Model Build
Manage Model
Satellite
MPC/IERS
Tools
Settings
Telescope pointing
Epoch
J2000
JNow
HA
05 53 39
5.8943
RA
23 53 58
23.8994
DEC
+89 34 10
89.5693
ALT
48.15
AZ
359.52
Tracking / Flip
Tracking
Follow
Flip
Lunar
Solar
Sidereal
Parking / Emergency stop
Mount Park
Mount Stop
Status and settings mount computer
Local Sidereal Time
05 44 03
HMS
DeltaT expires
2021-09-29
YMD
Clock sync offset
-
ms
Pierside is
EAST
Refraction temperature
+26.5
°C
Refraction pressure
942.7
hPa
Refraction correction
ON
Slew Rate
7
°/s
Time to Meridian
325
min
Time to Flip
385
min
Unattended Flip
OFF
Dual Axis Tracking
ON
WakeOnLan
On
GPS time synced
OFF
Connection is:
LAN
WebInterace
ON
Horizon Lim high
90
°
Horizon Lim low
-5
°
Flip Track Tolerance
15
°
Flip Slew Tolerance
15
°
Site Latitude
48N 07 59
DMS
Site Longitude
011E 34 59
DMS
Site Elevation
588.0
m



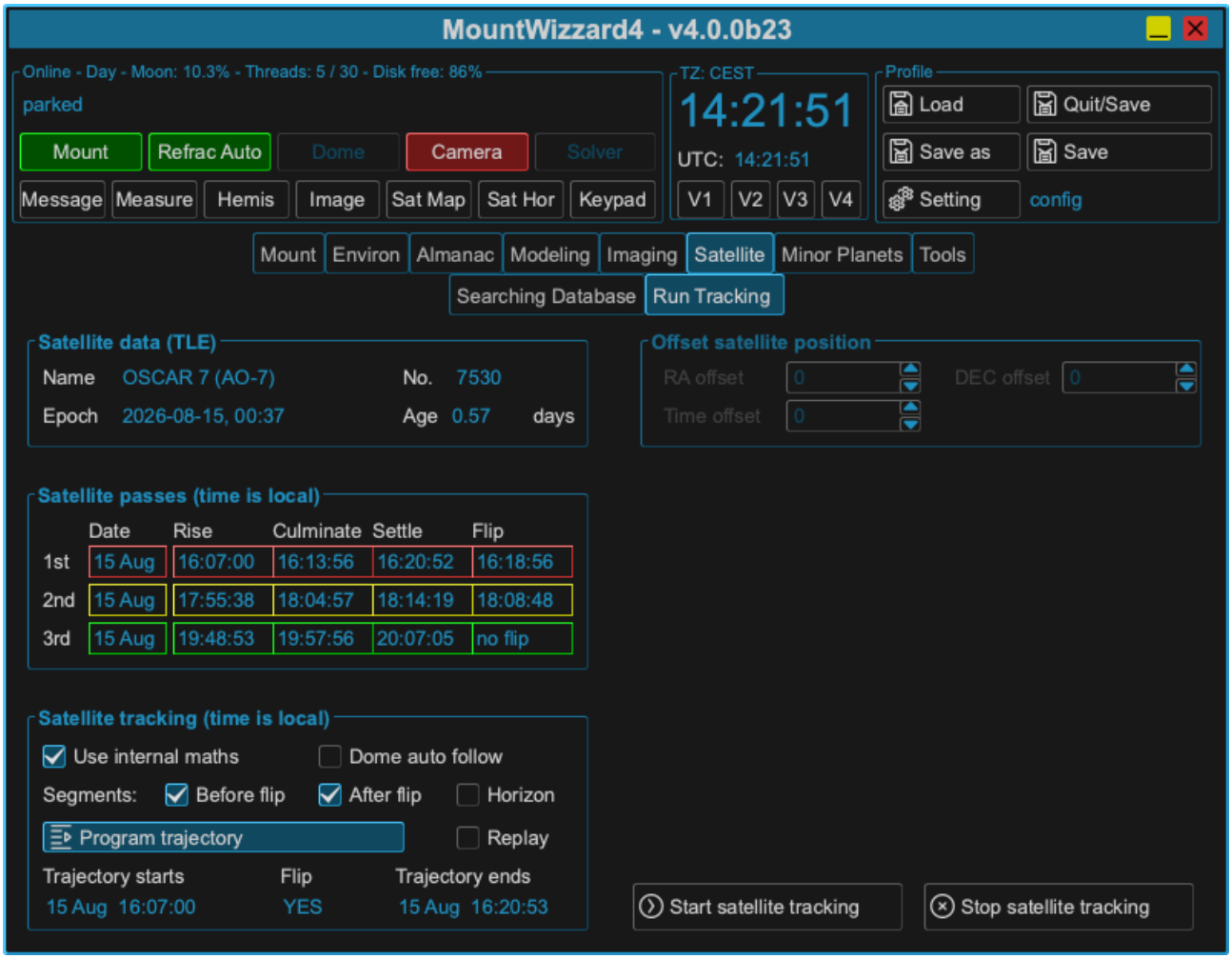

**⚠ Warning**

The meridian track limits have to be chosen carefully as the mount might hit your setup !

The biggest change in using satellite tracking with the new implementation is how the data is handled to the mount: whereas in classic mode only the TLE data has to be uploaded (which is quick) now the whole track coordinates have to be programmed . As this takes up to 10 seconds, MountWizzard4 does not automatically start the transfer. Once your setup (choice of segment, horizon etc.) is made, you have to start the programming by pushing the **Prog** button.




After a successful upload, the trajectory data is populated and the Start / Stop tracking buttons are enable like in classic mode.




## Tools for the Job

# Troubleshooting Q&A and hints

Based on many feedbacks and solved issues, please check first if your question already has some answers or at least some hints how to improve the situation.

## Installation

Despite MountWizzard4 will run on many platforms, your setup might have some special constrains which need to taken into account. In the following I try to refer to the important ones.

## Python

MountWizzard4 version 4 runs on python 3.10 - 3.13. On other versions scripts will fail and MountWizzard4 will not run.

On windows, you need to select if you are using 32bit or 64bit python depending on drivers you are using for your devices. 32bit and 64bit could not be mixed !

Normally you will use a preinstalled python version (if that fits) or use a python version from python.org. Please do not use other sources.

As MountWizzard4 lives in a virtual environment, updating python does not automatically update the virtual environment. If you need to update python for any reason there are two possibilities: New install of MountWizzard4 in a new work dir or deleting the venv folder in your actual work dir and running the install script again.

## MountWizzard4 App

There is no need for running MountWizzard4 with admin rights. If so, something is wrong. MountWizzard4 will run in a virtual environment. Please do not try to install MountWizzard4 as a system application as this might interfere with other installations.

MountWizzard4 does not behave as expected: please post a log and describe the procedure in steps. If you could add screen shots this helps a lot.

## MountWizzard4 Updating does fail

Since v2 MountWizzard4 should be able to handle all updates / downgrades with the internal updater. From v1 to v2 windows needs the MountWizzard4_Update.bat script as the internal updater can’t free used windows libraries. MacOSx and Linux should be fine.

## Scripts

Installation does not need admin rights. If so, please check the folder locations. In windows10 desktop and some other folder are not writable for applications.

The scripts do nothing special, you could use for many topics manual commands as well. Unfortunately the scripts could not manage all special setups, but feel free to change them accordingly.

# Mount connectivity

MountWizzard4 only supports IP links. As data latency is a critical topic, please use a wired connection to the mount. Wireless connection might have some drops in connections (you will see this with mount button switching red / green multiple times)

Please check your IP settings, gateways if first connections fail.

If your WOL does not work, please check MAC address, WOL being enabled. If you switched your mount manual off and cut the power supply, sometimes WOL does not work the first time. You need a redundant path if you are in a remote site!

Basically multiple instances of MountWizzard4 could be up and running, but MountWizzard4 take up to 6 parallel connections to the mount. The documentation allows in total 10 connection each of the two ports (3490, 3492). This might overload the 10 micron system.

# Device connectivity

## ASCOM Device does not work / connect

ASCOM uses different types of drivers for the devices. Some of them need an environment in 32bit or 64bit like you application. So if you are using an 64bit application for imaging your drivers will be 64bit compatible. In this case the python installation also needs to be 64bit. Otherwise the connection will fail.

Modern CMOS cameras with large sensors normally work on 64bit mode.

Many device driver only support one connection at the same time. So if you imaging application already *took* a device, MountWizzard4 might be not able to connect anymore.

Please test your setup running with ASCOM suite (included in ASCOM platform installation) or any other programs you good know to test device functions outside MountWizzard4.

## INDI Device does not work / connect

MountWizzard4 does not talk directly to any device. This is also true for USB devices used together with INDI. It talks to a device driver. INDI provides the driver interface by running an INDI server in the background. This could be done when with starting KStars - as it has a device server build in - or a dedicated server running in the background without having KStars installed and running. This server is called INDI WebManager.

# Model building

# Updating IERS/SAT/MPC data

## Data could not be fetched from internet

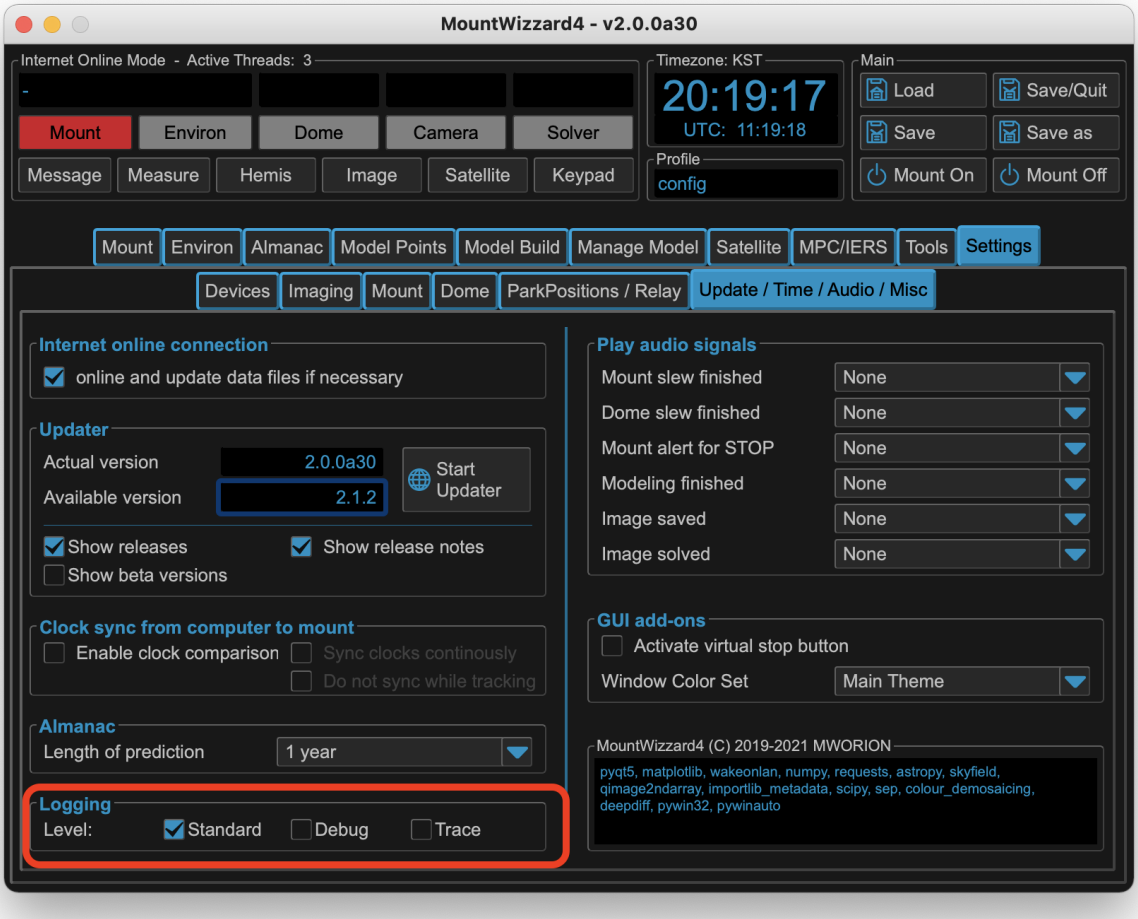
You need an internet connection and set MountWizzard4 in online mode to download new data for upload.

# Tracking satellites

# Logfiles and reports

## Where could I change the log level ?

The log level could be changed under settings misc. The default setting is warning. If you need analyzes, please go to debug. If a driver or mount connectivity is related as well, please go to trace. Please be aware that log file especially in trace mode could become big.




# Architectural topics and math

Within these pages I would explain how and why I made the architecture decisions for linking it with the 10micron mount computer. This might help for setting up or just explain the behavior you experience when using MountWizzard4. I do this also as my development documentation. There might be some faults and error in it. If you find one, please let me know. I would like to get MountWizzard4 from it's technical base as clean as possible.

## Handling time

One basic definition is that MountWizzard4 will use at any time the clock of the mount computer. Therefore MountWizzard4 polls julian date, difference utc - ut1, time sidereal. This allows full sync for any calculation to be made. No time from computer to mount is necessary, but could be done at any time (except during model build run). The mount mostly use the julian date representation except for model build where a local sidereal time (LST) is used. In this case MountWizzard4 just

stores the value and feed it back when the model is programmed. That's the reason why you should not change time during model run.

One important difference between MountWizzard4 and Mount exists. As I use skyfield as on of the frameworks with it's units for Angle, Coords, Time etc. I have to take the time definition of skyfield into account. Skyfield chooses TT (Terrestrial Time) as it's basic concept, whereas the mount uses UTC (Coordinated Universal Time) as reference. TT is a modern astronomical time standard defined by the International Astronomical Union. TT is distinct from the time scale often used as a basis for civil purposes, UTC. TT is indirectly the basis of UTC, via International Atomic Time (TAI).

## Precision of internal calculations

MountWizzard4 is using for all calculations the skyfield (https://rhodesmill.org/skyfield/) from Brandon Rhodes. As for the new command set offered with 10microns FW3.x it needs to calculate the alt/az coordinates for a satellite track each second for the entire track. As you would like to follow the as precise as possible I made some comparisons between the internal calculations done in 10micron mount and the results provided by skyfield.

In skyfield there is a chapter about satellite calculations and precision: https://rhodesmill.org/skyfield/earth-satellites.html#avoid-calling-the-observe-method Despite the fact that the observe method is expensive the difference in calculation time for a 900 step track is on my computer 120ms (using more precise observe method) to 7ms (using the less precise difference).

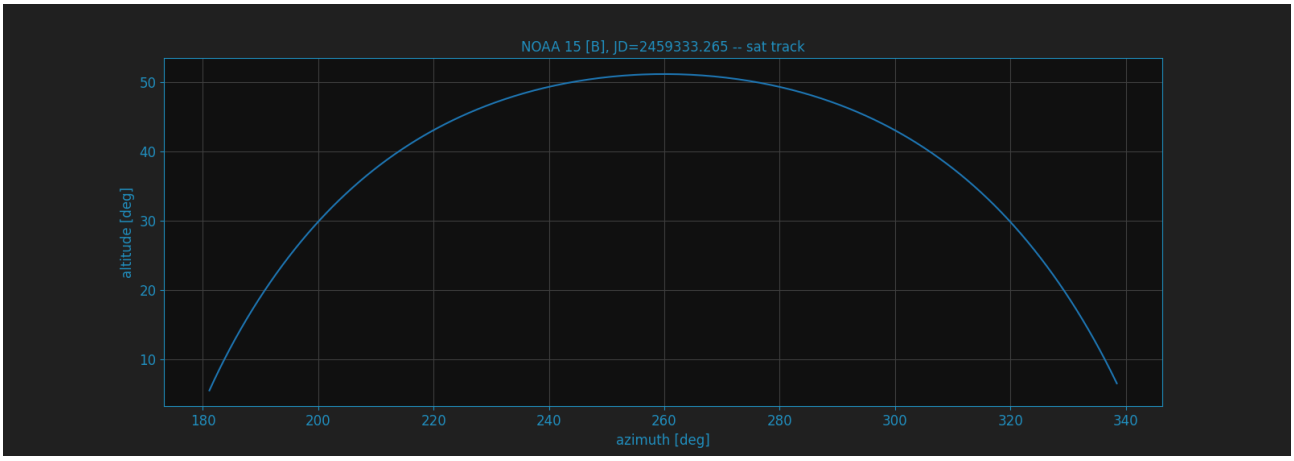
Brandon writes about it:

> While satellite positions are only accurate to about a kilometer anyway, accounting for light travel time only affected the position in this case by less than an additional tenth of a kilometer. This difference is not meaningful when compared to the uncertainty that is inherent in satellite positions to begin with, so you should neglect it and simply subtract GCRS-centered vectors instead as detailed above.

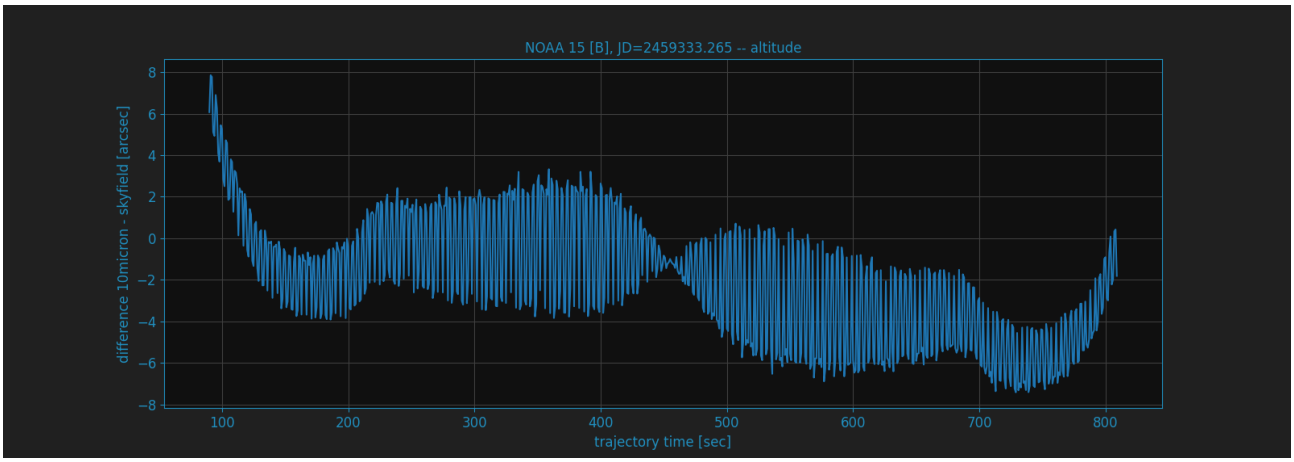
Here the charts for NOAA 15 [B] at julian date JD=2459333.26498 for the transit happening. The used TLE data was:

```
NOAA 15 [B]
1 25338U 98030A   21104.44658620  .00000027  00000-0  29723-4 0  9990
2 25338  98.6888 133.5239 0011555 106.3612 253.8839 14.26021970192127
```

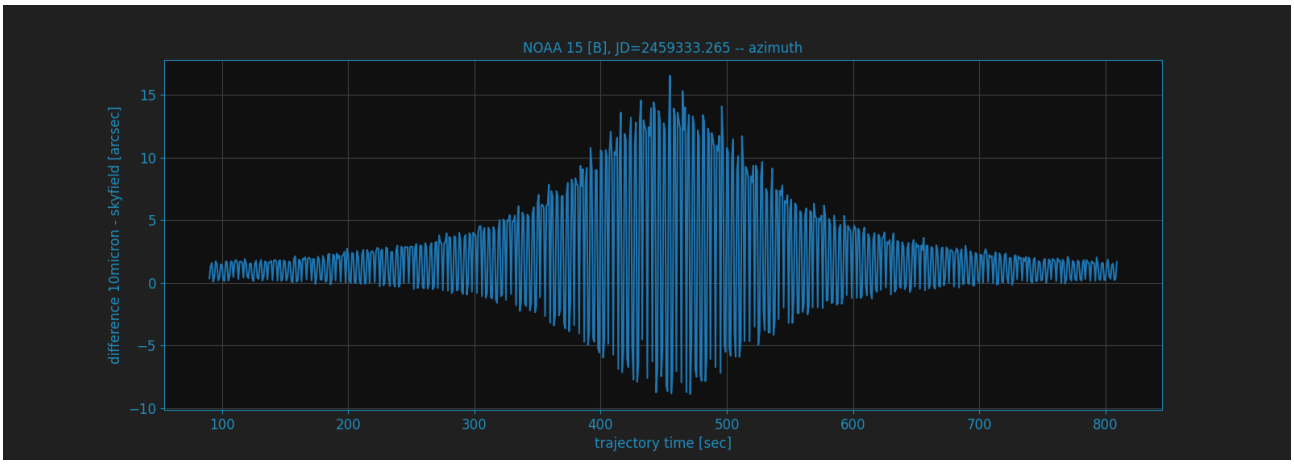
You could see the alt/az of the sat track.



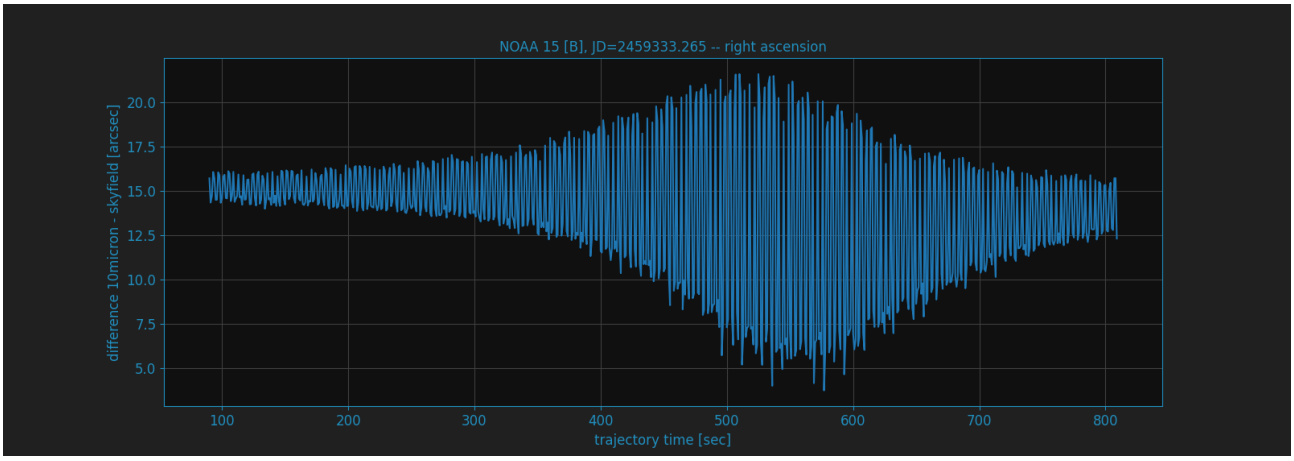

the difference for altitude between 10micron and skyfield



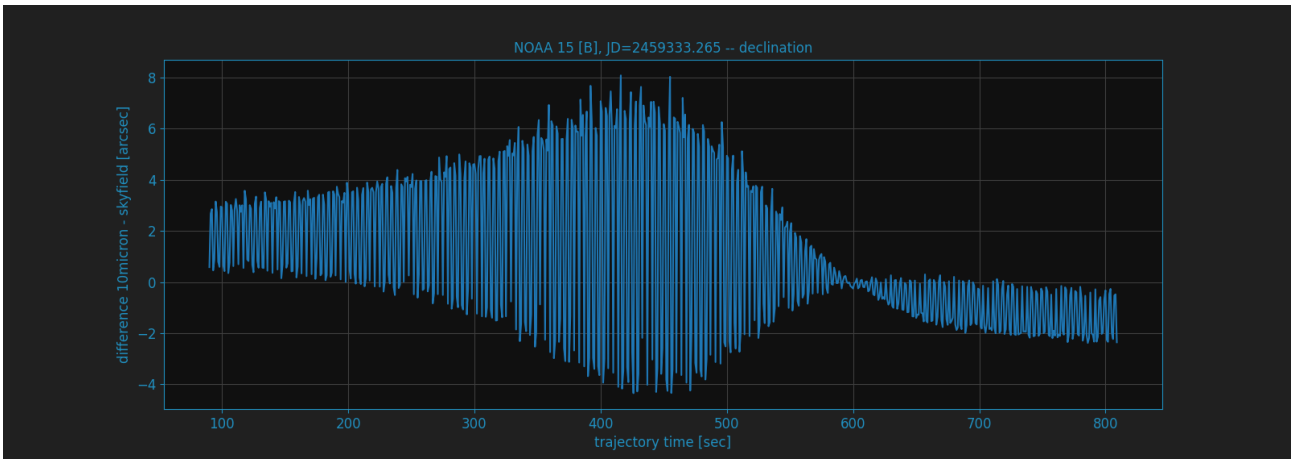

the difference for azimuth between 10micron and skyfield




the difference for right ascension between 10micron and skyfield



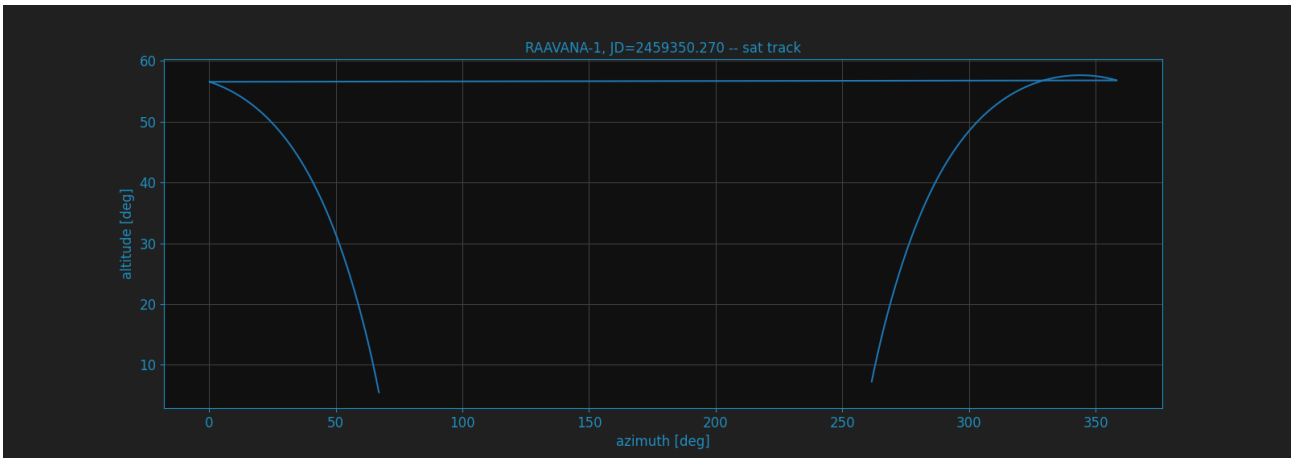

the difference for declination between 10micron and skyfield



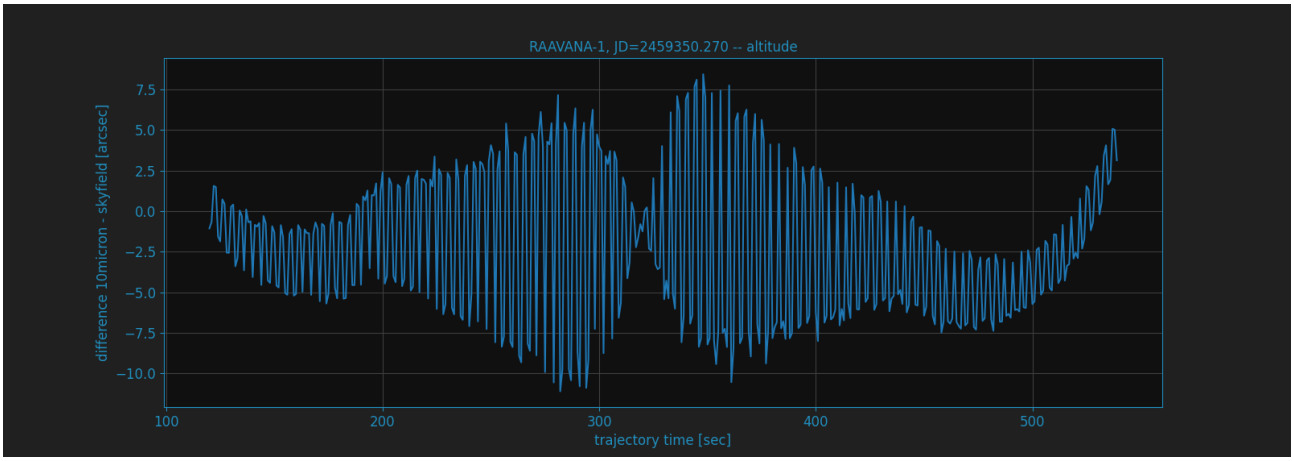

There is a set of plots for another satellite, which shows the same behavior. The used TLE data was:

```
RAAVANA-1
1 44329U 98067QE  21134.29933328  .00044698  00000-0  30736-3 0  9995
2 44329  51.6342 100.9674 0004554 122.3279 237.8162 15.74179130108776
```

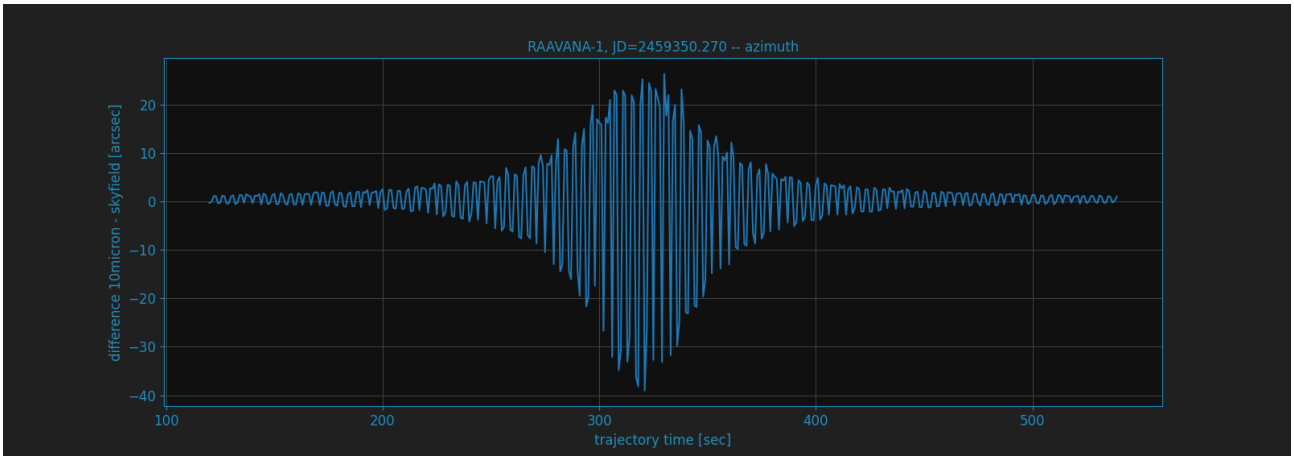
You could see the alt/az of the sat track.



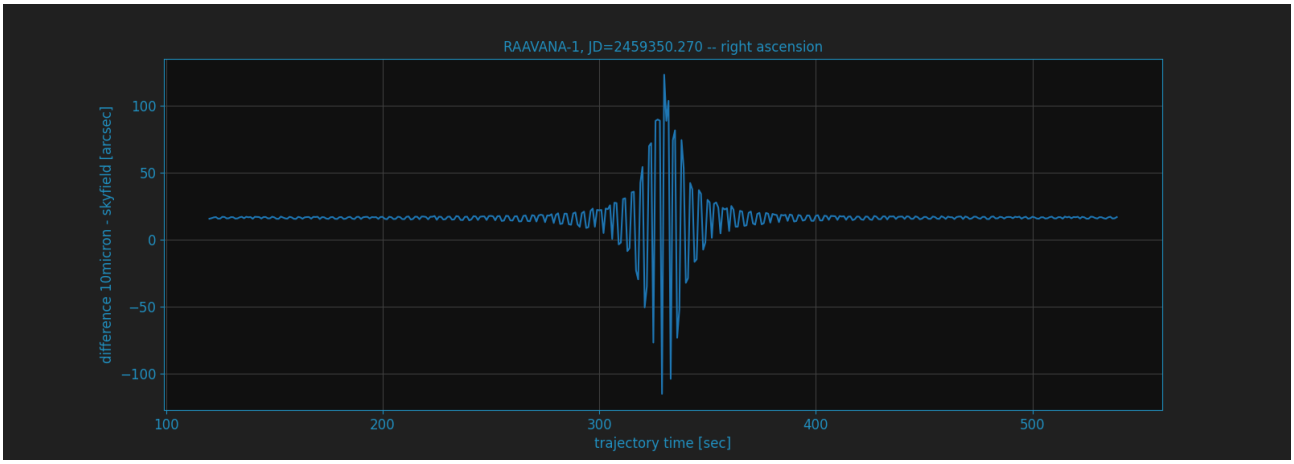

the difference for altitude between 10micron and skyfield



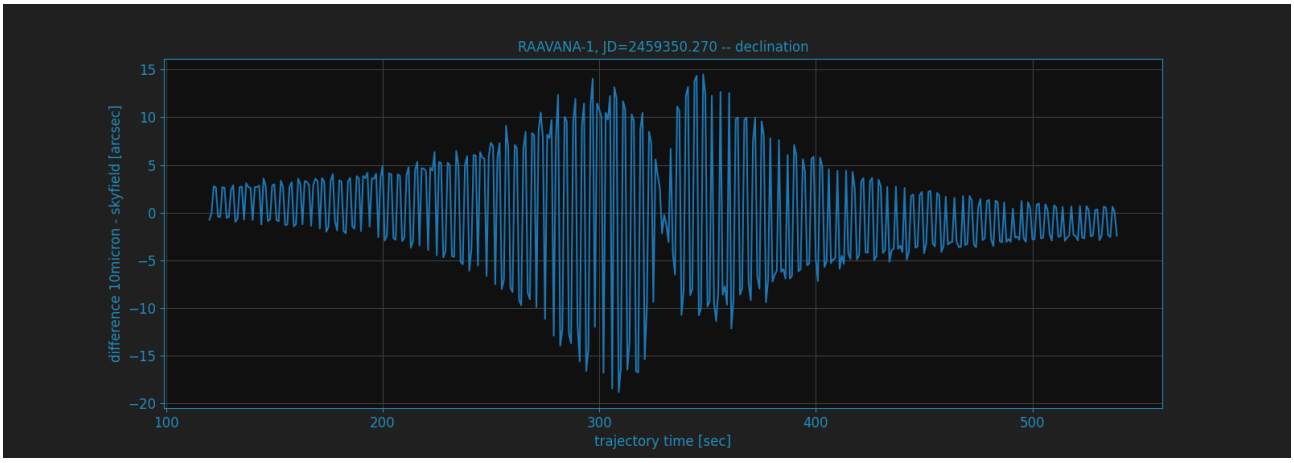

the difference for azimuth between 10micron and skyfield




the difference for right ascension between 10micron and skyfield




the difference for declination between 10micron and skyfield




For all calculations is valid:

- they are using refraction correction with the same values.
- the coordinates from 10micron are gathered with :TLEGEQJD#, :TLEGAZJD# commands
- julian date is in UTC time system
- 10micron firmware 3.0.4
- skyfield version 1.39

# Layering rules

The package layers are checked automatically ( `uv run lint-imports` , part of the CI lint job):

```
gui  ->  logic  ->  mountcontrol  ->  base
```

A layer may import only from the layers to its right. `mw4.mainApp` , `mw4.loader` and `mw4.deviceRegistry` are top-level modules that wire the layers together.

Inside `logic` :

- `controllers` (GUI independent logic, one package per GUI area) and `drivers` (device back-ends) do not import each other, and the controller areas do not import each other.
- `fits` , `profiles` , `file` and `databaseProcessing` are foundations: they import neither drivers nor controllers.
- `fits` , `profiles` , `databaseProcessing` and `controllers` do not use Qt, with two documented exceptions ( `QObject` workers with signals in `photometry` and `modelRun` ).

`QtWidgets`, `QtGui` and `QtCharts` may only be imported in `mw4.gui` and the application entry points (checked by ruff rule `TID251`). `QtCore` (`QObject`, `Signal`, `QThreadPool`) and `QtNetwork` are accepted in all layers.

# Changelog

## Beta versions

### Version 4.x

4.0.0b28

- this is a major release ! please try out in a separate work folder for test! for 2024 Hardware, please use the 3.2.5 or later firmware! MW4 will normally support only the latest 3 python versions.
- add: support for python 3.12 and 3.14, remove 3.7 - 3.11
- add: native support for Macs with Apple silicon
- add: windows automatic generation of Desktop link file for MW4 including dpi / scale parameters
- add: macOS automatic generation of .app bundle for MW4 including dpi / scale parameters
- add: linux automatic generation of .desktop file for MW4 including dpi / scale parameters
- add: show free disk space of work dir in the status bar
- add: comets / asteroids uploads now without 10micron installer
- add: support uploading mechanism for databases for macos and linux
- add: support for comet and asteroid data in extended downloads
- add: age parameter for MPC SAT databases before auto download
- add: split Satellite window into satellite world map and satellite horizon view
- add: support for extended json files for satellite data (new format)
- add: support for boltwood file reading
- add: support for compressed / uncompressed xisf files for image display (simple versions)
- add: support for compressed fits files for image display (gzip, bzip2)

- add: support for NINA alpaca devices, please install Alpaca plugin for NINA and enable in settings.
- add: new way of defining build point along celestial paths
- add: all settings are now available in separate windows -> easier to find and better overview
- add: GUI settings support transparency of windows if needed in all color modes
- add: new transparent mode for windows -> alpha value could be set 0..1
- add: volume setting for sound effects in settings, enable / disable sound effects in settings
- add: support for new 2024 HW - disabling WOL if not supported
- add: support for new 2024 HW - enabling auto power on setting if supported
- add: support for status of GPS module with PPS capability (no PPS functionality)
- improve: faster calculation of satellite track using internal calculations
- improve: faster handling of satellite sorting and filtering
- improve: reworked filter set with clear behavior auf selections
- improve: faster and more reliable uploading mechanism for databases
- improve: celestrak interface url's and retrieval strategy
- improve: optimized poll settings
- improve: refraction update only when needed
- improve: reduced cpu memory size of app
- improve: faster startup time
- improve: faster database loading
- improve: better visibility of editable fields
- improve: better status information
- improve: optimizing material look & feel for simulator
- improve: don't delete message list when color change
- improve: rewrite of the online documentation
- improve: no 10micron installer needed anymore for object data

- improve: schedule and threads in waiting times (slow machines)
- improve: change from complicated json dump to YAML format for config file
- improve: more detailed earth map in satellite view
- improve: optimized GUI (imaging settings together ect.)
- improve: environment devices (now 4 generic ones), online and boltwood are generic
- change: moving PyQt5 to major PySide6
- change: moving libraries to latest versions
- change: plate solving features more reliable
- change: database handling for MPC files
- change: move INDI support to indipyclient library
- change: move ASCOM / ALPACA support to official Alpyca library
- change: move installer to separate repo (InstallerMW4) - for v4 not needed anymore, but still available for v3.x
- remove: old windows automation as it is not needed anymore
- remove: embedded documentation and replace with online link
- remove: automation of 10micron installer, replaced with web interface
- remove: updater for MW4 (use uv installer instead)
- remove: automatic profile conversion from x.x to 4.x (too complex)
- remove: blending profiles as it did not work so far.
- remove: native support for NINA (replaced by ALPACA Pluging on NINA)
- fix: typos